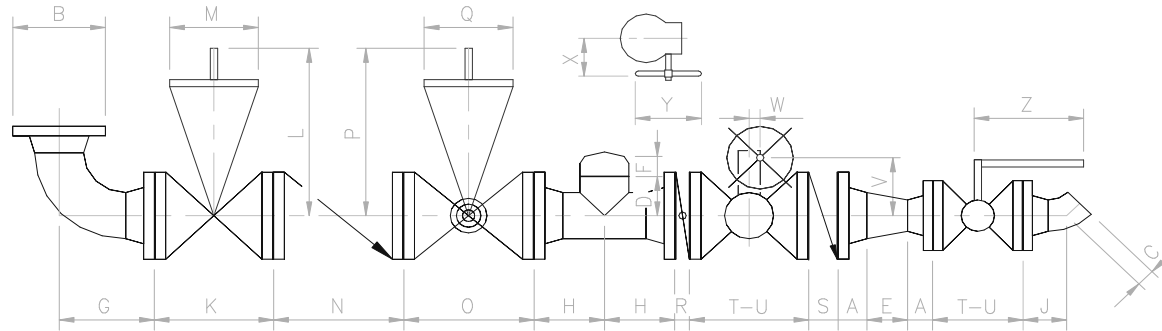


150# RF PIPING DIMENSIONS



SIZE	PIPE O.D.	WN FLG	OD FLG	45 ELL	TEE	REDUCER	CAP	LR ELL & WN FLG	TEE & WN FLG	45 ELL & WN FLG	GATE VALVE			CHECK VALVE	GLOBE VALVE			BUTTERFLY VALVE (1)	WAFER CHECK VALVE (2)	BALL VALVE (3)							SIZE	NO. BOLTS	STUD BOLT SIZE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
											K	L	M		N	O	P			Q	R	S	T	U	V	W			X	Y	Z	DIAMETER	LENGTH (4)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
		A	B	C	D	E	F	G	H	J																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														

(1) Posi-seal Butterfly Valves

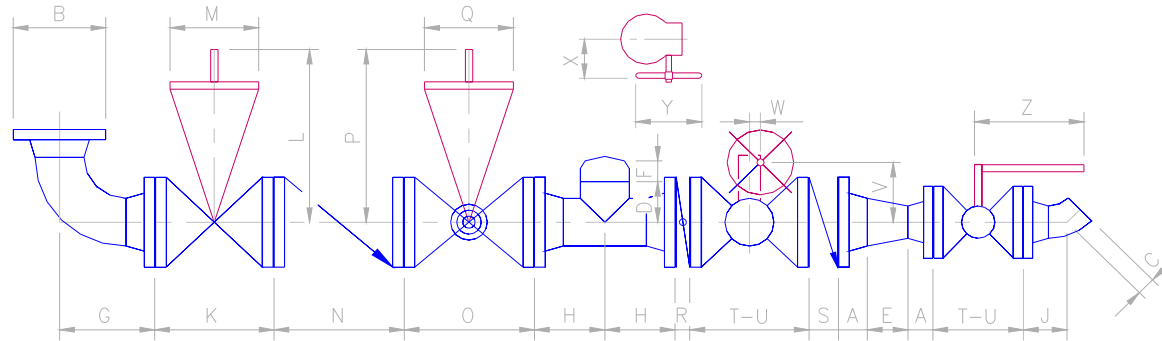
(2) Crane Duo-Chek II

(3) TK Ball Valves

(4) Length does not include crowns

CALC. BOX

300# RF PIPING DIMENSIONS



SIZE	PIPE O.D.	WN FLG	OD FLG	45 ELL	TEE	REDUCER	CAP	LR ELL & WN FLG	TEE & WN FLG	45 ELL & WN FLG	GATE VALVE			CHECK VALVE	GLOBE VALVE			WAFER BUTTERFLY VALVE (1)	WAFER CHECK VALVE (2)	BALL VALVE (3)							SIZE	NO. BOLTS	STUD BOLT SIZE	
											K	L	M		O	P	Q			FULL PORT	RED. PORT	VALVE TO HW (FB)	HW OFFSE T (FB)	HW PROJ (FB)	HW OD (FB)	LE VE R LE			DIAMETER	LENGTH (4)
1"	1 5/16	2 7/16	4 7/8	7/8	1 1/2	2	1 1/2	3 15/16	3 15/16	3 5/16										6 1/2							1"	4	5/8	3 1/4
1 1/2"	1 7/8	2 11/16	6 1/8	1 1/8	2 1/4	2 1/2	1 1/2	4 15/16	4 15/16	3 13/16	7 1/2	16 3/4	8							7 1/2							1 1/2"	4	3/4	3 3/4
2"	2 3/8	2 3/4	6 1/2	1 3/8	2 1/2	3	1 1/2	5 3/4	5 1/4	4 1/8	8 1/2	18	8	10 1/2	10 1/2	17 3/4	9		2 3/8	8 1/2	8 1/2	4 1/16				23	2"	8	5/8	3 1/2
2 1/2"	2 7/8	3	7 1/2	1 3/4	3	3 1/2	1 1/2	6 3/4	6	4 3/4	9 1/2	19	8	11 1/2	11 1/2	19	10		2 5/8								2 1/2"	8	3/4	4
3"	3 1/2	3 1/8	8 1/4	2	3 3/8	3 1/2	2	7 5/8	6 1/2	5 1/8	11 1/8	23 1/4	9	12 1/2	12 1/2	20 1/2	10	1 7/8	2 7/8	11 1/8	11 1/8	7 1/16	3 1/8	10 1/16	17 1/8	33	3"	8	3/4	4 1/4
4"	4 1/2	3 3/8	10	2 1/2	4 1/8	4	2 1/2	9 3/8	7 1/2	5 7/8	12	28 1/4	10	14	14	24 3/4	14	2 1/8	2 7/8	12	12	8	3 1/8	10	17 1/8	44	4"	8	3/4	4 1/2
6"	6 5/8	3 7/8	12 1/2	3 3/4	5 5/8	5 1/2	3 1/2	12 7/8	9 1/2	7 5/8	15 7/8	38 1/2	14	17 1/2	17 1/2	29 3/4	18	2 5/16	3 7/8	15 7/8	15 7/8	10 13/16	5	11 1/2	24		6"	12	3/4	5
8"	8 5/8	4 3/8	15	5	7	6	4	16 3/8	11 3/8	9 3/8	16 1/2	47	16	21	22	36 1/2	24	2 7/8	5	19 3/4	19 3/4	12 1/2	5	11 1/2	24		8"	12	7/8	5 1/2
10"	10 3/4	4 5/8	17 1/2	6 1/4	8 1/2	7	5	19 5/8	13 1/8	10 7/8	18	56 1/2	20	24 1/2				3 5/16	5 3/4	22 3/8	22 3/8	15 5/8	3 11/16	18 15/16	30 1/4		10"	16	1	6 1/4
12"	12 3/4	5 1/8	20 1/2	7 1/2	10	8	6	23 1/8	15 1/8	12 5/8	19 3/4	64 1/4	20	28				3 5/8	7 1/8	25 1/2	25 1/2	16 13/16	3 11/16	18 15/16	30 1/4		12"	16	1 1/8	6 3/4
14"	14	5 5/8	23	8 3/4	11	13	6 1/2	26 5/8	16 5/8	14 3/8	30	75 1/4	27					4 5/8	8 3/4	30	30	18 7/16	3 11/16	18 15/16	30 1/4		14"	20	1 1/8	7
16"	16	5 3/4	25 1/2	10	12	14	7	29 3/4	17 3/4	15 3/4	33	81	27					5 1/4	9 1/8	33	33	20 1/8	3 11/16	18 15/16	39		16"	20	1 1/4	7 1/2
18"	18	6 1/4	28	11 1/4	13 1/2	15	8	33 1/4	19 3/4	17 1/2	36	91 1/2	30					5 7/8	10 3/8	36		24 13/16	4 7/8	21 1/2	39		18"	24	1 1/4	7 3/4
20"	20	6 3/8	30 1/2	12 1/2	15	20	9	36 3/8	21 3/8	18 7/8	39	99 3/4	36					6 1/4	11 1/2	39	39	16 1/2	4 7/8	21 1/2	39		20"	24	1 1/4	8 1/4
24"	24	6 5/8	36	15	17	20	10 1/2	42 5/8	23 5/8	21 5/8	45	120 1/2	36					7	12 1/2	45	45	29 5/16	4 7/8	21 1/2	39		24"	24	1 1/2	9 1/4
26"	26	7 1/4	38 1/4	16	19 1/2	24	10 1/2	46 1/4	26 3/4	23 1/4									14	49		34 1/4	5 3/4	31 3/8	30 1/4		26"	28	1 5/8	10 1/4
28"	28	7 3/4	40 3/4	17 1/4	20 1/2	24	10 1/2	49 3/4	28 1/4	25										53		35 1/8	5 3/4	31 3/8	30 1/4		28"	28	1 5/8	10 3/4
30"	30	8 1/4	43	18 1/2	22	24	10 1/2	53 1/4	30 1/4	26 3/4								9 1/2	14 1/2	55		37 1/8	5 3/4	31 3/8	30 1/4		30"	28	1 3/4	11 1/2
36"	36	9 1/2	50	22 1/4	26 1/2	24	10 1/2	63 1/2	36	31 3/4								10 3/4	19	68		40 7/8	16	37 1/4	30 1/4		36"	32	2	13
42"	42	7 7/8	50 3/4	26	30	24	12	70 7/8	37 7/8	33 7/8									22 3/8								42"	36	2	14

(1) Posi-seal Butterfly Valves

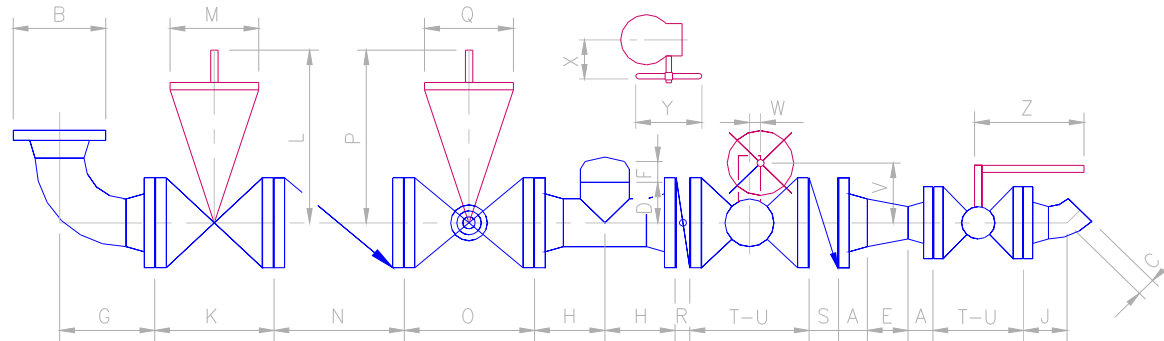
(2) Crane Duo-Chek II

(3) TK Ball Valves

(4) Length does not include crowns

CALC. BOX

600# RF PIPING DIMENSIONS



SIZE	PIPE O.D.	WN FLG	OD FLG	45 ELL	TEE	REDUCER	CAP	LR ELL & WN FLG	TEE & WN FLG	45 ELL & WN FLG	GATE VALVE			CHECK VALVE	GLOBE VALVE			BUTTERFLY VALVE (1)	WAFER CHECK VALVE (2)	BALL VALVE (3)							SIZE	NO. BOLTS	STUD BOLT SIZE	
																				FULL PORT	RED. PORT	VALVE TO HW	HW OFFSET	HW PROJ	HW OD	LEVER LENGT H			DIAMETER	LENGTH (4)
1"	1 5/16	2 11/16	4 7/8	7/8	1 1/2	2	1 1/2	4 3/16	4 3/16	3 9/16									8 1/2	U	V	W	X	Y	Z	1"	4	5/8	3 3/4	
1 1/2"	1 7/8	3	6 1/8	1 1/8	2 1/4	2 1/2	1 1/2	5 1/4	5 1/4	4 1/8			9 1/2					9 1/2		3 1/8					15	1 1/2"	4	3/4	4 1/4	
2"	2 3/8	3 1/8	6 1/2	1 3/8	2 1/2	3	1 1/2	6 1/8	5 5/8	4 1/2	11 1/2	18 1/4	8	11 1/2	11 1/2	19	10		2 3/8	11 1/2	11 1/2	5 13/16				23	2"	8	5/8	4 1/4
2 1/2"	2 7/8	3 3/8	7 1/2	1 3/4	3	3 1/2	1 1/2	7 1/8	6 3/8	5 1/8	13	22 1/4	9	13	13	21 1/4	12		2 5/8							2 1/2"	8	3/4	4 3/4	
3"	3 1/2	3 1/2	8 1/4	2	3 3/8	3 1/2	2	8	6 7/8	5 1/2	14	25 3/4	10	14	14	23 1/2	12	1 13/16	2 7/8	14	14	7 1/16	3 1/8	10 1/16	17 1/8	33	3"	8	3/4	5
4"	4 1/2	4 1/4	10 3/4	2 1/2	4 1/8	4	2 1/2	10 1/4	8 3/8	6 3/4	17	31 1/2	14	17	17	27 1/2	18	2 3/8	3 1/8	17	17	8	3 1/8	10 5/8	17 1/8	44	4"	8	7/8	5 3/4
6"	6 5/8	4 7/8	14	3 3/4	5 5/8	5 1/2	3 1/2	13 7/8	10 1/2	8 5/8	22	42 3/4	20	22	22	35	24	2 7/8	5 3/8	22	22	10 13/16	5	11 1/2	24		6"	12	1	6 3/4
8"	8 5/8	5 1/2	16 1/2	5	7	6	4	17 1/2	12 1/2	10 1/2	26	52 1/4	24	26				3 1/2	6 1/2	26	26	14	3 11/16	18 15/16	30 1/4		8"	12	1 1/8	7 3/4
10"	10 3/4	6 1/4	20	6 1/4	8 1/2	7	5	21 1/4	14 3/4	12 1/2	31	62 1/4	27	31				4 1/2	8 3/8	31	31	15 1/2	3 11/16	18 15/16	30 1/4		10"	16	1 1/4	8 1/2
12"	12 3/4	6 3/8	22	7 1/2	10	8	6	24 3/8	16 3/8	13 7/8	33	70	27	33				5 1/2	9	33	33	17 7/16	3 11/16	18 15/16	30 1/4		12"	20	1 1/4	8 3/4
14"	14	6 3/4	23 3/4	8 3/4	11	13	6 1/2	27 3/4	17 3/4	15 1/2	35	77 1/4	30					6 1/4	10 3/4	35	35	21 1/2	4 7/8	21 1/2	39		14"	20	1 3/8	9 1/4
16"	16	7 1/4	27	10	12	14	7	31 1/4	19 1/4	17 1/4	39	83 3/4	30					7	12	39	39	20 1/16	4 7/8	21 1/2	39		16"	20	1 1/2	10
18"	18	7 1/2	29 1/4	11 1/4	13 1/2	15	8	34 1/2	21	18 3/4	43	93 3/4	36					7 3/4	14 1/4	43	43	25 5/8	7 3/4	28	30 1/4		18"	20	1 5/8	10 3/4
20"	20	7 3/4	32	12 1/2	15	20	9	37 3/4	22 3/4	20 1/4	47	104 1/2	36					9	14 1/2	47	47	27 3/8	7 3/4	28	30 1/4		20"	24	1 5/8	11 1/2
24"	24	8 1/4	37	15 1/2	17	20	10 1/2	44 1/4	25 1/4	23 1/4	55	126	42					9	17 1/4	55	55	31 3/8	9 3/4	30	30 1/4		24"	24	1 7/8	13

(1) Posi-seal Butterfly Valves

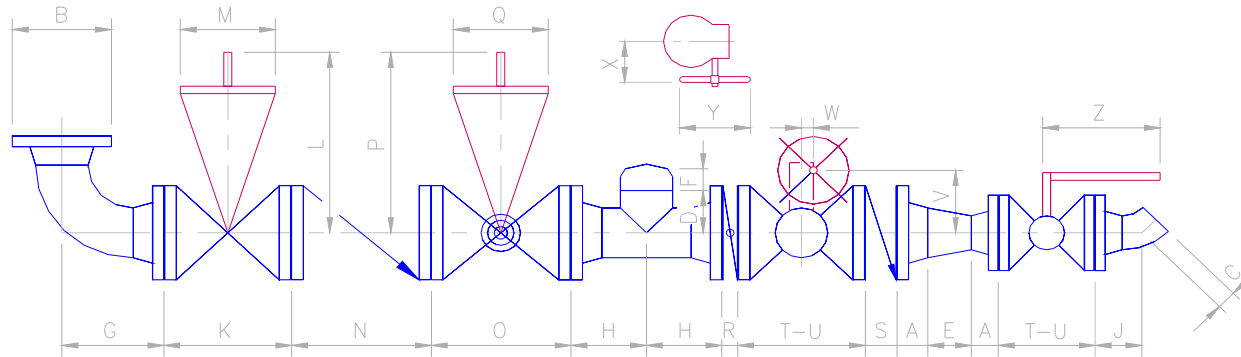
(2) Crane Duo-Chek II

(3) TK Ball Valves

(4) Length does not include crowns

CALC. BOX

900# RF PIPING DIMENSIONS



SIZE	PIPE O.D.	WN FLG	OD FLG	45 ELL	TEE	REDUCER	CAP	LR ELL & WN FLG	TEE & WN FLG	45 ELL & WN FLG	GATE VALVE			CHECK VALVE	GLOBE VALVE			WAFER BUTTERFLY VALVE (1)	WAFER CHECK VALVE (2)	BALL VALVE (3)							SIZE	NO. BOLTS	STUD BOLT SIZE	
											K	L	M		O	P	Q			T	U	V	W	X	Y	Z			DIAMETER	LENGTH (4)
1"	1 5/16	3 1/8	5 7/8	7/8	1 1/2	2	1 1/2	4 5/8	4 5/8	4										10						9	1"	4	7/8	5
1 1/2"	1 7/8	3 1/2	7	1 1/8	2 1/4	2 1/2	1 1/2	5 3/4	5 3/4	4 5/8										12						15	1 1/2"	4	1	5 1/2
2"	2 3/8	4 1/4	8 1/2	1 3/8	2 1/2	3	1 1/2	7 1/4	6 3/4	5 5/8								2 3/4	2 3/4	14 1/2	14 1/2					33	2"	8	7/8	5 3/4
2 1/2"	2 7/8	4 3/8	9 5/8	1 3/4	3	3 1/2	1 1/2	8 1/8	7 3/8	6 1/8								3 1/4	3 1/4								2 1/2"	8	1	6 1/4
3"	3 1/2	4 1/4	9 1/2	2	3 3/8	3 1/2	2	8 3/4	7 5/8	6 1/4	15	27 1/4	12	15	15	24	12	3 1/4	3 1/4	15	15	7 1/16	3 1/8	10 1/16	17 1/8	33	3"	8	7/8	5 3/4
4"	4 1/2	4 3/4	11 1/2	2 1/2	4 1/8	4	2 1/2	10 3/4	8 7/8	7 1/4	18	31 1/2	14	18	18	29 1/2	20	4	4	18	18	9 9/16	5	11 1/2	24	44	4"	8	1 1/8	6 3/4
6"	6 5/8	5 3/4	15	3 3/4	5 5/8	5 1/2	3 1/2	14 3/4	11 3/8	9 1/2	24	42 3/4	20	24	24	37 3/4	27	6 1/4	6 1/4	24	24	10 13/16	5	11 1/2	24		6"	12	1 1/8	7 3/4
8"	8 5/8	6 5/8	18 1/2	5	7	6	4	18 5/8	13 5/8	11 5/8	29	52 1/2	24	29				8 1/8	8 1/8	29	29	16	3 11/16	18 15/16	30 1/4		8"	12	1 3/8	8 3/4
10"	10 3/4	7 1/2	21 1/2	6 1/4	8 1/2	7	5	22 1/2	16	13 3/4	33	62 1/4	27					9 1/2	9 1/2	33	33	15 5/8	3 11/16	18 15/16	30 1/4		10"	16	1 3/8	9 1/4
12"	12 3/4	8 1/8	24	7 1/2	10	8	6	26 1/8	18 1/8	15 5/8	38	73 1/2	30					11 1/2	11 1/2	38	38	18 3/8	3 11/16	18 15/16	30 1/4		12"	20	1 3/8	10
14"	14	8 5/8	25 1/4	8 3/4	11	13	6 1/2	29 5/8	19 5/8	17 3/8	40 1/2	77 1/4	30					14	14	40 1/2	40 1/2	21 1/2	4 7/8	21 1/2	39		14"	20	1 1/2	10 3/4
16"	16	8 3/4	27 3/4	10	12	14	7	32 3/4	20 3/4	18 3/4	44 1/2	85 3/4	36					15 1/8	15 1/8	44 1/2	44 1/2	20 1/16	4 7/8	21 1/2	39		16"	20	1 5/8	11 1/4
18"	18	9 1/4	31	11 1/4	13 1/2	15	8	36 1/4	22 3/4	20 1/2								17 3/4	17 3/4	48		25 13/16	9 3/4	29	30 1/4		18"	20	1 7/8	12 3/4
20"	20	10	33 3/4	12 1/2	15	20	9	40	25	22 1/2								17 3/4	17 3/4	52	52	28 7/8	5 3/4	31 3/8	30 1/4		20"	20	2	13 1/2
24"	24	11 3/4	41	15	17	20	10 1/2	47 3/4	28 3/4	26 3/4								19 1/2	19 1/2	61	61	30 15/16	5 3/4	31 3/8	30 1/4		24"	20	2 1/2	17 1/4

(1) Posi-seal Butterfly Valves

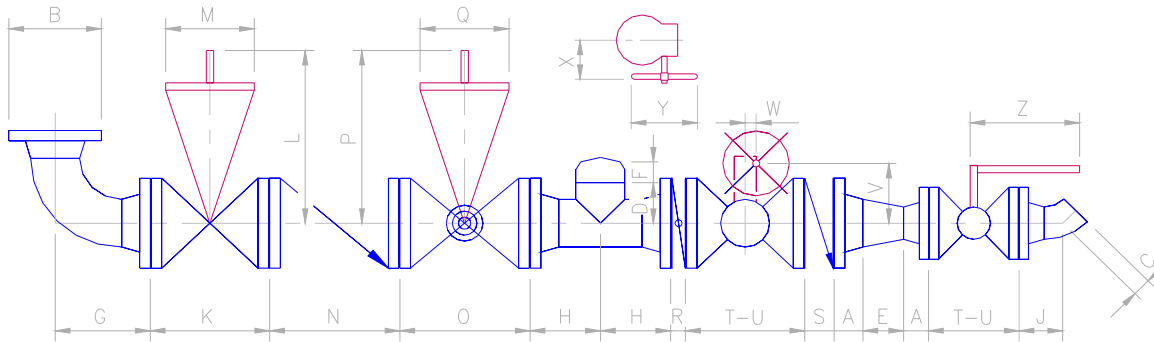
(2) Crane Duo-Chek II

(3) TK Ball Valves

(4) Length does not include crowns

CALC. BOX

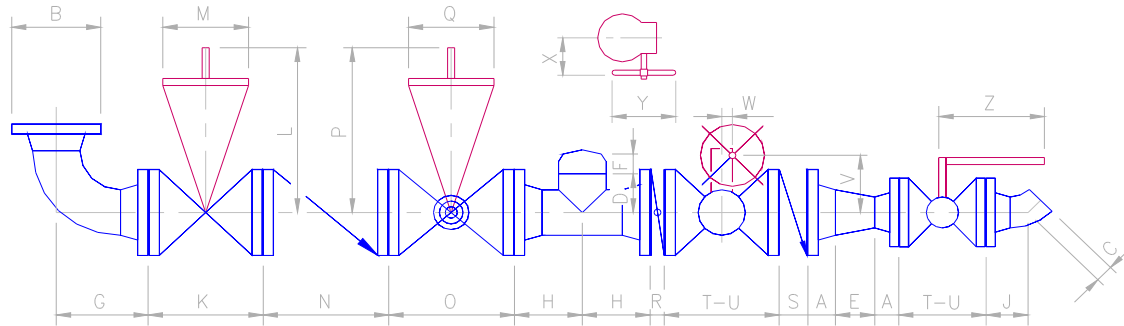
1500# RF PIPING DIMENSIONS



SIZE	PIPE O.D.	WN FLG	OD FLG	45 ELL	TEE	REDUCER	CAP	LR ELL & WN FLG	TEE & WN FLG	45 ELL & WN FLG	GATE VALVE			CHECK VALVE	GLOBE VALVE			WATER BUTTERFLY VALVE (1)	WAFER CHECK VALVE (2)	BALL VALVE (3)							SIZE	NO. BOLTS	STUD BOLT SIZE	
											K	L	M		O	P	Q			FULL PORT	RED. PORT	VALVE TO HW (FB)	HW OFFSE T (FB)	HW PROJ (FB)	HW OD (FB)	LE V E R L E			DIAMETER	LENGTH (4)
1"	1 5/16	3 1/8	5 7/8	7/8	1 1/2	2	1 1/2	4 5/8	4 5/8	4	10	16	8							10		3 1/8				9	1"	4	7/8	5
1 1/2"	1 7/8	3 1/2	7	1 1/8	2 1/4	2 1/2	1 1/2	5 3/4	5 3/4	4 5/8	12	20	9	12						12		5 13/32				15	1 1/2"	4	1	5 1/2
2"	2 3/8	4 1/4	8 1/2	1 3/8	2 1/2	3	1 1/2	7 1/4	6 3/4	5 5/8	14 1/2	22 1/8	10	14 1/2	14 1/2	25 1/8	14		2 3/4	14 1/2	14 1/2	6 1/2				33	2"	8	7/8	5 3/4
2 1/2"	2 7/8	4 3/8	9 5/8	1 3/4	3	3 1/2	1 1/2	8 1/8	7 3/8	6 1/8	16 1/2	26 3/8	12	16 1/2	16 1/2	28 1/8	18		3 1/4								2 1/2"	8	1	6 1/4
3"	3 1/2	4 7/8	10 1/2	2	3 3/8	3 1/2	2	9 3/8	8 1/4	6 7/8	18 1/2	28	14	18 1/2	18 1/2	33 1/2	24		3 1/4	18 1/2	18 1/2	7 3/8	3 1/8	10 1/4	17 1/8	33	3"	8	1 1/8	7
4"	4 1/2	5 1/8	12 1/4	2 1/2	4 1/8	4	2 1/2	11 1/8	9 1/4	7 5/8	21 1/2	33	16	21 1/2					4	21 1/2	21 1/2	9 9/16	5	11 1/2	24	44	4"	8	1 1/4	7 3/4
6"	6 5/8	7	15 1/2	3 3/4	5 5/8	5 1/2	3 1/2	16	12 5/8	10 3/4	27 3/4	47	24	27 3/4					6 1/4	27 3/4	27 3/4	12 9/16	5	18 15/16	30 1/4		6"	12	1 3/8	10 1/4
8"	8 5/8	8 5/8	19	5	7	6	4	20 5/8	15 5/8	13 5/8	32 3/4	55	27	32 3/4					8 1/8	32 3/4	32 3/4	14 1/2	3 11/16	18 15/16	30 1/4		8"	12	1 5/8	11 1/2
10"	10 3/4	10 1/4	23	6 1/4	8 1/2	7	5	25 1/4	18 3/4	16 1/2									9 3/4	39	39	18 7/8	4 7/8	21 1/2	39		10"	12	1 7/8	13 1/4
12"	12 3/4	11 3/8	26 1/2	7 1/2	10	8	6	29 3/8	21 3/8	18 7/8									12	44 1/2	44 1/2	21 11/16	4 7/8	21 1/2	39		12"	16	2	14 3/4
14"	14	12	29 1/2	8 3/4	11	13	6 1/2	33	23	20 3/4									14	49 1/2	49 1/2	23 3/4	9 3/4	29	39		14"	16	2 1/4	16
16"	16	12 1/2	32 1/2	10	12	14	7	36 1/2	24 1/2	22 1/2									15 1/8	54 1/2	54 1/2	25 3/4	9 3/4	29	30 1/4		16"	16	2 1/2	17 1/2
18"	18	13 1/8	36	11 1/4	13 1/2	15	8	40 1/8	26 5/8	24 3/8									18 7/16	60 1/2		30 1/2	5 3/4	31 3/8	30 1/4		18"	16	2 3/4	19 1/2
20"	20	14 1/4	38 3/4	12 1/2	15	20	9	44 1/4	29 1/4	26 3/4									21	65 1/2	65 1/2	31 1/4	5 3/4	31 3/8	30 1/4		20"	16	3	21 1/2
24"	24	16 1/4	46	15	17	20	10 1/2	52 1/4	33 1/4	31 1/4									22	76 1/2	76 1/2	37 7/8	16	37 1/4	30 1/4		24"	16	3 1/2	24 1/2

- (1) CALC. BOX
 (2) Crane Duo-Chek II
 (3) TK Ball Valves
 (4) Length does not include crowns

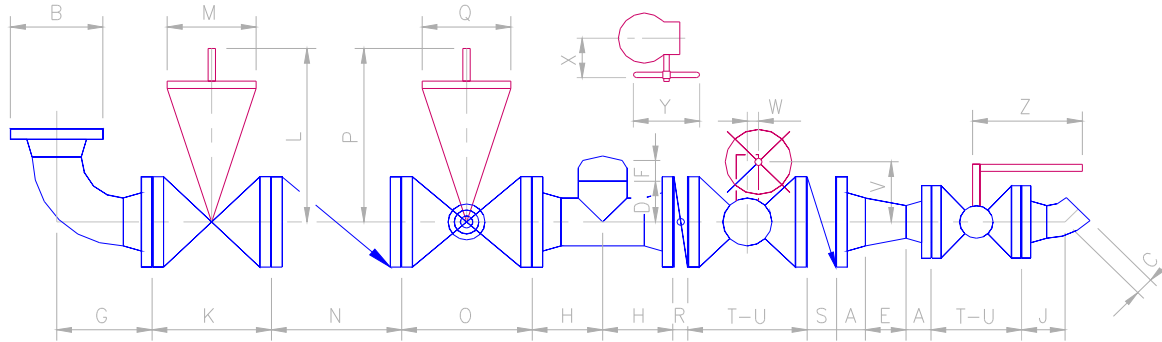
900# RTJ PIPING DIMENSIONS



SIZE	PIPE O.D.	WN FLG	OD FLG	45 ELL	TEE	REDUCER	CAP	LR ELL & WN FLG	TEE & WN FLG	45 ELL & WN FLG	GATE VALVE			CHECK VALVE	GLOBE VALVE				BALL VALVE (3)							SIZE	NO. BOLTS	STUD BOLT SIZE		RING INFO.		
																			FULL PORT	RED. PORT	VALVE TO HW (FB)	HW OFFSE T (FB)	HW PROJ (FB)	HW OD (FB)	LE VE R LE			DIAMETER	LENGTH (4)	RING NO.	GAP	
		A	B	C	D	E	F	G	H	J	K	L	M	N	O	P	Q	T	U	V	W	X	Y	Z								
1"	1 5/16	3 1/8	5 7/8	7/8	1 1/2	2	1 1/2	4 5/8	4 5/8	4								10						9	1"	4	7/8	5	R-16	5/32		
1 1/2"	1 7/8	3 1/2	7	1 1/8	2 1/4	2 1/2	1 1/2	5 3/4	5 3/4	4 5/8								12						15	1 1/2"	4	1	5 1/2	R-20	5/32		
2"	2 3/8	4 5/16	8 1/2	1 3/8	2 1/2	3	1 1/2	7 5/16	6 13/16	5 11/16				14 5/8	14 1/2	23	13 3/4	14 5/8	14 5/8					33	2"	8	7/8	5 3/4	R-24	1/8		
2 1/2"	2 7/8	4 7/16	9 5/8	1 3/4	3	3 1/2	1 1/2	8 3/16	7 7/16	6 3/16				16 5/8	16 1/2	24 1/2	15 1/2							2 1/2"	8	1	6 1/4	R-27	1/8			
3"	3 1/2	4 5/16	9 1/2	2	3 3/8	3 1/2	2	8 13/16	7 11/16	6 5/16	15 1/8	27 1/4	12	15 1/8	15 1/8	24	12	15 1/8	15 1/8	7 1/16	3 1/8	10 1/16	17 1/8	33	3"	8	7/8	6	R-31	5/32		
4"	4 1/2	4 13/16	11 1/2	2 1/2	4 1/8	4	2 1/2	10 13/16	8 15/16	7 5/16	18 1/8	31 1/2	14	18 1/8	18 1/8	29 1/2	20	18 1/8	18 1/8	9 9/16	5	11 1/2	24	44	4"	8	1 1/8	7	R-37	5/32		
6"	6 5/8	5 13/16	15	3 3/4	5 5/8	5 1/2	3 1/2	14 13/16	11 7/16	9 9/16	24 1/8	42 3/4	20	24 1/8	24 1/8	37 3/4	27	24 1/8	24 1/8	10 13/16	5	11 1/2	24		6"	12	1 1/8	7 3/4	R-45	5/32		
8"	8 5/8	6 11/16	18 1/2	5	7	6	4	18 11/16	13 11/16	11 11/16	29 1/8	52 1/2	24	29 1/8				29 1/8	29 1/8	16	3 11/16	18 15/16	30 1/4		8"	12	1 3/8	9	R-49	5/32		
10"	10 3/4	7 9/16	21 1/2	6 1/4	8 1/2	7	5	22 9/16	16 1/16	13 13/16	33 1/8	62 1/4	27					33 1/8	33 1/8	15 5/8	3 11/16	18 15/16	30 1/4		10"	16	1 3/8	9 1/2	R-53	5/32		
12"	12 3/4	8 3/16	24	7 1/2	10	8	6	26 3/16	18 3/16	15 11/16	38 1/8	73 1/2	30					38 1/8	38 1/8	18 3/8	3 11/16	18 15/16	30 1/4		12"	20	1 3/8	10 1/4	R-57	5/32		
14"	14	8 13/16	25 1/4	8 3/4	11	13	6 1/2	29 13/16	19 13/16	17 9/16	40 5/8	77 1/4	30					40 7/8	40 7/8	21 1/2	4 7/8	21 1/2	39		14"	20	1 1/2	11 1/4	R-62	5/32		
16"	16	8 15/16	27 3/4	10	12	14	7	32 15/16	20 15/16	18 15/16	44 5/8	85 3/4	36					44 1/2	44 1/2	20 1/16	4 7/8	21 1/2	39		16"	20	1 5/8	11 3/4	R-66	5/32		
18"	18			11 1/4	13 1/2	15	8	36 1/2	23	20 3/4								48 1/2		25 13/16	9 3/4	29	30 1/4		18"	20	1 7/8	13 1/2	R-70	3/16		
20"	20	10 1/4	33 3/4	12 1/2	15	20	9	40 1/4	25 1/4	22 3/4								52 1/2	52 1/2	28 7/8	5 3/4	31 3/8	30 1/4		20"	20	2	14 1/4	R-74	3/16		
24"	24	12 1/8	41	15	17	20	10 1/2	48 1/8	29 1/8	27 1/8								61 3/4	61 3/4	30 15/16	5 3/4	31 3/8	30 1/4		24"	20	2 1/2	17 3/4	R-78	7/32		

- (1) CALC. BOX
 (2)
 (3) TK Ball Valves
 (4) Length does not include crowns

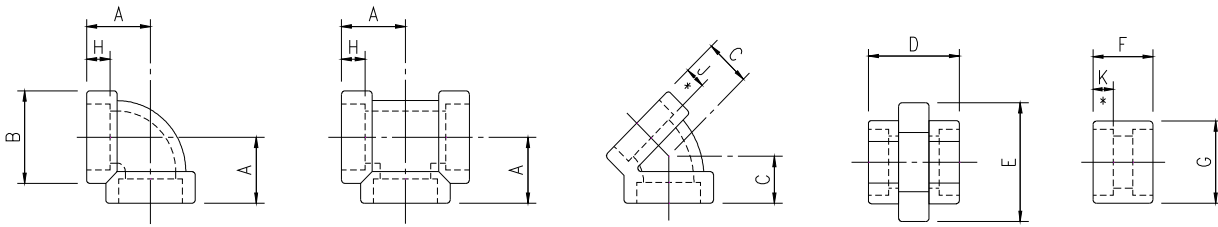
1500# RTJ PIPING DIMENSIONS



SIZE	PIPE O.D.	WN FLG	OD FLG	45 ELL	TEE	REDUCER	CAP	LR ELL & WN FLG	TEE & WN FLG	45 ELL & WN FLG	GATE VALVE				CHECK VALVE	GLOBE VALVE				BALL VALVE (3)							SIZE	NO. BOLTS	STUD BOLT SIZE		RING INFO.	
																				FULL PORT	RED. PORT	VALVE TO HW (FB)	HW OFFSET (FB)	HW PROJ (FB)	HW OD (FB)	LEVER			DIAMETER	LENGTH (4)	RING NO.	GAP
1"	1 5/16	3 1/8	5 7/8	7/8	1 1/2	2	1 1/2	4 5/8	4 5/8	4	10 1/8	16	8				10		3 1/8				9	1"	4	7/8	5	R-16	5/32			
1 1/2"	1 7/8	3 1/2	7	1 1/8	2 1/4	2 1/2	1 1/2	5 3/4	5 3/4	4 5/8	12 1/8	20	9	12			12		5 13/32				15	1 1/2"	4	1	5 1/2	R-20	5/32			
2"	2 3/8	4 5/16	8 1/2	1 3/8	2 1/2	3	1 1/2	7 5/16	6 13/16	5 11/16	14 5/8	22 1/8	10	14 5/8	14 1/2	25 1/8	14	14 5/8	14 5/8	6 1/2			33	2"	8	7/8	5 3/4	R-24	1/8			
2 1/2"	2 7/8	4 7/16	9 5/8	1 3/4	3	3 1/2	1 1/2	8 3/16	7 7/16	6 3/16	16 5/8	26 3/8	12	16 5/8	16 1/2	28 1/8	18						2 1/2"	8	1	6 1/4	R-27	1/8				
3"	3 1/2	4 15/16	10 1/2	2	3 3/8	3 1/2	2	9 7/16	8 5/16	6 15/16	18 5/8	28	14	18 5/8	18 1/2	33 1/2	24	18 5/8	18 5/8	7 3/8	3 1/8	10 1/4	17 1/8	33	3"	8	1 1/8	7	R-35	1/8		
4"	4 1/2	5 3/16	12 1/4	2 1/2	4 1/8	4	2 1/2	11 3/16	9 5/16	7 11/16	21 5/8	33	16	21 5/8				21 5/8	21 5/8	9 9/16	5	11 1/2	24	44	4"	8	1 1/4	7 3/4	R-39	1/8		
6"	6 5/8	7 1/8	15 1/2	3 3/4	5 5/8	5 1/2	3 1/2	16 1/8	12 3/4	10 7/8	27 7/8	47	24	28				28	28	12 9/16	5	18 15/16	30 1/4		6"	12	1 3/8	10 1/2	R-46	1/8		
8"	8 5/8	8 13/16	19	5	7	6	4	20 13/16	15 13/16	13 13/16	32 7/8	55	27	32 7/8				33 1/8	33 1/8	14 1/2	3 11/16	18 15/16	30 1/4		8"	12	1 5/8	12	R-50	5/32		
10"	10 3/4	10 7/16	23	6 1/4	8 1/2	7	5	25 7/16	18 15/16	16 11/16							39 3/8	39 3/8	18 7/8	4 7/8	21 1/2	39		10"	12	1 7/8	13 3/4	R-54	5/32			
12"	12 3/4	11 11/16	26 1/2	7 1/2	10	8	6	29 11/16	21 11/16	19 3/16							45 1/8	45 1/8	21 11/16	4 7/8	21 1/2	39		12"	16	2	15 1/2	R-58	3/16			
14"	14	12 3/8	29 1/2	8 3/4	11	13	6 1/2	33 3/8	23 3/8	21 1/8							50 1/4	50 1/4	23 3/4	9 3/4	29	39		14"	16	2 1/4	17	R-63	7/32			
16"	16	12 15/16	32 1/2	10	12	14	7	36 15/16	24 15/16	22 15/16							55 3/8	55 3/8	25 3/4	9 3/4	29	30 1/4		16"	16	2 1/2	18 1/2	R-67	5/16			
18"	18	13 9/16	36	11 1/4	13 1/2	15	8	40 9/16	27 1/16	24 13/16							61 3/8		30 1/2	5 3/4	31 3/8	30 1/4		18"	16	2 3/4	20 1/2	R-71	5/16			
20"	20	14 11/16	38 3/4	12 1/2	15	20	9	44 11/16	29 11/16	27 3/16							66 3/8	66 3/8	31 1/4	5 3/4	31 3/8	30 1/4		20"	16	3	22 1/2	R-75	3/8			
24"	24	16 13/16	46	15	17	20	10 1/2	52 13/16	33 13/16	31 13/16							77 5/8	77 5/8	37 7/8	16	37 1/4	30 1/4		24"	16	3 1/2	25 3/4	R-79	7/16			

- (1)
- (2) Crane Duo-Chek II
- (3) TK Ball Valves
- (4) Length does not include crowns

SCREWED FITTINGS



3000#

SIZE	A	B	C	D	E	F	G	H*	J*	K*
1/2	1 5/16	1 1/2	1	2 1/8	2 3/16	1 7/8	1 1/8	1/2	1/2	1/2
3/4	1 1/2	1 13/16	1 1/8	2 5/16	2 9/16	2	1 3/8	9/16	9/16	9/16
1	1 3/4	2 3/16	1 5/16	2 1/2	2 15/16	2 3/8	1 3/4	11/16	11/16	11/16
1 1/2	2 3/8	3	1 11/16	3 1/8	3 11/16	3 1/8	2 1/2	11/16	11/16	11/16
2	2 1/2	3 5/16	1 3/4	3 1/2	4 9/16	3 3/8	3	3/4	3/4	3/4
2 1/2	3 1/4	4	2 1/16	4 5/8	5 1/4	3 5/8	3 5/8	15/16	15/16	15/16

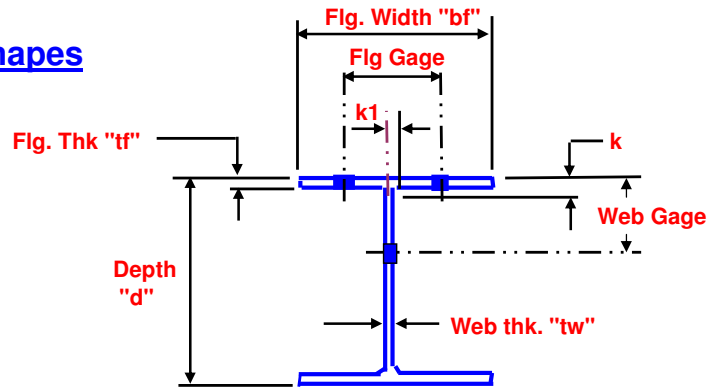
6000#

SIZE	A	B	C	D	E	F	G	H*	J*	K*
1/2	1 1/2	1 13/16	1 1/8	2 7/8	2 13/16	1 7/8	1 1/2	1/2	1/2	1/2
3/4	1 3/4	2 3/16	1 5/16	3 3/8	3 1/8	2	1 3/4	9/16	9/16	9/16
1	2	2 7/16	1 3/8	3 5/8	3 13/16	2 3/8	2 1/4	11/16	11/16	11/16
1 1/2	2 1/2	3 5/16	1 3/4	4 3/16	4 3/4	3 1/8	3	11/16	11/16	11/16
2	3 1/4	4	2 1/16	4 5/8	5 1/4	3 3/8	3 5/8	3/4	3/4	3/4
2 1/2	3 3/4	4 3/4	2 1/2	-	-	3 5/8	4 1/4	15/16	15/16	15/16

* Thread engagement typical for H, J & K

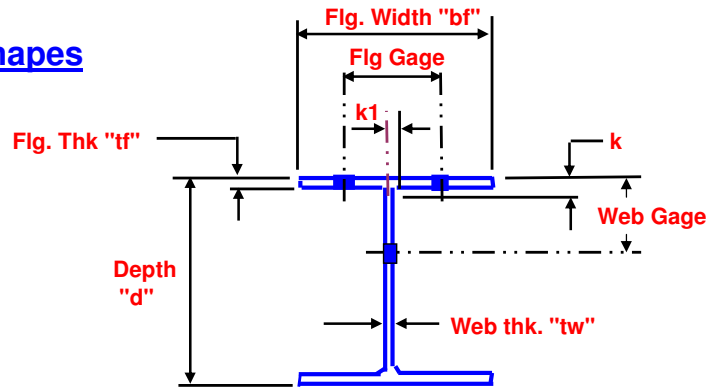
WELDOLET																		
CENTER OF RUN TO FACE OF FITTING																		
RUN SIZE	WT. OR SCH.	OUTLET SIZE																
		1/2	3/4	1	1 1/2	2	2 1/2	3	4	6	8	10	12	14	16	18	20	24
1/2	STD	1 3/16																
	XS	1 3/16																
	160&XXS	1 1/2																
3/4	STD	1 1/4	1 7/16															
	XS	1 1/4	1 7/16															
	160&XXS	1 5/8	1 3/4															
1	STD	1 3/8	1 1/2	1 11/16														
	XS	1 3/8	1 1/2	1 11/16														
	160&XXS	1 3/4	1 7/8	2 1/8														
1 1/2	STD	1 11/16	1 13/16	2	2 1/4													
	XS	1 11/16	1 13/16	2	2 1/4													
	160&XXS	2 1/16	2 3/16	2 7/16	2 15/16													
2	STD	1 15/16	2 1/16	2 1/4	2 1/2	3 5/8												
	XS	1 15/16	2 1/16	2 1/4	2 1/2	2 11/16												
	160&XXS	2 5/16	2 7/16	2 11/16	3 3/16	3 3/8												
2 1/2	STD	2 3/16	2 5/16	2 1/2	2 3/4	2 15/16	3 1/16											
	XS	2 3/16	2 5/16	2 1/2	2 3/4	2 15/16	3 1/16											
	160&XXS	2 9/16	2 11/16	2 15/16	3 7/16	3 5/8	3 7/8											
3	STD	2 1/2	2 5/8	2 13/16	3 1/16	3 1/4	3 3/8	3 1/2										
	XS	2 1/2	2 5/8	2 13/16	3 1/16	3 1/4	3 3/8	3 1/2										
	160&XXS	2 7/8	3	3 1/4	3 3/4	3 15/16	4 3/16	4 5/8										
4	STD	3	3 1/8	3 5/16	3 9/16	3 3/4	3 7/8	4	4 1/4									
	XS	3	3 1/8	3 5/16	3 9/16	3 3/4	3 7/8	4	4 1/4									
	160&XXS	3 3/8	3 1/2	3 3/4	4 1/4	4 7/16	4 11/16	5 1/8	5 9/16									
6	STD	4 1/16	4 3/16	4 3/8	4 5/8	4 13/16	4 15/16	5 1/16	5 5/16	5 11/16								
	XS	4 1/16	4 3/16	4 3/8	4 5/8	4 13/16	4 15/16	5 1/16	5 5/16	6 3/8								
	160&XXS	4 7/16	4 9/16	4 13/16	5 5/16	5 1/2	5 3/4	6 3/16	6 5/8	7 7/16								
8	STD	5 1/16	5 3/16	5 3/8	5 5/8	5 13/16	5 15/16	6 1/16	6 5/16	6 11/16	7 1/16							
	XS	5 1/16	5 3/16	5 3/8	5 5/8	5 13/16	5 15/16	6 1/16	6 5/16	7 3/8	8 3/16							
	160&XXS					6 1/2	6 3/4	7 3/16	7 5/8	8 7/16	*							
10	STD	6 1/8	6 1/4	6 7/16	6 11/16	6 7/8	7	7 1/8	7 3/8	7 3/4	8 1/8	8 7/16						
	XS	6 1/8	6 1/4	6 7/16	6 11/16	6 7/8	7	7 1/8	7 3/8	8 7/16	9 1/4	9 1/8						
	160&XXS					7 9/16	7 13/16	8 1/4	8 11/16	9 1/2	*	*						
12	STD	7 1/8	7 1/4	7 7/16	7 11/16	7 7/8	8	8 1/8	8 3/8	8 3/4	9 1/8	9 7/16	9 3/4					
	XS	7 1/8	7 1/4	7 7/16	7 11/16	7 7/8	8	8 1/8	8 3/8	9 7/16	10 1/4	10 1/16	10 5/16					
	160&XXS					8 9/16	8 13/16	9 1/4	9 11/16	10 1/2	*	*	*					
14	STD	7 3/4	7 7/8	8 1/16	8 5/16	8 1/2	8 5/8	8 3/4	9	9 3/8	9 3/4	10 1/16	10 3/8	10 1/2				
	XS	7 3/4	7 7/8	8 1/16	8 5/16	8 1/2	8 5/8	8 3/4	9	10 1/16	10 7/8	10 11/16	11 1/16	11 1/8				
	160&XXS					9 3/16	9 7/16	9 7/8	10 5/16	11 1/8	*	*	*	*				
16	STD	8 3/4	8 7/8	9 1/16	9 5/16	9 1/2	9 5/8	9 3/4	10	10 3/8	10 3/4	11 1/16	11 3/8	11 1/2	11 11/16			
	XS	8 3/4	8 7/8	9 1/16	9 5/16	9 1/2	9 5/8	9 3/4	10	11 1/16	11 7/8	11 11/16	12 1/16	11 15/16	12 7/16			
	160&XXS					10 3/16	10 7/16	10 7/8	11 5/16	12 1/8	*	*	*	*	*			
18	STD	9 3/4	9 7/8	10 1/16	10 5/16	10 1/2	10 5/8	10 3/4	11	11 3/8	11 3/4	12 1/16	12 3/8	12 1/2	12 11/16	13 1/16		
	XS	9 3/4	9 7/8	10 1/16	10 5/16	10 1/2	10 5/8	10 3/4	11	12 1/16	12 7/8	12 11/16	13 1/16	12 15/16	13 3/16	13 11/16		
	160&XXS					11 3/16	11 7/16	11 7/8	12 5/16	13 1/8	*	*	*	*	*	*		
20	STD	10 3/4	10 7/8	11 1/16	11 5/16	11 1/2	11 5/8	11 3/4	12	12 3/8	12 3/4	13 1/16	13 3/8	13 1/2	13 11/16	13 13/16	14 5/8	
	XS	10 3/4	10 7/8	11 1/16	11 5/16	11 1/2	11 5/8	11 3/4	12	13 1/16	13 7/8	13 11/16	14 1/16	13 15/16	14 3/16	14 3/8	15	
	160&XXS					12 3/16	12 7/16	12 7/8	13 5/16	14 1/8	*	*	*	*	*	*	*	
24	STD	12 3/4	12 7/8	13 1/16	13 5/16	13 1/2	13 5/8	13 3/4	14	14 3/8	14 3/4	15 1/16	15 3/8	15 1/2	15 11/16	15 13/16	16	17 3/8
	XS	12 3/4	12 7/8	13 1/16	13 5/16	13 1/2	13 5/8	13 3/4	14	15 1/16	15 7/8	15 11/16	16 1/16	15 15/16	16 3/16	16 3/8	16 11/16	17 1/2
	160&XXS					14 3/16	14 7/16	14 7/8	15 5/16	16 1/8	*	*	*	*	*	*	*	*

* USE HEVY WALL WELDOLET

W Shapes

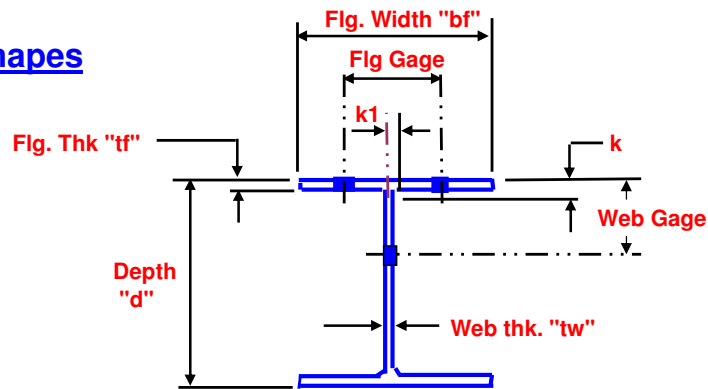
Shapes in shaded rows are not available from domestic producers

Beam Size	Depth d	Web Thk. tw	Fig. Width bf	Fig. Thk. tf	k	k1	Flange Bolt Gage	Web Bolt Gage	Sq. Ft.- Surf Area - per Ft	Sq.M.- Surf Area - per M	Weight Kg per M
W44x285	44	1	11 3/4	1 3/4	2 11/16	1 3/8			11.08	3.378	424.127
W44x248	43 5/8	7/8	11 3/4	1 9/16	2 1/2	1 5/16			11.04	3.366	369.065
W44x224	43 1/4	13/16	11 3/4	1 7/16	2 5/16	1 5/16			10.99	3.350	333.349
W44x198	42 7/8	11/16	11 3/4	1 1/4	2 1/8	1 1/4			10.95	3.337	294.656
W40x328	40	15/16	17 7/8	1 3/4	3 1/8	1 11/16			12.47	3.800	488.118
W40x298	39 3/4	13/16	17 7/8	1 9/16	3	1 5/8			12.45	3.794	443.473
W40x268	39 3/8	3/4	17 3/4	1 7/16	2 13/16	1 9/16			12.35	3.766	398.828
W40x244	39	11/16	17 3/4	1 1/4	2 5/8	1 9/16			12.30	3.750	363.112
W40x221	38 5/8	11/16	17 3/4	1 1/16	2 7/16	1 9/16			12.24	3.731	328.884
W40x192	38 1/4	11/16	17 3/4	13/16	2 1/4	1 9/16			12.18	3.712	285.727
W40x655	43 5/8	2	16 7/8	3 9/16	4 15/16	2 1/4			12.56	3.829	974.747
W40x593	43	1 13/16	16 3/4	3 1/4	4 5/8	2 1/8			12.45	3.794	882.481
W40x531	42 3/8	1 5/8	16 1/2	2 15/16	4 5/16	2			12.29	3.747	790.215
W40x480	41 3/4	1 7/16	16 3/8	2 5/8	4	2			12.18	3.712	714.319
W40x436	41 3/8	1 5/16	16 1/4	2 3/8	3 13/16	1 15/16			12.09	3.686	648.840
W40x397	41	1 1/4	16 1/8	2 3/16	3 5/8	1 7/8			12.00	3.658	590.801
W40x362	40 1/2	1 1/8	16	2	3 3/8	1 13/16			11.90	3.626	538.715
W40x324	40 1/8	1	15 7/8	1 13/16	3 3/16	1 3/4			11.81	3.600	482.165
W40x297	39 7/8	15/16	15 7/8	1 5/8	3 1/16	1 11/16			11.78	3.591	441.985
W40x277	39 3/4	13/16	15 7/8	1 9/16	3	1 5/8			11.78	3.591	412.221
W40x249	39 3/8	3/4	15 3/4	1 7/16	2 13/16	1 9/16			11.69	3.562	370.553
W40x215	39	5/8	15 3/4	1 1/4	2 5/8	1 9/16			11.65	3.550	319.955
W40x199	38 5/8	5/8	15 3/4	1 1/16	2 7/16	1 9/16			11.58	3.531	296.145
W40x183	39	5/8	11 3/4	1 1/4	2 5/8	1 9/16			10.31	3.143	272.334
W40x167	38 5/8	5/8	11 3/4	1	2 7/16	1 9/16			10.25	3.124	248.523
W40x149	38 1/4	5/8	11 3/4	13/16	2 1/4	1 1/2			10.19	3.105	221.736
W36x848	42 1/2	2 1/2	18 1/8	4 1/2	5 11/16	2 1/4			12.71	3.874	1261.963
W36x798	42	2 3/8	18	4 5/16	5 7/16	2 3/16			12.60	3.842	1187.555
W36x720	41 1/4	2 3/16	17 3/4	3 7/8	5 1/16	2 1/16			12.43	3.788	1071.478
W36x650	40 1/2	2	17 5/8	3 9/16	4 11/16	2			12.29	3.747	967.307
W36x588	39 7/8	1 13/16	17 3/8	3 1/4	4 3/8	1 7/8			12.14	3.699	875.040
W36x527	39 1/4	1 5/8	17 1/4	2 15/16	4 1/16	1 3/4			12.02	3.664	784.262
W36x485	38 3/4	1 1/2	17 1/8	2 11/16	3 13/16	1 3/4			11.92	3.632	721.760
W36x439	38 1/4	1 3/8	17	2 7/16	3 9/16	1 5/8			11.81	3.600	653.304
W36x393	37 3/4	1 1/4	16 7/8	2 3/16	3 5/16	1 5/8			11.71	3.569	584.848
W36x359	37 3/8	1 1/8	16 3/4	2	3 1/8	1 9/16			11.63	3.543	534.251
W36x328	37 1/8	1	16 5/8	1 7/8	3	1 1/2			11.56	3.524	488.118
W36x300	36 3/4	15/16	16 5/8	1 11/16	2 13/16	1 1/2	5 1/2	3 3/4	11.51	3.508	446.449
W36x280	36 1/2	7/8	16 5/8	1 9/16	2 11/16	1 1/2	5 1/2	3 3/4	11.48	3.499	416.686
W36x260	36 1/4	13/16	16 1/2	1 7/16	2 9/16	1 1/2	5 1/2	3 1/2	11.41	3.477	386.923
W36x245	36 1/8	13/16	16 1/2	1 3/8	2 1/2	1 7/16	5 1/2	3 1/2	11.39	3.470	364.600
W36x230	35 7/8	3/4	16 1/2	1 1/4	2 3/8	1 7/16	5 1/2	3 1/2	11.35	3.461	342.278
W36x256	37 3/8	1	12 1/4	1 3/4	2 5/8	1 5/16	5 1/2		10.15	3.092	380.970
W36x232	37 1/8	7/8	12 1/8	1 9/16	2 1/2	1 1/4	5 1/2		10.08	3.073	345.254
W36x210	36 3/4	13/16	12 1/8	1 3/8	2 5/16	1 1/4	5 1/2	3 1/2	10.03	3.058	312.514
W36x194	36 1/2	3/4	12 1/8	1 1/4	2 3/16	1 3/16	5 1/2	3 1/2	10.00	3.048	288.704
W36x182	36 3/8	3/4	12 1/8	1 3/16	2 1/8	1 3/16	5 1/2	3 1/4	9.98	3.042	270.846
W36x170	36 1/8	11/16	12	1 1/8	2	1 3/16	5 1/2	3 1/4	9.91	3.019	252.988
W36x160	36	5/8	12	1	1 15/16	1 1/8	5 1/2	3 1/4	9.90	3.016	238.106
W36x150	35 7/8	5/8	12	15/16	1 7/8	1 1/8	5 1/2	3	9.88	3.010	223.225
W36x135	35 1/2	5/8	12	13/16	1 11/16	1 1/8	5 1/2	3	9.81	2.991	200.902

W Shapes

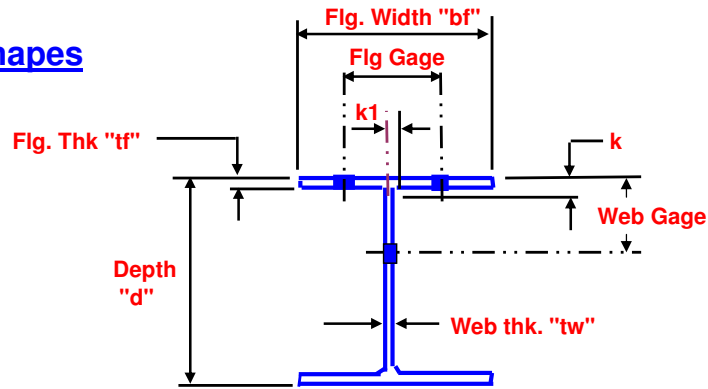
Shapes in shaded rows are not available from domestic producers

Beam Size	Depth d	Web Thk. tw	Flg. Width bf	Flg. Thk. tf	k	k1	Flange Bolt Gage	Web Bolt Gage	Sq. Ft.- Surf Area - per Ft	Sq.M.- Surf Area - per M	Weight Kg per M
W33x619	38 1/2	2	16 7/8	3 9/16	4 3/8	1 3/4			11.71	3.569	921.174
W33x567	37 7/8	1 13/16	16 3/4	3 1/4	4 1/16	1 11/16			11.59	3.534	843.789
W33x515	37 3/8	1 5/8	16 5/8	3	3 13/16	1 5/8			11.50	3.505	766.404
W33x468	36 3/4	1 1/2	16 1/2	2 3/4	3 1/2	1 9/16			11.38	3.467	696.461
W33x424	36 3/8	1 3/8	16 3/8	2 1/2	3 5/16	1 7/16			11.29	3.442	630.982
W33x387	36	1 1/4	16 1/4	2 1/4	3 1/8	1 3/8			11.21	3.416	575.919
W33x354	35 1/2	1 3/16	16 1/8	2 1/16	2 7/8	1 3/8			11.09	3.381	526.810
W33x318	35 1/8	1 1/16	16	1 7/8	2 11/16	1 5/16			11.01	3.356	473.236
W33x291	34 7/8	1	15 7/8	1 3/4	2 9/16	1 1/4			10.94	3.334	433.056
W33x263	34 1/2	7/8	15 3/4	1 9/16	2 3/8	1 3/16			10.85	3.308	391.387
W33x241	34 1/8	13/16	15 7/8	1 3/8	2 3/16	1 3/16	5 1/2	3 1/2	10.84	3.305	358.648
W33x221	33 7/8	3/4	15 3/4	1 1/4	2 1/16	1 3/16	5 1/2	3 1/2	10.77	3.283	328.884
W33x201	33 5/8	11/16	15 3/4	1 1/8	1 15/16	1 1/8	5 1/2	3 1/4	10.74	3.273	299.121
W33x169	33 7/8	11/16	11 1/2	1 1/4	2 1/16	1 1/8			9.36	2.854	251.500
W33x152	33 1/2	5/8	11 5/8	1 1/16	1 7/8	1 1/8	5 1/2	3 1/4	9.35	2.851	226.201
W33x141	33 1/4	5/8	11 1/2	15/16	1 3/4	1 1/16	5 1/2	3	9.27	2.826	209.831
W33x130	33 1/8	9/16	11 1/2	7/8	1 11/16	1 1/16	5 1/2	3	9.26	2.823	193.461
W33x118	32 7/8	9/16	11 1/2	3/4	1 9/16	1 1/16	5 1/2	2 3/4	9.22	2.810	175.603
W30x581	35 3/8	2	16 1/4	3 9/16	4 5/16	1 11/16			10.98	3.346	864.623
W30x526	34 3/4	1 13/16	16	3 1/4	4	1 5/8			10.82	3.299	782.774
W30x477	34 1/4	1 5/8	15 7/8	3	3 3/4	1 9/16			10.73	3.270	709.854
W30x433	33 5/8	1 1/2	15 3/4	2 11/16	3 7/16	1 1/2			10.60	3.232	644.375
W30x391	33 1/4	1 3/8	15 5/8	2 7/16	3 1/4	1 7/16			10.52	3.207	581.872
W30x357	32 3/4	1 1/4	15 1/2	2 1/4	3	1 3/8			10.42	3.175	531.275
W30x326	32 3/8	1 1/8	15 3/8	2 1/16	2 13/16	1 5/16			10.33	3.150	485.141
W30x292	32	1	15 1/4	1 7/8	2 5/8	1 1/4			10.25	3.124	434.544
W30x261	31 5/8	15/16	15 1/8	1 5/8	2 7/16	1 3/16			10.16	3.096	388.411
W30x235	31 1/4	13/16	15	1 1/2	2 1/4	1 1/8			10.07	3.070	349.719
W30x211	31	3/4	15 1/8	1 5/16	2 1/8	1 1/8	5 1/2	3 1/2	10.08	3.073	314.003
W30x191	30 5/8	11/16	15	1 3/16	1 15/16	1 1/16	5 1/2	3 1/4	9.99	3.045	284.239
W30x173	30 1/2	5/8	15	1 1/16	1 7/8	1 1/16	5 1/2	3 1/4	9.98	3.042	257.452
W30x148	30 5/8	5/8	10 1/2	1 3/16	2	1			8.50	2.591	220.248
W30x132	30 1/4	5/8	10 1/2	1	1 3/4	1 1/16	5 1/2	3	8.44	2.572	196.438
W30x124	30 1/8	9/16	10 1/2	15/16	1 11/16	1	5 1/2	3	8.43	2.569	184.532
W30x116	30	9/16	10 1/2	7/8	1 5/8	1	5 1/2	3	8.41	2.562	172.627
W30x108	29 7/8	9/16	10 1/2	3/4	1 9/16	1	5 1/2	3	8.39	2.556	160.722
W30x99	29 5/8	1/2	10 1/2	11/16	1 7/16	1	5 1/2	2 3/4	8.35	2.546	147.328
W30x90	29 1/2	1/2	10 3/8	9/16	1 5/16	1			8.29	2.527	133.935
W27x539	32 1/2	2	15 1/4	3 9/16	4 1/4	1 5/8			10.17	3.099	802.120
W27x494	32	1 13/16	15 1/8	3 1/4	4	1 9/16			10.07	3.070	735.153
W27x448	31 3/8	1 5/8	15	3	3 11/16	1 1/2			9.96	3.035	666.697
W27x407	30 7/8	1 1/2	14 3/4	2 3/4	3 7/16	1 7/16			9.81	2.991	605.683
W27x368	30 3/8	1 3/8	14 5/8	2 1/2	3 3/16	1 5/16			9.71	2.959	547.644
W27x336	30	1 1/4	14 1/2	2 1/4	3	1 5/16			9.63	2.934	500.023
W27x307	29 5/8	1 3/16	14 1/2	2 1/16	2 13/16	1 1/4			9.57	2.918	575.919
W27x281	29 1/4	1 1/16	14 3/8	1 15/16	2 5/8	1 3/16			9.49	2.892	418.174
W27x258	29	1	14 1/4	1 3/4	2 1/2	1 1/8			9.42	2.870	383.946
W27x235	28 5/8	15/16	14 1/4	1 5/8	2 5/16	1 1/8			9.36	2.854	349.719
W27x217	28 3/8	13/16	14 1/8	1 1/2	2 3/16	1 1/16			9.30	2.835	322.932
W27x194	28 1/8	3/4	14	1 5/16	2 1/16	1			9.23	2.813	288.704
W27x178	27 3/4	3/4	14 1/8	1 3/16	1 7/8	1 1/16	5 1/2	3 1/4	9.21	2.807	264.893
W27x161	27 5/8	11/16	14	1 1/16	1 13/16	1	5 1/2	3 1/4	9.16	2.791	239.594
W27x146	27 3/8	5/8	14	1	1 11/16	1	5 1/2	3	9.13	2.781	217.272

W Shapes

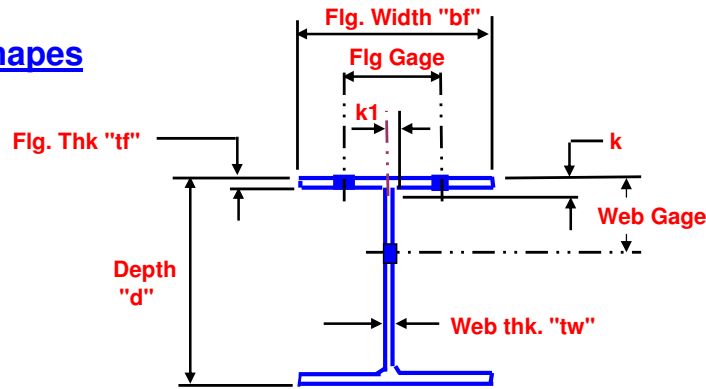
Shapes in shaded rows are not available from domestic producers

Beam Size	Depth d	Web Thk. tw	Flg. Width bf	Flg. Thk. tf	k	k1	Flange Bolt Gage	Web Bolt Gage	Sq. Ft.- Surf Area - per Ft	Sq.M.- Surf Area - per M	Weight Kg per M
W27x129	27 5/8	5/8	10	1 1/8	1 13/16	15/16	5 1/2		7.83	2.388	191.973
W27x114	27 1/4	9/16	10 1/8	15/16	1 5/8	15/16	5 1/2	3	7.82	2.384	169.651
W27x102	27 1/8	1/2	10	13/16	1 9/16	15/16	5 1/2	3	7.77	2.369	151.793
W27x94	26 7/8	1/2	10	3/4	1 7/16	15/16	5 1/2	2 3/4	7.73	2.356	139.887
W27x84	26 3/4	7/16	10	5/8	1 3/8	15/16	5 1/2	2 3/4	7.72	2.353	125.006
W24x492	29 5/8	2	14 1/8	3 9/16	4 5/16	1 9/16			9.31	2.838	732.177
W24x450	29 1/8	1 13/16	14	3 1/4	4 1/16	1 1/2			9.22	2.810	669.674
W24x408	28 1/2	1 5/8	13 3/4	3	3 3/4	1 3/8			9.06	2.762	607.171
W24x370	28	1 1/2	13 5/8	2 3/4	3 1/2	1 5/16			8.96	2.731	550.621
W24x335	27 1/2	1 3/8	13 1/2	2 1/2	3 1/4	1 1/4			8.85	2.699	498.535
W24x306	27 1/8	1 1/4	13 3/8	2 1/4	3 1/16	1 3/16			8.77	2.673	455.378
W24x279	26 3/4	1 3/16	13 1/4	2 1/16	2 7/8	1 1/8			8.68	2.645	415.198
W24x250	26 3/8	1 1/16	13 1/8	1 7/8	2 11/16	1 1/8			8.59	2.619	372.041
W24x229	26	1	13 1/8	1 3/4	2 1/2	1			8.54	2.604	340.790
W24x207	25 3/4	7/8	13	1 9/16	2 3/8	1			8.48	2.584	308.050
W24x192	25 1/2	13/16	13	1 7/16	2 1/4	1			8.45	2.575	285.727
W24x176	25 1/4	3/4	12 7/8	1 5/16	2 1/8	15/16			8.38	2.553	261.917
W24x162	25	11/16	13	1 1/4	2	1 1/16	5 1/2	3 1/4	8.39	2.556	241.083
W24x146	24 3/4	5/8	12 7/8	1 1/16	1 7/8	1 1/16	5 1/2	3 1/4	8.31	2.534	217.272
W24x131	24 1/2	5/8	12 7/8	15/16	1 3/4	1 1/16	5 1/2	3	8.27	2.521	194.949
W24x117	24 1/4	9/16	12 3/4	7/8	1 5/8	1	5 1/2	3	8.20	2.499	174.115
W24x104	24	1/2	12 3/4	3/4	1 1/2	1	5 1/2	2 3/4	8.17	2.489	154.769
W24x103	24 1/2	9/16	9	1	1 3/4	13/16	5 1/2		6.99	2.130	153.281
W24x94	24 1/4	1/2	9 1/8	7/8	1 5/8	1	5 1/2	3	7.00	2.134	139.887
W24x84	24 1/8	1/2	9	3/4	1 9/16	15/16	5 1/2	3	6.94	2.115	125.006
W24x76	23 7/8	7/16	9	11/16	1 7/16	15/16	5 1/2	2 3/4	6.91	2.105	113.100
W24x68	23 3/4	7/16	9	9/16	1 3/8	15/16	5 1/2	2 3/4	6.89	2.099	101.195
W24x62	23 3/4	7/16	7	9/16	1 3/8	15/16	3 1/2	2 3/4	6.22	1.895	92.266
W24x55	23 5/8	3/8	7	1/2	1 5/16	15/16	3 1/2	2 3/4	6.21	1.892	81.849
W21x402	26	1 3/4	13 3/8	3 1/8	3 7/8	1 7/16			8.50	2.591	598.242
W21x364	25 1/2	1 9/16	13 1/4	2 7/8	3 5/8	1 3/8			8.41	2.562	541.692
W21x333	25	1 7/16	13 1/8	2 5/8	3 3/8	1 5/16			8.30	2.530	495.559
W21x300	24 1/2	1 5/16	13	2 3/8	3 1/8	1 1/4			8.20	2.499	446.449
W21x275	24 1/8	1 1/4	12 7/8	2 3/16	3	1 3/16			8.10	2.470	409.245
W21x248	23 3/4	1 1/8	12 3/4	2	2 3/4	1 1/8			8.02	2.445	369.065
W21x223	23 3/8	1	12 5/8	1 13/16	2 9/16	1 1/16			7.94	2.419	331.861
W21x201	23	15/16	12 5/8	1 5/8	2 3/8	1			7.89	2.403	299.121
W21x182	22 3/4	13/16	12 1/2	1 1/2	2 1/4	1			7.82	2.384	270.846
W21x166	22 1/2	3/4	12 3/8	1 3/8	2 1/8	15/16			7.75	2.362	247.035
W21x147	22	3/4	12 1/2	1 1/8	1 7/8	1 1/16	5 1/2	3 1/4	7.71	2.350	218.760
W21x132	21 7/8	5/8	12 1/2	1 1/16	1 13/16	1	5 1/2	3	7.71	2.350	196.438
W21x122	21 5/8	5/8	12 3/8	15/16	1 11/16	1	5 1/2	3	7.63	2.324	181.556
W21x111	21 1/2	9/16	12 3/8	7/8	1 5/8	15/16	5 1/2	2 3/4	7.61	2.321	165.186
W21x101	21 3/8	1/2	12 1/4	13/16	1 9/16	15/16	5 1/2	2 3/4	7.56	2.305	150.305
W21x93	21 5/8	9/16	8 3/8	15/16	1 11/16	1	5 1/2	3	6.30	1.921	138.399
W21x83	21 3/8	1/2	8 3/8	13/16	1 9/16	15/16	5 1/2	2 3/4	6.27	1.911	123.518
W21x73	21 1/4	7/16	8 1/4	3/4	1 1/2	15/16	5 1/2	2 3/4	6.22	1.895	108.636
W21x68	21 1/8	7/16	8 1/4	11/16	1 7/16	7/8	5 1/2	2 3/4	6.20	1.889	101.195
W21x62	21	3/8	8 1/4	5/8	1 3/8	7/8	5 1/2	2 1/2	6.19	1.886	92.266

W Shapes

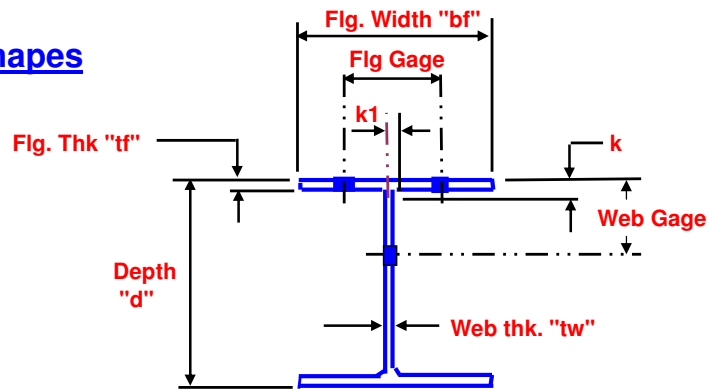
Shapes in shaded rows are not available from domestic producers

Beam Size	Depth d	Web Thk. tw	Fig. Width bf	Fig. Thk. tf	k	k1	Flange Bolt Gage	Web Bolt Gage	Sq. Ft.- Surf Area - per Ft	Sq.M.- Surf Area - per M	Weight Kg per M
W21x57	21	3/8	6 1/2	5/8	1 3/8	7/8	3 1/2	2 3/4	5.60	1.708	84.825
W21x50	20 7/8	3/8	6 1/2	9/16	1 5/16	7/8	3 1/2	2 1/2	5.58	1.702	74.408
W21x44	20 5/8	3/8	6 1/2	7/16	1 3/16	7/8	3 1/2	2 1/2	5.54	1.689	65.479
W18x311	22 3/8	1 1/2	12	2 3/4	3 7/16	1 3/16			7.48	2.280	462.819
W18x283	21 7/8	1 3/8	11 7/8	2 1/2	3 3/16	1 3/16			7.38	2.248	421.150
W18x258	21 1/2	1 1/4	11 3/4	2 5/16	3	1 1/8			7.29	2.223	383.946
W18x234	21	1 3/16	11 5/8	2 1/8	2 3/4	1			7.18	2.188	348.230
W18x211	20 5/8	1 1/16	11 1/2	1 15/16	2 9/16	1			7.09	2.162	314.003
W18x192	20 3/8	1	11 1/2	1 3/4	2 7/16	15/16			7.06	2.153	285.727
W18x175	20	7/8	11 3/8	1 9/16	2 1/4	7/8			6.98	2.127	260.429
W18x158	19 3/4	13/16	11 1/4	1 7/16	2 1/8	7/8			6.91	2.105	235.130
W18x143	19 1/2	3/4	11 1/4	1 5/16	2	13/16			6.88	2.096	212.807
W18x130	19 1/4	11/16	11 1/8	1 3/16	1 7/8	13/16			6.80	2.073	193.461
W18x119	19	5/8	11 1/4	1 1/16	1 3/4	15/16	5 1/2	3	6.81	2.076	177.092
W18x106	18 3/4	9/16	11 1/4	15/16	1 5/8	15/16	5 1/2	3	6.78	2.067	157.745
W18x97	18 5/8	9/16	11 1/8	7/8	1 9/16	7/8	5 1/2	2 3/4	6.72	2.048	144.352
W18x86	18 3/8	1/2	11 1/8	3/4	1 7/16	7/8	5 1/2	2 3/4	6.69	2.038	127.982
W18x76	18 1/4	7/16	11	11/16	1 3/8	13/16	5 1/2	2 3/4	6.64	2.022	113.100
W18x71	18 1/2	1/2	7 5/8	13/16	1 1/2	7/8	3 1/2	2 3/4	5.54	1.689	105.660
W18x65	18 3/8	7/16	7 5/8	3/4	1 7/16	7/8	3 1/2	2 3/4	5.53	1.686	96.731
W18x60	18 1/4	7/16	7 1/2	11/16	1 3/8	13/16	3 1/2	2 3/4	5.47	1.667	89.290
W18x55	18 1/8	3/8	7 1/2	5/8	1 5/16	13/16	3 1/2	2 3/4	5.46	1.664	81.849
W18x50	18	3/8	7 1/2	9/16	1 1/4	13/16	3 1/2	2 1/2	5.44	1.657	74.408
W18x46	18	3/8	6	5/8	1 1/4	13/16	3 1/2	2 1/2	4.94	1.505	68.456
W18x40	17 7/8	5/16	6	1/2	1 3/16	13/16	3 1/2	2 1/2	4.93	1.502	59.527
W18x35	17 3/4	5/16	6	7/16	1 1/8	3/4	3 1/2	2 1/2	4.91	1.495	52.086
W16x100	17	9/16	10 3/8	1	1 11/16	15/16	5 1/2	3	6.20	1.889	148.816
W16x89	16 3/4	1/2	10 3/8	7/8	1 9/16	7/8	5 1/2	2 3/4	6.17	1.880	132.447
W16x77	16 1/2	7/16	10 1/4	3/4	1 7/16	7/8	5 1/2	2 3/4	6.09	1.857	114.589
W16x67	16 3/8	3/8	10 1/4	11/16	1 3/8	13/16	5 1/2	2 3/4	6.08	1.854	99.707
W16x57	16 3/8	7/16	7 1/8	11/16	1 3/8	7/8	3 1/2	2 3/4	5.03	1.534	84.825
W16x50	16 1/4	3/8	7 1/8	5/8	1 5/16	13/16	3 1/2	2 3/4	5.02	1.530	74.408
W16x45	16 1/8	3/8	7	9/16	1 1/4	13/16	3 1/2	2 1/2	4.96	1.511	66.967
W16x40	16	5/16	7	1/2	1 3/16	13/16	3 1/2	2 1/2	4.95	1.508	59.527
W16x36	15 7/8	5/16	7	7/16	1 1/8	3/4	3 1/2	2 1/2	4.93	1.502	53.574
W16x31	15 7/8	1/4	5 1/2	7/16	1 1/8	3/4	2 3/4	2 1/2	4.44	1.353	46.133
W16x26	15 3/4	1/4	5 1/2	3/8	1 1/16	3/4	2 3/4	2 1/4	4.42	1.346	38.692
W14x730	22 3/8	3 1/16	17 7/8	4 15/16	5 9/16	2 3/16	na	na	9.18	2.797	1086.360
W14x665	21 5/8	2 13/16	17 5/8	4 1/2	5 3/16	2 1/16	na	na	9.01	2.746	989.629
W14x605	20 7/8	2 5/8	17 3/8	4 3/16	4 13/16	1 15/16	na	na	8.83	2.692	900.339
W14x550	20 1/4	2 3/8	17 1/4	3 13/16	4 1/2	1 13/16	na	na	8.73	2.661	818.490
W14x500	19 5/8	2 3/16	17	3 1/2	4 3/16	1 3/4	na	na	8.57	2.613	744.082
W14x455	19	2	16 7/8	3 3/16	3 7/8	1 5/8	na	na	8.46	2.578	677.115

W Shapes

Shapes in shaded rows are not available from domestic producers

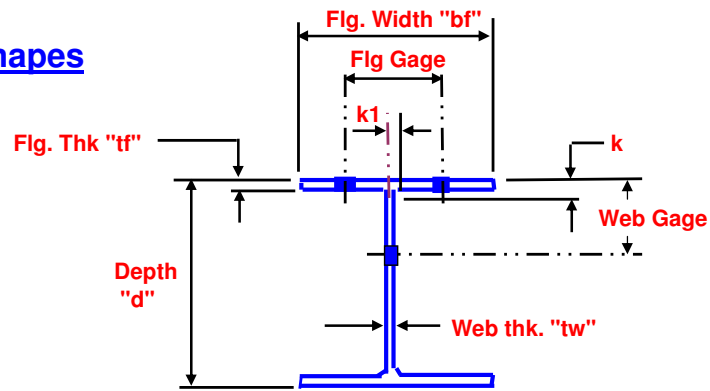
Beam Size	Depth d	Web Thk. tw	Fig. Width bf	Fig. Thk. tf	k	k1	Flange Bolt Gage	Web Bolt Gage	Sq. Ft.- Surf Area - per Ft	Sq.M.- Surf Area - per M	Weight Kg per M
W14x426	18 5/8	1 7/8	16 3/4	3 1/16	3 11/16	1 9/16	na	na	8.38	2.553	633.958
W14x398	18 1/4	1 3/4	16 5/8	2 7/8	3 1/2	1 1/2	na	na	8.29	2.527	592.289
W14x370	17 7/8	1 5/8	16 1/2	2 11/16	3 5/16	1 7/16	na	na	8.21	2.502	550.621
W14x342	17 1/2	1 9/16	16 3/8	2 1/2	3 1/8	1 3/8	na	na	8.11	2.473	508.952
W14x311	17 1/8	1 7/16	16 1/4	2 1/4	2 15/16	1 5/16	na	na	8.03	2.448	462.819
W14x283	16 3/4	1 5/16	16 1/8	2 1/16	2 3/4	1 1/4	na	na	7.95	2.423	421.150
W14x257	16 3/8	1 3/16	16	1 7/8	2 9/16	1 3/16	na	na	7.86	2.397	382.458
W14x233	16	1 1/16	15 7/8	1 3/4	2 3/8	1 3/16	na	na	7.78	2.372	346.742
W14x211	15 3/4	1	15 3/4	1 9/16	2 1/4	1 1/8	na	na	7.71	2.350	314.003
W14x193	15 1/2	7/8	15 3/4	1 7/16	2 1/8	1 1/16	na	na	7.69	2.343	287.216
W14x176	15 1/4	13/16	15 5/8	1 5/16	2	1 1/16	na	na	7.61	2.321	261.917
W14x159	15	3/4	15 5/8	1 3/16	1 7/8	1	na	na	7.58	2.311	236.618
W14x145	14 3/4	11/16	15 1/2	1 1/16	1 3/4	1	na	na	7.51	2.289	215.784
W14x132	14 5/8	5/8	14 3/4	1	1 11/16	15/16	5 1/2	3	7.25	2.210	196.438
W14x120	14 1/2	9/16	14 5/8	15/16	1 5/8	15/16	5 1/2	3	7.20	2.194	178.580
W14x109	14 3/8	1/2	14 5/8	7/8	1 9/16	7/8	5 1/2	2 3/4	7.19	2.191	162.210
W14x99	14 1/8	1/2	14 5/8	3/4	1 7/16	7/8	5 1/2	2 3/4	7.15	2.178	147.328
W14x90	14	7/16	14 1/2	11/16	1 3/8	7/8	5 1/2	2 3/4	7.09	2.162	133.935
W14x82	14 1/4	1/2	10 1/8	7/8	1 5/8	1	5 1/2	2 3/4	5.67	1.727	122.029
W14x74	14 1/8	7/16	10 1/8	13/16	1 9/16	15/16	5 1/2	2 3/4	5.66	1.724	110.124
W14x68	14	7/16	10	3/4	1 1/2	15/16	5 1/2	2 3/4	5.59	1.705	101.195
W14x61	13 7/8	3/8	10	5/8	1 7/16	15/16	5 1/2	2 3/4	5.58	1.702	90.778
W14x53	13 7/8	3/8	8	11/16	1 7/16	15/16	5 1/2	2 3/4	4.92	1.499	78.873
W14x48	13 3/4	5/16	8	5/8	1 3/8	7/8	5 1/2	2 1/2	4.91	1.495	71.432
W14x43	13 5/8	5/16	8	1/2	1 5/16	7/8	5 1/2	2 1/2	4.89	1.489	63.991
W14x38	14 1/8	5/16	6 3/4	1/2	1 1/16	5/8	3 1/2	2 1/2	4.55	1.387	56.550
W14x34	14	5/16	6 3/4	7/16	1	5/8	3 1/2	2 1/2	4.53	1.381	50.598
W14x30	13 7/8	1/4	6 3/4	3/8	15/16	5/8	3 1/2	2 1/2	4.52	1.378	44.645
W14x26	13 7/8	1/4	5	7/16	15/16	9/16	2 3/4	2 1/2	3.94	1.200	38.692
W14x22	13 3/4	1/4	5	5/16	7/8	9/16	2 3/4	2 1/4	3.92	1.194	32.740
W12x336	16 7/8	1 3/4	13 3/8	2 15/16	3 11/16	1 1/2	na	na	6.98	2.127	500.023
W12x305	16 3/8	1 5/8	13 1/4	2 11/16	3 7/16	1 7/16	na	na	6.88	2.096	453.890
W12x279	15 7/8	1 1/2	13 1/8	2 1/2	3 3/16	1 3/8	na	na	6.77	2.064	415.198
W12x252	15 3/8	1 3/8	13	2 1/4	2 15/16	1 5/16	na	na	6.67	2.032	375.017
W12x230	15	1 5/16	12 7/8	2 1/16	2 3/4	1 1/4	na	na	6.57	2.003	342.278
W12x210	14 3/4	1 3/16	12 3/4	1 7/8	2 5/8	1 1/4	na	na	6.51	1.984	312.514
W12x190	14 3/8	1 1/16	12 5/8	1 3/4	2 7/16	1 3/16	na	na	6.43	1.959	282.751
W12x170	14	15/16	12 5/8	1 9/16	2 1/4	1 1/8	na	na	6.39	1.946	252.988
W12x152	13 3/4	7/8	12 1/2	1 3/8	2 1/8	1 1/16	na	na	6.31	1.924	226.201
W12x136	13 3/8	13/16	12 3/8	1 1/4	1 15/16	1	na	na	6.22	1.895	202.390
W12x120	13 1/8	11/16	12 3/8	1 1/8	1 13/16	1	5 1/2	3	6.20	1.889	178.580
W12x106	12 7/8	5/8	12 1/4	1	1 11/16	15/16	5 1/2	3	6.13	1.867	157.745
W12x96	12 3/4	9/16	12 1/8	7/8	1 5/8	7/8	5 1/2	3	6.07	1.851	142.864
W12x87	12 1/2	1/2	12 1/8	13/16	1 1/2	7/8	5 1/2	2 3/4	6.04	1.842	129.470
W12x79	12 3/8	1/2	12 1/8	3/4	1 7/16	7/8	5 1/2	2 3/4	6.02	1.835	117.565
W12x72	12 1/4	7/16	12	11/16	1 3/8	7/8	5 1/2	2 3/4	5.97	1.819	107.148
W12x65	12 1/8	3/8	12	5/8	1 5/16	13/16	5 1/2	2 1/2	5.96	1.816	96.731
W12x58	12 1/4	3/8	10	5/8	1 3/8	13/16	5 1/2	2 3/4	5.31	1.619	86.314
W12x53	12	3/8	10	9/16	1 1/4	13/16	5 1/2	2 1/2	5.27	1.607	78.873

W Shapes

Shapes in shaded rows are not available from domestic producers

Beam Size	Depth d	Web Thk. tw	Fig. Width bf	Fig. Thk. tf	k	k1	Flange Bolt Gage	Web Bolt Gage	Sq. Ft.- Surf Area - per Ft	Sq.M.- Surf Area - per M	Weight Kg per M
W12x50	12 1/4	3/8	8 1/8	5/8	1 3/8	13/16	5 1/2	2 3/4	4.69	1.429	74.408
W12x45	12	5/16	8	9/16	1 1/4	13/16	5 1/2	2 1/2	4.61	1.407	66.967
W12x40	12	5/16	8	1/2	1 1/4	3/4	5 1/2	2 1/2	4.61	1.407	59.527
W12x35	12 1/2	5/16	6 1/2	1/2	1	9/16	3 1/2	2 1/2	4.20	1.280	52.086
W12x30	12 3/8	1/4	6 1/2	7/16	15/16	1/2	3 1/2	2 1/2	4.19	1.276	44.645
W12x26	12 1/4	1/4	6 1/2	3/8	7/8	1/2	3 1/2	2 1/2	4.17	1.270	38.692
W12x22	12 1/4	1/4	4	7/16	7/8	1/2	2 1/4	2 1/2	3.33	1.016	32.740
W12x19	12 1/8	1/4	4	3/8	13/16	1/2	2 1/4	2 1/4	3.31	1.010	28.275
W12x16	12	1/4	4	1/4	3/4	1/2	2 1/4	2 1/4	3.29	1.003	23.811
W12x14	11 7/8	3/16	4	1/4	11/16	1/2	2 1/4	2 1/4	3.28	1.000	20.834
W10x112	11 3/8	3/4	10 3/8	1 1/4	1 7/8	15/16	na	na	5.23	1.594	166.674
W10x100	11 1/8	11/16	10 3/8	1 1/8	1 3/4	7/8	na	na	5.20	1.584	148.816
W10x88	10 7/8	5/8	10 1/4	1	1 5/8	13/16	5 1/2	2 3/4	5.13	1.562	130.958
W10x77	10 5/8	1/2	10 1/4	7/8	1 1/2	13/16	5 1/2	2 3/4	5.10	1.556	114.589
W10x68	10 3/8	1/2	10 1/8	3/4	1 3/8	3/4	5 1/2	2 3/4	5.02	1.530	101.195
W10x60	10 1/4	7/16	10 1/8	11/16	1 5/16	3/4	5 1/2	2 1/2	5.01	1.527	89.290
W10x54	10 1/8	3/8	10	5/8	1 1/4	11/16	5 1/2	2 1/2	4.96	1.511	80.361
W10x49	10	5/16	10	9/16	1 3/16	11/16	5 1/2	2 1/2	4.95	1.508	72.920
W10x45	10 1/8	3/8	8	5/8	1 1/4	11/16	5 1/2	2 1/2	4.29	1.308	66.967
W10x39	9 7/8	5/16	8	1/2	1 1/8	11/16	5 1/2	2 1/2	4.26	1.299	58.038
W10x33	9 3/4	5/16	8	7/16	1 1/16	11/16	5 1/2	2 1/4	4.24	1.292	49.109
W10x30	10 1/2	5/16	5 3/4	1/2	15/16	1/2	2 3/4	2 1/2	3.61	1.102	44.645
W10x26	10 3/8	1/4	5 3/4	7/16	7/8	1/2	2 3/4	2 1/4	3.60	1.099	38.692
W10x22	10 1/8	1/4	5 3/4	3/8	3/4	1/2	2 3/4	2 1/4	3.56	1.086	32.740
W10x19	10 1/4	1/4	4	3/8	13/16	1/2	2 1/4	2 1/4	3.00	0.914	28.275
W10x17	10 1/8	1/4	4	5/16	3/4	1/2	2 1/4	2 1/4	2.98	0.908	25.299
W10x15	10	1/4	4	1/4	11/16	7/16	2 1/4	2 1/4	2.96	0.902	22.322
W10x12	9 7/8	3/16	4	3/16	5/8	7/16	2 1/4	2	2.95	0.899	17.858
W8x67	9	9/16	8 1/4	15/16	1 7/16	11/16	na	na	4.16	1.267	99.707
W8x58	8 3/4	1/2	8 1/4	13/16	1 5/16	11/16	5 1/2	2 3/4	4.13	1.257	86.314
W8x48	8 1/2	3/8	8 1/8	11/16	1 3/16	5/8	5 1/2	2 1/2	4.06	1.238	71.432
W8x40	8 1/4	3/8	8 1/8	9/16	1 1/16	5/8	5 1/2	2 1/2	4.02	1.226	59.527
W8x35	8 1/8	5/16	8	1/2	1	9/16	5 1/2	2 1/4	3.97	1.210	52.086
W8x31	8	5/16	8	7/16	15/16	9/16	5 1/2	2 1/4	3.95	1.203	46.133
W8x28	8	5/16	6 1/2	7/16	15/16	9/16	3 1/2	2 1/4	3.45	1.051	41.669
W8x24	7 7/8	1/4	6 1/2	3/8	7/8	9/16	3 1/2	2 1/4	3.44	1.048	35.716
W8x21	8 1/4	1/4	5 1/4	3/8	13/16	1/2	2 3/4	2 1/4	3.08	0.940	31.251
W8x18	8 1/8	1/4	5 1/4	5/16	3/4	7/16	2 3/4	2 1/4	3.06	0.933	26.787
W8x15	8 1/8	1/4	4	5/16	3/4	1/2	2 1/4	2 1/4	2.65	0.806	22.322
W8x13	8	1/4	4	1/4	11/16	7/16	2 1/4	2 1/4	2.63	0.800	19.346
W8x10	7 7/8	3/16	4	3/16	5/8	7/16	2 1/4	2	2.61	0.797	14.882
W6x25	6 3/8	5/16	6 1/8	7/16	13/16	7/16	3 1/2	2 1/4	3.05	0.930	37.204
W6x20	6 1/4	1/4	6	3/8	3/4	7/16	3 1/2	2 1/4	3.00	0.914	29.763
W6x15	6	1/4	6	1/4	5/8	3/8	3 1/2	2 1/4	2.96	0.902	22.322

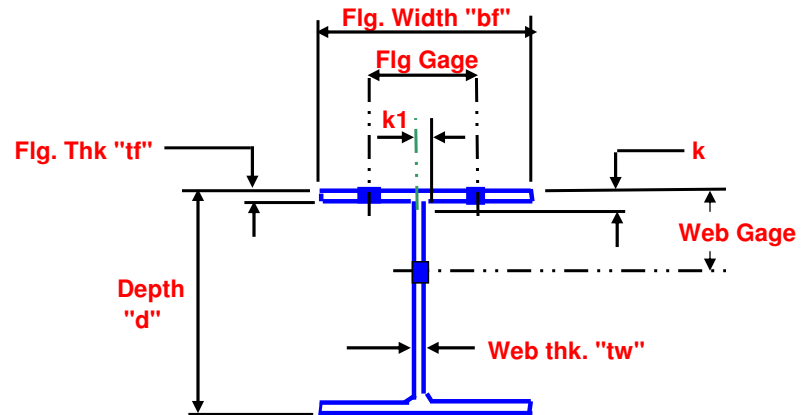
W Shapes



Shapes in shaded rows are not available from domestic producers

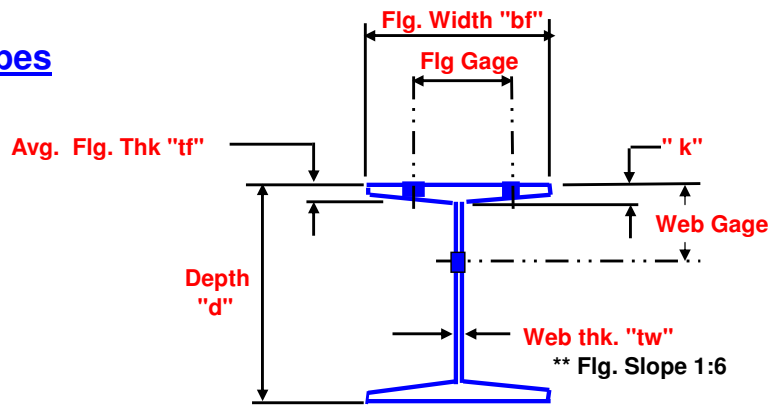
Beam Size	Depth d	Web Thk. tw	Fig. Width bf	Fig. Thk. tf	k	k1	Flange Bolt Gage	Web Bolt Gage	Sq. Ft.- Surf Area - per Ft	Sq.M.- Surf Area - per M	Weight Kg per M
W6x16	6 1/4	1/4	4	3/8	3/4	7/16	2 1/4	2 1/4	2.33	0.711	23.811
W6x12	6	1/4	4	1/4	5/8	3/8	2 1/4	2 1/4	2.29	0.699	17.858
W6x9	5 7/8	3/16	4	3/16	9/16	3/8	2 1/4	2	2.28	0.695	13.393
W5x19	5 1/8	1/4	5	7/16	13/16	7/16	2 3/4	2 1/4	2.48	0.756	28.275
W5x16	5	1/4	5	3/8	3/4	7/16	2 3/4	2 1/4	2.46	0.749	23.811
W4x13	4 1/8	1/4	4	3/8	11/16	7/16	2 1/4	2	1.98	0.603	19.346

M Shapes



Beam Size	Depth d	Web Thk. tw	Flg. Width bf	Flg. Thk. tf	k	Max Bolt	Flange Bolt Gage	Web Bolt Gage	Sq. Ft.- Surf Area - per Ft	Sq.M.- Surf Area - per M	Weight Kg per M
M14x18	14	3/16	4	1/4	5/8	3/4			3.64	1.108	26.787
M12x11.8	12	3/16	3 1/8	1/4	9/16	-			3.01	0.918	17.560
M12x10.8	12	3/16	3 1/8	1/4	1/2	1/2			3.01	0.918	16.072
M12x10	12	3/16	3 1/4	3/16	1/2				3.05	0.930	14.882
M10x9	10	3/16	2 3/4	3/16	9/16	-			2.55	0.778	13.393
M10x8	10	3/16	2 3/4	3/16	7/16	3/8			2.55	0.778	11.905
M10x7.5	10	1/8	2 3/4	3/16	7/16	3/8			2.56	0.781	11.161
M8x6.5	8	1/8	2 1/4	3/16	1/2	-			2.06	0.629	9.673
M6x4.4	6	1/8	1 7/8	3/16	3/16	-			1.60	0.489	6.548
M5x18.9	5	5/16	5	7/16	7/8	7/8			2.45	0.746	28.126

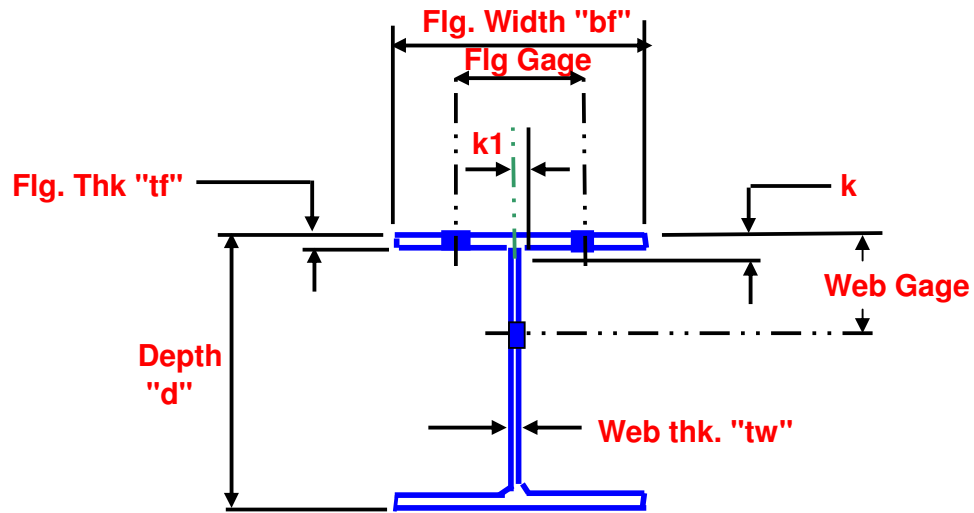
S Shapes



Beam Size	Depth d	Web Thk. tw	Fig. Width bf	*Fig. Thk. tf	k	Grip	Max Bolt	Flange Bolt Gage	Web Bolt Gage	Sq. Ft.- Surf Area - per Ft	Sq.M.- Surf Area - per M	Weight Kg per M
S24x121	24 1/2	13/16	8	1 1/16	2	1 1/8	1			6.61	2.016	180.068
S24x106	24 1/2	5/8	7 7/8	1 1/16	2	1 1/8	1			6.60	2.013	157.745
S24x100	24	3/4	7 1/4	7/8	1 3/4	7/8	1	4	3	6.29	1.918	148.816
S24x90	24	5/8	7 1/8	7/8	1 3/4	7/8	1	4	3	6.27	1.911	133.935
S24x80	24	1/2	7	7/8	1 3/4	7/8	1			6.25	1.905	119.053
S20x96	20 1/4	13/16	7 1/4	15/16	1 3/4	15/16	1			5.66	1.724	142.864
S20x86	20 1/4	11/16	7	15/16	1 3/4	15/16	1			5.59	1.705	127.982
S20x75	20	5/8	6 3/8	13/16	1 5/8	13/16	7/8	3 1/2	3	5.35	1.632	111.612
S20x66	20	1/2	6 1/4	13/16	1 5/8	13/16	7/8			5.33	1.626	98.219
S18x70	18	11/16	6 1/4	11/16	1 1/2	11/16	7/8	3 1/2	2 3/4	4.97	1.514	104.171
S18x54.7	18	7/16	6	11/16	1 1/2	11/16	7/8	3 1/2	2 3/4	4.93	1.502	81.403
S15x50	15	9/16	5 5/8	5/8	1 3/8	9/16	3/4	3 1/2	2 3/4	4.28	1.305	74.408
S15x42.9	15	7/16	5 1/2	5/8	1 3/8	9/16	3/4	3 1/2	2 3/4	4.26	1.299	63.842
S12x50	12	11/16	5 1/2	11/16	1 7/16	11/16	3/4	3	2 3/4	3.72	1.133	74.408
S12x40.8	12	7/16	5 1/4	11/16	1 7/16	5/8	3/4	3	2 3/4	3.68	1.121	60.717
S12x35	12	7/16	5 1/8	9/16	1 3/16	1/2	3/4	3	2 1/2	3.64	1.108	52.086
S12x31.8	12	3/8	5	9/16	1 3/16	1/2	3/4	3	2 1/2	3.60	1.099	47.324
S10x35	10	5/8	5	1/2	1 1/8	1/2	3/4	2 3/4	2 1/2	3.23	0.984	52.086
S10x25.4	10	5/16	4 5/8	1/2	1 1/8	1/2	3/4	2 3/4	2 1/2	3.16	0.962	37.799
S8x23	8	7/16	4 1/8	7/16	1	7/16	3/4	2 1/4	2 1/2	2.64	0.803	34.228
S8x18.4	8	1/4	4	7/16	1	7/16	3/4	2 1/4	2 1/2	2.63	0.800	27.382
S7x20	7	7/16	3 7/8	3/8	15/16	3/8	5/8			2.39	0.727	29.763
S7x15.3	7	1/4	3 5/8	3/8	15/16	3/8	5/8	2 1/4	2 1/2	2.33	0.711	22.769
S6x17.25	6	7/16	3 5/8	3/8	7/8	3/8	5/8	2	2 1/4	2.14	0.651	25.671
S6x12.5	6	1/4	3 3/8	3/8	7/8	3/8	-	-	2 1/4	2.08	0.635	18.602
S5x14.75	5	1/2	3 1/4	5/16	13/16	5/16	-			1.83	0.559	21.950
S5x10	5	3/16	3	5/16	13/16	5/16	-	-	2 1/4	1.80	0.549	14.882
S4x9.5	4	5/16	2 3/4	5/16	3/4	5/16	-	-	2	1.53	0.467	14.138
S4x7.7	4	3/16	2 5/8	5/16	3/4	5/16	-	-	2	1.51	0.460	11.459
S3x7.5	3	3/8	2 1/2	1/4	11/16	1/4	-	-	-	1.27	0.387	11.161
S3x5.7	3	3/16	2 3/8	1/4	11/16	1/4	-	-	-	1.26	0.384	8.483

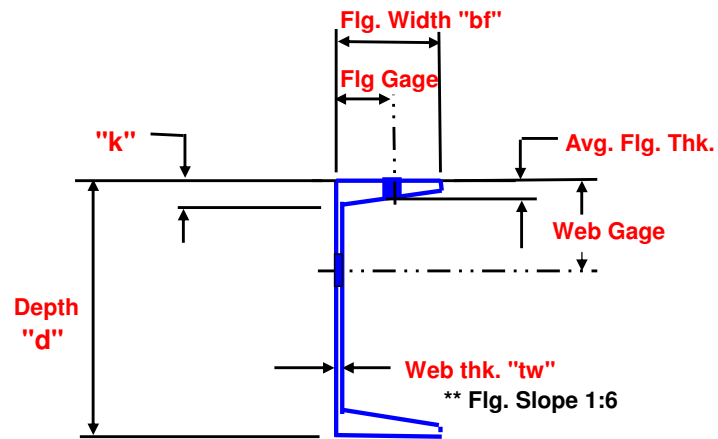
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HP Shapes



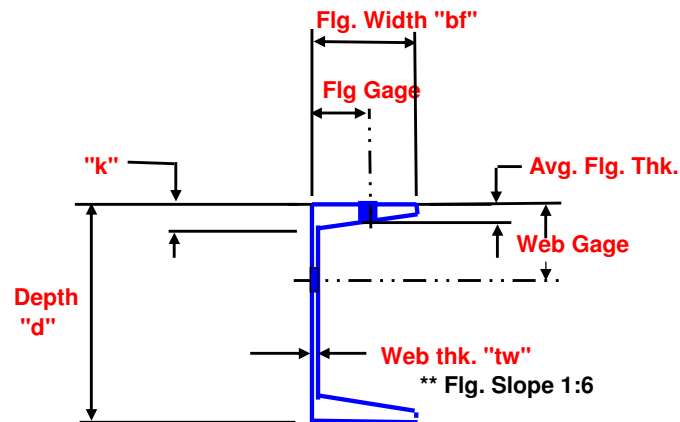
Beam Size	Depth d	Web Thk. tw	Flg. Width bf	Flg. Thk. tf	k	k1	Sq. Ft.- Surf Area - per Ft	Sq.M.- Surf Area - per M	Weight Kg per M
HP14x117	14 1/4	13/16	14 7/8	13/16	1 1/2	1	7.20	2.194	174.115
HP14x102	14	11/16	14 3/4	11/16	1 3/8	1	7.14	2.175	151.793
HP14x89	13 7/8	5/8	14 3/4	5/8	1 5/16	15/16	7.13	2.172	132.447
HP14x73	13 5/8	1/2	14 5/8	1/2	1 3/16	7/8	7.06	2.153	108.636
HP13x100	13 1/8	3/4	13 1/4	3/4	1 7/16	1	6.48	1.975	148.816
HP13x87	13	11/16	13 1/8	11/16	1 3/8	15/16	6.43	1.959	129.470
HP13x73	12 3/4	9/16	13	9/16	1 1/4	15/16	6.36	1.940	108.636
HP13x60	12 1/2	7/16	12 7/8	7/16	1 1/8	7/8	6.30	1.921	89.290
HP12x84	12 1/4	11/16	12 1/4	11/16	1 3/8	1	6.01	1.832	125.006
HP12x74	12 1/8	5/8	12 1/4	5/8	1 5/16	15/16	6.00	1.829	110.124
HP12x63	12	1/2	12 1/8	1/2	1 1/4	7/8	5.96	1.816	93.754
HP12x53	11 3/4	7/16	12	7/16	1 1/8	7/8	5.89	1.794	78.873
HP10x57	10	9/16	10 1/4	9/16	1 3/16	13/16	4.99	1.521	84.825
HP10x42	9 3/4	7/16	10 1/8	7/16	1 1/16	3/4	4.93	1.502	62.503
HP8x36	8	7/16	8 1/8	7/16	15/16	5/8	3.97	1.210	53.574

Channels Standard



Channel Size	Depth d	Web Thk. tw	Flg. Width bf	Avg. Flg. Thk. tf	k	Grip	Max. Bolt	Flange Bolt Gage	Web Bolt Gage	Sq. Ft.- Surf Area - per Ft	Sq.M.- Surf Area - per M	Weight Kg per M
C15x50	15	11/16	3 3/4	5/8	1 7/16	5/8	1	2 1/4	2 3/4	3.53	1.076	74.408
C15x40	15	1/2	3 1/2	5/8	1 7/16	5/8	1	2	2 3/4	3.48	1.060	59.527
C15x33.9	15	3/8	3 3/8	5/8	1 7/16	5/8	1	2	2 3/4	3.46	1.054	50.449
C12x30	12	1/2	3 1/8	1/2	1 1/8	1/2	7/8	1 3/4	2 1/2	2.88	0.876	44.645
C12x25	12	3/8	3	1/2	1 1/8	1/2	7/8	1 3/4	2 1/2	2.85	0.870	37.204
C12x20.7	12	5/16	3	1/2	1 1/8	1/2	7/8	1 3/4	2 1/2	2.86	0.873	30.805
C10x30	10	11/16	3	7/16	1	7/16	3/4	1 3/4	2 1/2	2.48	0.756	44.645
C10x25	10	1/2	2 7/8	7/16	1	7/16	3/4	1 3/4	2 1/2	2.47	0.752	37.204
C10x20	10	3/8	2 3/4	7/16	1	7/16	3/4	1 1/2	2 1/2	2.45	0.746	29.763
C10x15.3	10	1/4	2 5/8	7/16	1	7/16	3/4	1 1/2	2 1/2	2.43	0.740	22.769
C9x20	9	7/16	2 5/8	7/16	15/16	7/16	3/4			2.23	0.679	29.763
C9x15	9	5/16	2 1/2	7/16	15/16	7/16	3/4	1 3/8	2 1/2	2.21	0.673	22.322
C9x13.4	9	1/4	2 3/8	7/16	15/16	7/16	3/4	1 3/8	2 1/2	2.18	0.664	19.941
C8x18.75	8	1/2	2 1/2	3/8	15/16	3/8	3/4	1 1/2	2 1/2	2.02	0.616	27.903
C8x13.75	8	5/16	2 3/8	3/8	15/16	3/8	3/4	1 3/8	2 1/2	2.01	0.613	20.462
C8x11.5	8	1/4	2 1/4	3/8	15/16	3/8	3/4	1 3/8	2 1/2	1.98	0.603	17.114
C7x14.75	7	7/16	2 1/4	3/8	7/8	3/8	5/8	1 1/4	2 1/2	1.78	0.543	21.950
C7x12.25	7	5/16	2 1/4	3/8	7/8	3/8	5/8	1 1/4	2 1/2	1.80	0.549	18.230
C7x9.8	7	3/16	2 1/8	3/8	7/8	3/8	5/8	1 1/4	2 1/4	1.78	0.543	14.584
C6x13	6	7/16	2 1/8	5/16	13/16	5/16	5/8	1 3/8	2 1/4	1.58	0.483	19.346
C6x10.5	6	5/16	2	5/16	13/16	3/8	5/8	1 1/8	2 1/4	1.56	0.476	15.626
C6x8.2	6	3/16	1 7/8	5/16	13/16	5/16	5/8	1 1/8	2 1/4	1.54	0.470	12.203
C5x9	5	5/16	1 7/8	5/16	3/4	5/16	5/8	1 1/8	2 1/4	1.35	0.413	13.393
C5x6.7	5	3/16	1 3/4	5/16	3/4	-	-	-	2 1/4	1.33	0.406	9.971
C4x7.25	4	5/16	1 3/4	5/16	11/16	5/16	5/8	1	2	1.15	0.349	10.789
C4x5.4	4	3/16	1 5/8	5/16	11/16	-	-	-	2	1.13	0.343	8.036
C3x6	3	3/8	1 5/8	1/4	11/16	-	-	-	-	0.94	0.286	8.929
C3x5	3	1/4	1 1/2	1/4	11/16	-	-	-	-	0.92	0.279	7.441
C3x4.1	3	3/16	1 3/8	1/4	11/16	-	-	-	-	0.89	0.270	6.101

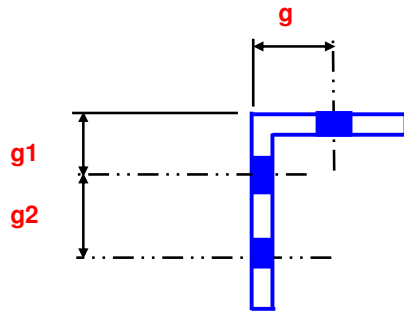
Channels Miscellaneous



Channel Size	Depth d	Web Thk. tw	Flg. Width bf	Avg. Flg. Thk. tf	k	Grip	Max. Bolt	Flange Bolt Gage	Web Bolt Gage	Sq. Ft.- Surf Area - per Ft	Sq.M.- Surf Area - per M	Weight Kg per M
MC18x58	18	11/16	4 1/4	5/8	1 3/8	5/8	1			4.20	1.280	86.314
MC18x51.9	18	5/8	4 1/8	5/8	1 3/8	5/8	1			4.17	1.270	77.236
MC18x45.8	18	1/2	4	5/8	1 3/8	5/8	1			4.15	1.264	68.158
MC18x42.7	18	7/16	4	5/8	1 3/8	5/8	1			4.16	1.267	63.545
MC13x50	13	13/16	4 3/8	5/8	1 3/8	5/8	1			3.39	1.032	74.408
MC13x40	13	9/16	4 1/8	5/8	1 3/8	9/16	1			3.34	1.019	59.527
MC13x35	13	7/16	4 1/8	5/8	1 3/8	9/16	1			3.36	1.026	52.086
MC13x31.8	13	3/8	4	5/8	1 3/8	9/16	1			3.33	1.016	47.324
MC12x50	12	13/16	4 1/8	11/16	1 5/16	11/16	1			3.13	0.953	74.408
MC12x45	12	11/16	4	11/16	1 5/16	11/16	1			3.10	0.946	66.967
MC12x40	12	9/16	3 7/8	11/16	1 5/16	11/16	1			3.08	0.940	59.527
MC12x35	12	7/16	3 3/4	11/16	1 5/16	11/16	1			3.06	0.933	52.086
MC12x31	12	3/8	3 5/8	11/16	1 5/16	11/16	1			3.03	0.924	46.133
MC12x10.6	12	3/16	1 1/2	5/16	11/16	-	-			2.42	0.737	15.775
MC10x41.1	10	13/16	4 3/8	9/16	1 1/4	9/16	7/8			2.90	0.883	61.164
MC10x33.6	10	9/16	4 1/8	9/16	1 1/4	9/16	7/8			2.85	0.870	50.002
MC10x28.5	10	7/16	4	9/16	1 1/4	9/16	7/8			2.83	0.864	42.413
MC10x25	10	3/8	3 3/8	9/16	1 1/4	9/16	7/8			2.64	0.803	37.204
MC10x22	10	5/16	3 3/8	9/16	1 1/4	9/16	7/8			2.65	0.806	32.740
MC10x8.4	10	3/16	1 1/2	1/4	11/16	-	-			2.09	0.638	12.501
MC10x6.5	10	1/8	1 1/8	3/16	7/16	-	-			1.99	0.606	9.673
MC9x25.4	9	7/16	3 1/2	9/16	1 3/16	9/16	7/8			2.50	0.762	37.799
MC9x23.9	9	3/8	3 1/2	9/16	1 3/16	9/16	7/8			2.51	0.765	35.567
MC8x22.8	8	7/16	3 1/2	1/2	1 3/16	1/2	7/8			2.34	0.714	33.930
MC8x21.4	8	3/8	3 1/2	1/2	1 3/16	1/2	7/8			2.35	0.718	31.847
MC8x20	8	3/8	3	1/2	1 1/8	1/2	7/8			2.19	0.667	29.763
MC8x18.7	8	3/8	3	1/2	1 3/16	1/2	7/8			2.19	0.667	27.829
MC8x8.5	8	3/16	1 7/8	5/16	3/4	5/16	5/8			1.88	0.572	12.649
MC7x22.7	7	1/2	3 5/8	1/2	1 1/8	1/2	7/8			2.21	0.673	33.781
MC7x19.1	7	3/8	3 1/2	1/2	1 1/8	1/2	7/8			2.19	0.667	28.424
MC6x18	6	3/8	3 1/2	1/2	1 1/16	1/2	7/8			2.02	0.616	26.787
MC6x15.3	6	5/16	3 1/2	3/8	7/8	3/8	7/8			2.05	0.625	22.769
MC6x16.3	6	3/8	3	1/2	1 1/16	1/2	3/4			1.85	0.565	24.257
MC6x15.1	6	5/16	3	1/2	1 1/16	1/2	3/4			1.86	0.568	22.471
MC6x12	6	5/16	2 1/2	3/8	13/16	3/8	5/8			1.72	0.524	17.858

Angles

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Usual Gages For Angles , Inches

Leg	8	7	6	5	4	3 1/2	3	2 1/2	2	1 3/4	1 3/8	1 1/4	1
g	4 1/2	4	3 1/2	3	2 1/2	2	1 3/4	1 3/8	1 1/8	1	7/8	3/4	5/8
g1	3	2 1/2	2 1/4	2									
g2	3	3	2 1/2	1 3/4									

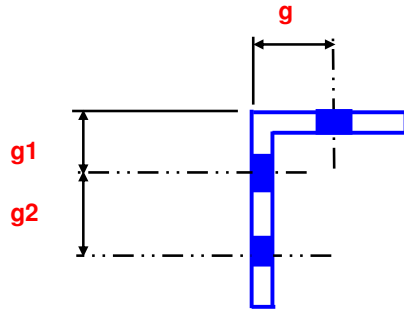
Angles in shaded rows may not be readily available

Size	Thk.	k	Weight lbs/ft	Weight Kg per M	Surf. Area Sq.Ft-Ft.	Surf. Area Sq.M-M.
L9x4	5/8	1 1/8	26.30	39.139	2.167	0.660
L9x4	9/16	1 1/16	23.80	35.418	2.167	0.660
L9x4	1/2	1	21.30	31.698	2.167	0.660
L8x8	1 1/8	1 3/4	56.90	84.677	2.667	0.813
L8x8	1	1 5/8	51.00	75.896	2.667	0.813
L8x8	7/8	1 1/2	45.00	66.967	2.667	0.813
L8x8	3/4	1 3/8	38.90	57.890	2.667	0.813
L8x8	5/8	1 1/4	32.70	48.663	2.667	0.813
L8x8	9/16	1 3/16	29.60	44.050	2.667	0.813
L8x8	1/2	1 1/8	26.40	39.288	2.667	0.813
L8x6	1	1 1/2	44.20	65.777	2.333	0.711
L8x6	7/8	1 3/8	39.10	58.187	2.333	0.711
L8x6	3/4	1 1/4	33.80	50.300	2.333	0.711
L8x6	5/8	1 1/8	28.50	42.413	2.333	0.711
L8x6	9/16	1 1/16	25.70	38.246	2.333	0.711
L8x6	1/2	1	23.00	34.228	2.333	0.711
L8x6	7/16	15/16	20.20	30.061	2.333	0.711
L8x4	1	1 1/2	37.40	55.657	2.000	0.610
L8x4	3/4	1 1/4	28.70	42.710	2.000	0.610
L8x4	9/16	1 1/16	21.90	32.591	2.000	0.610
L8x4	1/2	1	19.60	29.168	2.000	0.610
L7x4	3/4	1 1/4	26.20	38.990	1.833	0.559
L7x4	5/8	1 1/8	22.10	32.888	1.833	0.559
L7x4	1/2	1	17.90	26.638	1.833	0.559
L7x4	3/8	7/8	13.60	20.239	1.833	0.559
L6x6	1	1 1/2	37.40	55.657	2.000	0.610
L6x6	7/8	1 3/8	33.10	49.258	2.000	0.610
L6x6	3/4	1 1/4	28.70	42.710	2.000	0.610
L6x6	5/8	1 1/8	24.20	36.014	2.000	0.610
L6x6	9/16	1 1/16	21.90	32.591	2.000	0.610
L6x6	1/2	1	19.60	29.168	2.000	0.610
L6x6	7/16	15/16	17.20	25.596	2.000	0.610
L6x6	3/8	7/8	14.90	22.174	2.000	0.610
L6x6	5/16	13/16	12.40	18.453	2.000	0.610
L6x4	7/8	1 3/8	27.20	40.478	1.667	0.508
L6x4	3/4	1 1/4	23.60	35.121	1.667	0.508
L6x4	5/8	1 1/8	20.00	29.763	1.667	0.508
L6x4	9/16	1 1/16	18.10	26.936	1.667	0.508
L6x4	1/2	1	16.20	24.108	1.667	0.508
L6x4	7/16	15/16	14.30	21.281	1.667	0.508
L6x4	3/8	7/8	12.30	18.304	1.667	0.508
L6x4	5/16	13/16	10.30	15.328	1.667	0.508
L6x3 1/2	1/2	1	15.30	22.769	1.583	0.483
L6x3 1/2	3/8	7/8	11.70	17.412	1.583	0.483
L6x3 1/2	5/16	13/16	9.80	14.584	1.583	0.483
L5x5	7/8	1 3/8	27.20	40.478	1.667	0.508
L5x5	3/4	1 1/4	23.60	35.121	1.667	0.508
L5x5	5/8	1 1/8	20.00	29.763	1.667	0.508
L5x5	1/2	1	16.20	24.108	1.667	0.508
L5x5	7/16	15/16	14.30	21.281	1.667	0.508
L5x5	3/8	7/8	12.30	18.304	1.667	0.508

Size	Thk.	k	Weight lbs/ft	Weight Kg per M	Surf. Area Sq.Ft-Ft.	Surf. Area Sq.M-M.
L4x3 1/2	1/2	15/16	11.90	17.709	1.250	0.381
L4x3 1/2	7/16	7/8	10.60	15.775	1.250	0.381
L4x3 1/2	3/8	13/16	9.10	13.542	1.250	0.381
L4x3 1/2	5/16	3/4	7.70	11.459	1.250	0.381
L4x3 1/2	1/4	11/16	6.20	9.227	1.250	0.381
L4x3	1/2	15/16	11.10	16.519	1.167	0.356
L4x3	7/16	7/8	9.80	14.584	1.167	0.356
L4x3	3/8	13/16	8.50	12.649	1.167	0.356
L4x3	5/16	3/4	7.20	10.715	1.167	0.356
L4x3	1/4	11/16	5.80	8.631	1.167	0.356
L3 1/2x3 1/2	1/2	7/8	11.10	16.519	1.167	0.356
L3 1/2x3 1/2	7/16	13/16	9.80	14.584	1.167	0.356
L3 1/2x3 1/2	3/8	3/4	8.50	12.649	1.167	0.356
L3 1/2x3 1/2	5/16	11/16	7.20	10.715	1.167	0.356
L3 1/2x3 1/2	1/4	5/8	5.80	8.631	1.167	0.356
L3 1/2x3	1/2	15/16	10.20	15.179	1.083	0.330
L3 1/2x3	7/16	7/8	9.10	13.542	1.083	0.330
L3 1/2x3	3/8	13/16	7.90	11.756	1.083	0.330
L3 1/2x3	5/16	3/4	6.60	9.822	1.083	0.330
L3 1/2x3	1/4	11/16	5.40	8.036	1.083	0.330
L3 1/2x2 1/2	1/2	15/16	9.40	13.989	1.000	0.305
L3 1/2x2 1/2	7/16	7/8	8.30	12.352	1.000	0.305
L3 1/2x2 1/2	3/8	13/16	7.20	10.715	1.000	0.305
L3 1/2x2 1/2	5/16	3/4	6.10	9.078	1.000	0.305
L3 1/2x2 1/2	1/4	11/16	4.90	7.292	1.000	0.305
L3x3	1/2	13/16	9.40	13.989	1.000	0.305
L3x3	7/16	3/4	8.30	12.352	1.000	0.305
L3x3	3/8	11/16	7.20	10.715	1.000	0.305
L3x3	5/16	5/8	6.10	9.078	1.000	0.305
L3x3	1/4	9/16	4.90	7.292	1.000	0.305
L3x3	3/16	1/2	3.71	5.521	1.000	0.305
L3x2 1/2	1/2	7/8	8.50	12.649	0.917	0.279
L3x2 1/2	7/16	13/16	7.60	11.310	0.917	0.279
L3x2 1/2	3/8	3/4	6.60	9.822	0.917	0.279
L3x2 1/2	5/16	11/16	5.60	8.334	0.917	0.279
L3x2 1/2	1/4	5/8	4.50	6.697	0.917	0.279
L3x2 1/2	3/16	9/16	3.39	5.045	0.917	0.279
L3x2	1/2	13/16	7.70	11.459	0.833	0.254
L3x2	7/16	3/4	6.80	10.120	0.833	0.254
L3x2	3/8	11/16	5.90	8.780	0.833	0.254
L3x2	5/16	5/8	5.00	7.441	0.833	0.254
L3x2	1/4	9/16	4.10	6.101	0.833	0.254
L3x2	3/16	1/2	3.07	4.569	0.833	0.254
L2 1/2x2 1/2	1/2	13/16	7.70	11.459	0.833	0.254
L2 1/2x2 1/2	3/8	11/16	5.90	8.780	0.833	0.254
L2 1/2x2 1/2	5/16	5/8	5.00	7.441	0.833	0.254
L2 1/2x2 1/2	1/4	9/16	4.10	6.101	0.833	0.254
L2 1/2x2 1/2	3/16	1/2	3.07	4.569	0.833	0.254
L2 1/2x2	3/8	11/16	5.30	7.887	0.750	0.229
L2 1/2x2	5/16	5/8	4.50	6.697	0.750	0.229

Angles

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Usual Gages For Angles , Inches

Leg	8	7	6	5	4	3 1/2	3	2 1/2	2	1 3/4	1 3/8	1 1/4	1
g	4 1/2	4	3 1/2	3	2 1/2	2	1 3/4	1 3/8	1 1/8	1	7/8	3/4	5/8
g1	3	2 1/2	2 1/4	2									
g2	3	3	2 1/2	1 3/4									

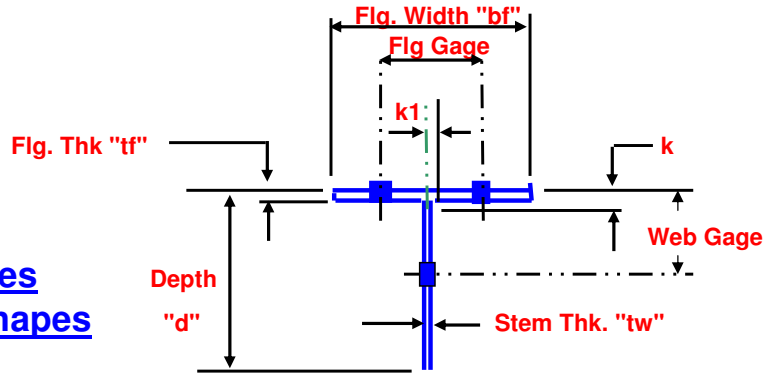
Angles in shaded rows may not be readily available

Size	Thk.	k	Weight lbs/ft	Weight Kg per M	Surf. Area Sq.Ft-Ft.	Surf. Area Sq.M-M.
L5x5	5/16	13/16	10.30	15.328	1.667	0.508
L5x3 1/2	3/4	1 1/4	19.80	29.466	1.417	0.432
L5x3 1/2	5/8	1 1/8	16.80	25.001	1.417	0.432
L5x3 1/2	1/2	1	13.60	20.239	1.417	0.432
L5x3 1/2	7/16	15/16	12.00	17.858	1.417	0.432
L5x3 1/2	3/8	7/8	10.40	15.477	1.417	0.432
L5x3 1/2	5/16	13/16	8.70	12.947	1.417	0.432
L5x3 1/2	1/4	3/4	7.00	10.417	1.417	0.432
L5x3	5/8	1	15.70	23.364	1.333	0.406
L5x3	1/2	1	12.80	19.048	1.333	0.406
L5x3	7/16	15/16	11.30	16.816	1.333	0.406
L5x3	3/8	7/8	9.80	14.584	1.333	0.406
L5x3	5/16	13/16	8.20	12.203	1.333	0.406
L5x3	1/4	3/4	6.60	9.822	1.333	0.406
L4x4	3/4	1 1/8	18.50	27.531	1.333	0.406
L4x4	5/8	1	15.70	23.364	1.333	0.406
L4x4	1/2	7/8	12.80	19.048	1.333	0.406
L4x4	7/16	13/16	11.30	16.816	1.333	0.406
L4x4	3/8	3/4	9.80	14.584	1.333	0.406
L4x4	5/16	11/16	8.20	12.203	1.333	0.406
L4x4	1/4	5/8	6.60	9.822	1.333	0.406

Size	Thk.	k	Weight lbs/ft	Weight Kg per M	Surf. Area Sq.Ft-Ft.	Surf. Area Sq.M-M.
L2 1/2x2	1/4	9/16	3.62	5.387	0.750	0.229
L2 1/2x2	3/16	1/2	2.75	4.092	0.750	0.229
L2x2	3/8	5/8	4.70	6.994	0.667	0.203
L2x2	5/16	9/16	3.92	5.834	0.667	0.203
L2x2	1/4	1/2	3.19	4.747	0.667	0.203
L2x2	3/16	7/16	2.44	3.631	0.667	0.203
L2x2	1/8	3/8	1.65	2.455	0.667	0.203
L1 3/4x1 3/4	1/4	1/2	2.77	4.122	0.583	0.178
L1 3/4x1 3/4	3/16	7/16	2.12	3.155	0.583	0.178
L1 1/2x1 1/2	1/4	7/16	2.34	3.482	0.500	0.152
L1 1/2x1 1/2	3/16	3/8	1.80	2.679	0.500	0.152
L1 1/4x1 1/4	1/4	7/16	1.92	2.857	0.417	0.127
L1 1/4x1 1/4	3/16	3/8	1.48	2.202	0.417	0.127
L1 1/8x1 1/8	1/8	7/32	0.90	1.339	0.375	0.114
L1x1	1/8	1/4	0.80	1.191	0.333	0.102

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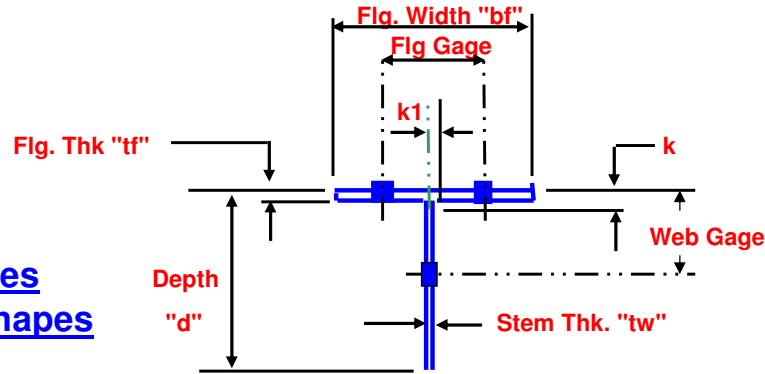
Structural Tees Cut from W shapes



WTee Size	Depth d	Stem Thk. tw	Flg. Width bf	Flg. Thk. tf	k	k1	Flange Bolt Gage	Web Bolt Gage	Sq. Ft.- Surf Area - per Ft	Sq.M.- Surf Area - per M	Weight Kg per M
WT18x179.5	18 11/16	1 1/8	16 3/4	2	3 1/8				5.91	1.800	267.125
WT18x164	18 9/16	1	16 5/8	1 7/8	3				5.86	1.788	244.059
WT18x150	18 3/8	15/16	16 5/8	1 11/16	2 13/16	1 1/2	5 1/2	3 3/4	5.83	1.778	223.225
WT18x140	18 1/4	7/8	16 5/8	1 9/16	2 11/16	1 1/2	5 1/2	3 3/4	5.81	1.772	208.343
WT18x130	18 1/8	13/16	16 1/2	1 7/16	2 9/16	1 1/2	5 1/2	3 1/2	5.77	1.759	193.461
WT18x122.5	18	13/16	16 1/2	1 3/8	2 1/2	1 7/16	5 1/2	3 1/2	5.75	1.753	182.300
WT18x115	18	3/4	16 1/2	1 1/4	2 3/8	1 7/16	5 1/2	3 1/2	5.75	1.753	171.139
WT18x128	18 11/16	1	12 1/4	1 3/4	2 5/8				5.16	1.572	190.485
WT18x116	18 9/16	7/8	12 1/8	1 9/16	2 1/2				5.11	1.559	172.627
WT18x105	18 3/8	13/16	12 1/8	1 3/8	2 5/16	1 1/4	5 1/2	3 1/2	5.08	1.549	156.257
WT18x97	18 1/4	3/4	12 1/8	1 1/4	2 3/16	1 3/16	5 1/2	3 1/2	5.06	1.543	144.352
WT18x91	18 1/8	3/4	12 1/8	1 3/16	2 1/8	1 3/16	5 1/2	3 1/4	5.04	1.537	135.423
WT18x85	18 1/8	11/16	12	1 1/8	2	1 3/16	5 1/2	3 1/4	5.02	1.530	126.494
WT18x80	18	5/8	12	1	1 15/16	1 1/8	5 1/2	3 1/4	5.00	1.524	119.053
WT18x75	17 7/8	5/8	12	15/16	1 7/8	1 1/8	5 1/2	3	4.98	1.518	111.612
WT18x67.5	17 3/4	5/8	12	13/16	1 11/16	1 1/8	5 1/2	3	4.96	1.511	100.451
WT16.5x177	17 3/4	1 3/16	16 1/8	2 1/16	2 7/8				5.65	1.721	263.405
WT16.5x159	17 9/16	1 1/16	16	1 7/8	2 11/16				5.59	1.705	236.618
WT16.5x145.5	17 7/16	1	15 7/8	1 3/4	2 9/16				5.55	1.692	216.528
WT16.5x131.5	17 1/4	7/8	15 3/4	1 9/16	2 3/8				5.50	1.676	195.694
WT16.5x120.5	17 1/8	13/16	15 7/8	1 3/8	2 3/16	1 1/4	5 1/2	3 1/2	5.50	1.676	179.324
WT16.5x110.5	17	3/4	15 3/4	1 1/4	2 1/16	1 3/16	5 1/2	3 1/2	5.46	1.664	164.442
WT16.5x100.5	16 7/8	11/16	15 3/4	1 1/8	1 15/16	1 1/8	5 1/2	3 1/4	5.44	1.657	149.560
WT16.5x84.5	16 15/16	11/16	11 1/2	1 1/4	2 1/16				4.74	1.445	125.750
WT16.5x76	16 3/4	5/8	11 5/8	1 1/16	1 7/8	1 1/8	5 1/2	3 1/4	4.73	1.441	113.100
WT16.5x70.5	16 5/8	5/8	11 1/2	15/16	1 3/4	1 1/16	5 1/2	3	4.69	1.429	104.916
WT16.5x65	16 1/2	9/16	11 1/2	7/8	1 11/16	1 1/16	5 1/2	3	4.67	1.422	96.731
WT16.5x59	16 3/8	9/16	11 1/2	3/4	1 9/16	1 1/16	5 1/2	2 3/4	4.65	1.416	87.802
WT15x117.5	15 5/8	13/16	15	1 1/2	2 1/4				5.10	1.556	174.859
WT15x105.5	15 1/2	3/4	15 1/8	1 5/16	2 1/8	1 1/8	5 1/2	3 1/2	5.10	1.556	157.001
WT15x95.5	15 3/8	11/16	15	1 3/16	1 15/16	1 1/16	5 1/2	3 1/4	5.06	1.543	142.120
WT15x86.5	15 1/4	5/8	15	1 1/16	1 7/8	1 1/16	5 1/2	3 1/4	5.04	1.537	128.726
WT15x74	15 5/16	5/8	10 1/2	1 3/16	2				4.30	1.311	110.124
WT15x66	15 1/8	5/8	10 1/2	1	1 3/4	1 1/16	5 1/2	3	4.27	1.302	98.219
WT15x62	15 1/8	9/16	10 1/2	15/16	1 11/16	1	5 1/2	3	4.27	1.302	92.266
WT15x58	15	9/16	10 1/2	7/8	1 5/8	1	5 1/2	3	4.25	1.295	86.314
WT15x54	14 7/8	9/16	10 1/2	3/4	1 9/16	1	5 1/2	3	4.23	1.289	80.361
WT15x49.5	14 7/8	1/2	10 1/2	11/16	1 7/16	1	5 1/2	2 3/4	4.23	1.289	73.664
WT13.5x108.5	14 3/16	13/16	14 1/8	1 1/2	2 3/16				4.72	1.438	161.466
WT13.5x97	14 1/16	3/4	14	1 5/16	2 1/16				4.68	1.426	144.352
WT13.5x89	13 7/8	3/4	14 1/8	1 3/16	1 7/8	1 1/16	5 1/2	3 1/4	4.67	1.422	132.447
WT13.5x80.5	13 3/4	11/16	14	1 1/16	1 13/16	1	5 1/2	3 1/4	4.63	1.410	119.797
WT13.5x73	13 3/4	5/8	14	1	1 11/16	1	5 1/2	3	4.63	1.410	108.636
WT13.5x64.5	13 13/16	5/8	10	1 1/8	1 13/16				3.97	1.210	95.987

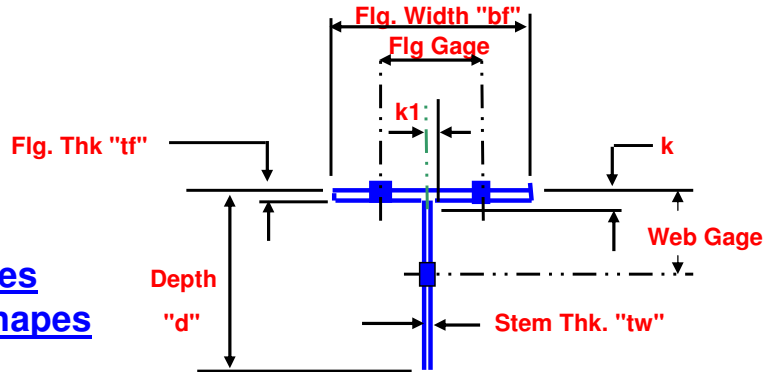
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Structural Tees Cut from W shapes



WTee Size	Depth d	Stem Thk. tw	Fig. Width bf	Fig. Thk. tf	k	k1	Flange Bolt Gage	Web Bolt Gage	Sq. Ft.- Surf Area - per Ft	Sq.M.- Surf Area - per M	Weight Kg per M
WT13.5x57	13 5/8	9/16	10 1/8	15/16	1 5/8	15/16	5 1/2	3	3.96	1.207	84.825
WT13.5x51	13 1/2	1/2	10	13/16	1 9/16	15/16	5 1/2	3	3.92	1.194	75.896
WT13.5x47	13 1/2	1/2	10	3/4	1 7/16	15/16	5 1/2	2 3/4	3.92	1.194	69.944
WT13.5x42	13 3/8	7/16	10	5/8	1 3/8	15/16	5 1/2	2 3/4	3.90	1.187	62.503
WT12x88	12 5/8	3/4	12 7/8	1 5/16	2 1/8				4.25	1.295	130.958
WT12x81	12 1/2	11/16	13	1 1/4	2	1 1/16	5 1/2	3 1/4	4.25	1.295	120.541
WT12x73	12 3/8	5/8	12 7/8	1 1/16	1 7/8	1 1/16	5 1/2	3 1/4	4.21	1.283	108.636
WT12x65.5	12 1/4	5/8	12 7/8	15/16	1 3/4	1 1/16	5 1/2	3	4.19	1.276	97.475
WT12x58.5	12 1/8	9/16	12 3/4	7/8	1 5/8	1	5 1/2	3	4.15	1.264	87.058
WT12x52	12	1/2	12 3/4	3/4	1 1/2	1	5 1/2	2 3/4	4.13	1.257	77.385
WT12x51.5	12 1/4	9/16	9	1	1 3/4				3.54	1.080	76.640
WT12x47	12 1/8	1/2	9 1/8	7/8	1 5/8	1	5 1/2	3	3.54	1.080	69.944
WT12x42	12	1/2	9	3/4	1 9/16	15/16	5 1/2	3	3.50	1.067	62.503
WT12x38	12	7/16	9	11/16	1 7/16	15/16	5 1/2	2 3/4	3.50	1.067	56.550
WT12x34	11 7/8	7/16	9	9/16	1 3/8	15/16	5 1/2	2 3/4	3.48	1.060	50.598
WT12x31	11 7/8	7/16	7	9/16	1 3/8	15/16	3 1/2	2 3/4	3.15	0.959	46.133
WT12x27.5	11 3/4	3/8	7	1/2	1 5/16	15/16	3 1/2	2 3/4	3.13	0.953	40.925
WT10.5x83	11 1/4	3/4	12 3/8	1 3/8	2 1/8				3.94	1.200	123.518
WT10.5x73.5	11	3/4	12 1/2	1 1/8	1 7/8	1 1/16	5 1/2	3 1/4	3.92	1.194	109.380
WT10.5x66	10 7/8	5/8	12 1/2	1 1/16	1 13/16	1	5 1/2	3	3.90	1.187	98.219
WT10.5x61	10 7/8	5/8	12 3/8	15/16	1 11/16	1	5 1/2	3	3.88	1.181	90.778
WT10.5x55.5	10 3/4	9/16	12 3/8	7/8	1 5/8	15/16	5 1/2	2 3/4	3.85	1.175	82.593
WT10.5x50.5	10 5/8	1/2	12 1/4	13/16	1 9/16	15/16	5 1/2	2 3/4	3.81	1.162	75.152
WT10.5x46.5	10 3/4	9/16	8 3/8	15/16	1 11/16	1	5 1/2	3	3.19	0.972	69.200
WT10.5x41.5	10 3/4	1/2	8 3/8	13/16	1 9/16	15/16	5 1/2	2 3/4	3.19	0.972	61.759
WT10.5x36.5	10 5/8	7/16	8 1/4	3/4	1 1/2	15/16	5 1/2	2 3/4	3.15	0.959	54.318
WT10.5x34	10 5/8	7/16	8 1/4	11/16	1 7/16	7/8	5 1/2	2 3/4	3.15	0.959	50.598
WT10.5x31	10 1/2	3/8	8 1/4	5/8	1 3/8	7/8	5 1/2	2 1/2	3.13	0.953	46.133
WT10.5x28.5	10 1/2	3/8	6 1/2	5/8	1 3/8	7/8	3 1/2	2 3/4	2.83	0.864	42.413
WT10.5x25	10 3/8	3/8	6 1/2	9/16	1 5/16	7/8	3 1/2	2 1/2	2.81	0.857	37.204
WT10.5x22	10 3/8	3/8	6 1/2	7/16	1 3/16	7/8	3 1/2	2 1/2	2.81	0.857	32.740
WT9x71.5	9 3/4	3/4	11 1/4	1 5/16	2				3.50	1.067	106.404
WT9x65	9 5/8	11/16	11 1/8	1 3/16	1 7/8				3.46	1.054	96.731
WT9x59.5	9 1/2	5/8	11 1/4	1 1/16	1 3/4	15/16	5 1/2	3	3.46	1.054	88.546
WT9x53	9 3/8	9/16	11 1/4	15/16	1 5/8	15/16	5 1/2	3	3.44	1.048	78.873
WT9x48.5	9 1/4	9/16	11 1/8	7/8	1 9/16	7/8	5 1/2	2 3/4	3.40	1.035	72.176
WT9x43	9 1/4	1/2	11 1/8	3/4	1 7/16	7/8	5 1/2	2 3/4	3.40	1.035	63.991
WT9x38	9 1/8	7/16	11	11/16	1 3/8	13/16	5 1/2	2 3/4	3.35	1.022	56.550
WT9x35.5	9 1/4	1/2	7 5/8	13/16	1 1/2	7/8	3 1/2	2 3/4	2.81	0.857	52.830
WT9x32.5	9 1/8	7/16	7 5/8	3/4	1 7/16	7/8	3 1/2	2 3/4	2.79	0.851	48.365
WT9x30	9 1/8	7/16	7 1/2	11/16	1 3/8	13/16	3 1/2	2 3/4	2.77	0.845	44.645
WT9x27.5	9	3/8	7 1/2	5/8	1 5/16	13/16	3 1/2	2 3/4	2.75	0.838	40.925
WT9x25	9	3/8	7 1/2	9/16	1 1/4	13/16	3 1/2	2 1/2	2.75	0.838	37.204

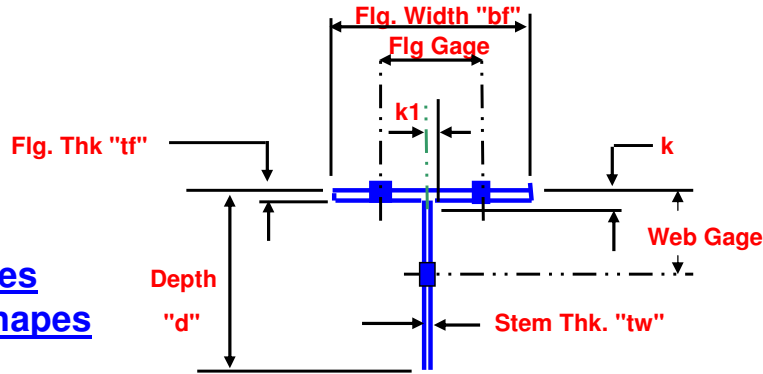
Structural Tees Cut from W shapes



WTee Size	Depth d	Stem Thk. tw	Flg. Width bf	Flg. Thk. tf	k	k1	Flange Bolt Gage	Web Bolt Gage	Sq. Ft.- Surf Area - per Ft	Sq.M.- Surf Area - per M	Weight Kg per M
WT9x23	9	3/8	6	5/8	1 1/4	13/16	3 1/2	2 1/2	2.50	0.762	34.228
WT9x20	9	5/16	6	1/2	1 3/16	13/16	3 1/2	2 1/2	2.50	0.762	29.763
WT9x17.5	8 7/8	5/16	6	7/16	1 1/8	3/4	3 1/2	2 1/2	2.48	0.756	26.043
WT8x50	8 1/2	9/16	10 3/8	1	1 11/16	15/16	5 1/2	3	3.15	0.959	74.408
WT8x44.5	8 3/8	1/2	10 3/8	7/8	1 9/16	7/8	5 1/2	2 3/4	3.13	0.953	66.223
WT8x38.5	8 1/4	7/16	10 1/4	3/4	1 7/16	7/8	5 1/2	2 3/4	3.08	0.940	57.294
WT8x33.5	8 1/8	3/8	10 1/4	11/16	1 3/8	13/16	5 1/2	2 3/4	3.06	0.933	49.853
WT8x28.5	8 1/4	7/16	7 1/8	11/16	1 3/8	7/8	3 1/2	2 3/4	2.56	0.781	42.413
WT8x25	8 1/8	3/8	7 1/8	5/8	1 5/16	13/16	3 1/2	2 3/4	2.54	0.775	37.204
WT8x22.5	8 1/8	3/8	7	9/16	1 1/4	13/16	3 1/2	2 1/2	2.52	0.768	33.484
WT8x20	8	5/16	7	1/2	1 3/16	13/16	3 1/2	2 1/2	2.50	0.762	29.763
WT8x18	7 7/8	5/16	7	7/16	1 1/8	3/4	3 1/2	2 1/2	2.48	0.756	26.787
WT8x15.5	8	1/4	5 1/2	7/16	1 1/8	3/4	2 3/4	2 1/2	2.25	0.686	23.067
WT8x13	7 7/8	1/4	5 1/2	3/8	1 1/16	3/4	2 3/4	2 1/4	2.23	0.679	19.346
WT7x365	11 1/4	3 1/16	17 7/8	4 15/16	5 9/16	2 3/16	na	na	4.85	1.480	543.180
WT7x332.5	10 7/8	2 13/16	17 5/8	4 1/2	5 3/16	2 1/16	na	na	4.75	1.448	494.815
WT7x302.5	10 1/2	2 5/8	17 3/8	4 3/16	4 13/16	1 15/16	na	na	4.65	1.416	450.170
WT7x275	10 1/8	2 3/8	17 1/4	3 13/16	4 1/2	1 13/16	na	na	4.56	1.391	409.245
WT7x250	9 3/4	2 3/16	17	3 1/2	4 3/16	1 3/4	na	na	4.46	1.359	372.041
WT7x227.5	9 1/2	2	16 7/8	3 3/16	3 7/8	1 5/8	na	na	4.40	1.340	338.557
WT7x213	9 3/8	1 7/8	16 3/4	3 1/16	3 11/16	1 9/16	na	na	4.35	1.327	316.979
WT7x199	9 1/8	1 3/4	16 5/8	2 7/8	3 1/2	1 1/2	na	na	4.29	1.308	296.145
WT7x185	9	1 5/8	16 1/2	2 11/16	3 5/16	1 7/16	na	na	4.25	1.295	275.310
WT7x171	8 3/4	1 9/16	16 3/8	2 1/2	3 1/8	1 3/8	na	na	4.19	1.276	254.476
WT7x155.5	8 1/2	1 7/16	16 1/4	2 1/4	2 15/16	1 5/16	na	na	4.13	1.257	231.410
WT7x141.5	8 3/8	1 5/16	16 1/8	2 1/16	2 3/4	1 1/4	na	na	4.08	1.245	210.575
WT7x128.5	8 1/4	1 3/16	16	1 7/8	2 9/16	1 3/16	na	na	4.04	1.232	191.229
WT7x116.5	8	1 1/16	15 7/8	1 3/4	2 3/8	1 3/16	na	na	3.98	1.213	173.371
WT7x105.5	7 7/8	1	15 3/4	1 9/16	2 1/4	1 1/8	na	na	3.94	1.200	157.001
WT7x96.5	7 3/4	7/8	15 3/4	1 7/16	2 1/8	1 1/16	na	na	3.92	1.194	143.608
WT7x88	7 5/8	13/16	15 5/8	1 5/16	2	1 1/16	na	na	3.88	1.181	130.958
WT7x79.5	7 1/2	3/4	15 5/8	1 3/16	1 7/8	1	na	na	3.85	1.175	118.309
WT7x72.5	7 3/8	11/16	15 1/2	1 1/16	1 3/4	1	na	na	3.81	1.162	107.892
WT7x66	7 3/8	5/8	14 3/4	1	1 11/16	15/16	5 1/2	3	3.69	1.124	98.219
WT7x60	7 1/4	9/16	14 5/8	15/16	1 5/8	15/16	5 1/2	3	3.65	1.111	89.290
WT7x54.5	7 1/8	1/2	14 5/8	7/8	1 9/16	7/8	5 1/2	2 3/4	3.63	1.105	81.105
WT7x49.5	7 1/8	1/2	14 5/8	3/4	1 7/16	7/8	5 1/2	2 3/4	3.63	1.105	73.664
WT7x45	7	7/16	14 1/2	11/16	1 3/8	7/8	5 1/2	2 3/4	3.58	1.092	66.967
WT7x41	7 1/8	1/2	10 1/8	7/8	1 5/8	1	5 1/2	2 3/4	2.88	0.876	61.015
WT7x37	7 1/8	7/16	10 1/8	13/16	1 9/16	15/16	5 1/2	2 3/4	2.88	0.876	55.062
WT7x34	7	7/16	10	3/4	1 1/2	15/16	5 1/2	2 3/4	2.83	0.864	50.598
WT7x30.5	7	3/8	10	5/8	1 7/16	15/16	5 1/2	2 3/4	2.83	0.864	45.389
WT7x26.5	7	3/8	8	11/16	1 7/16	15/16	5 1/2	2 3/4	2.50	0.762	39.436

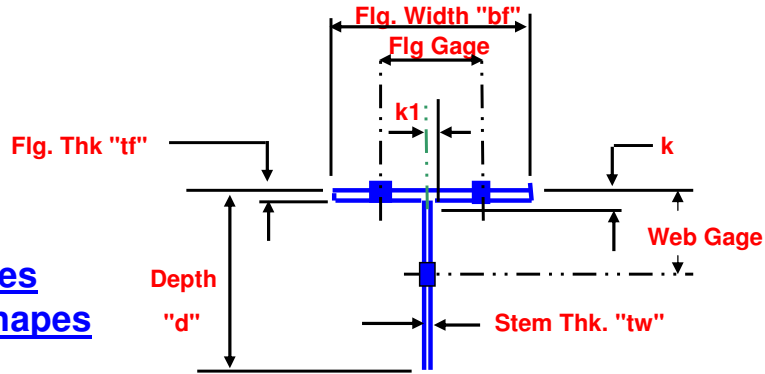
12/29/2008

Structural Tees Cut from W shapes



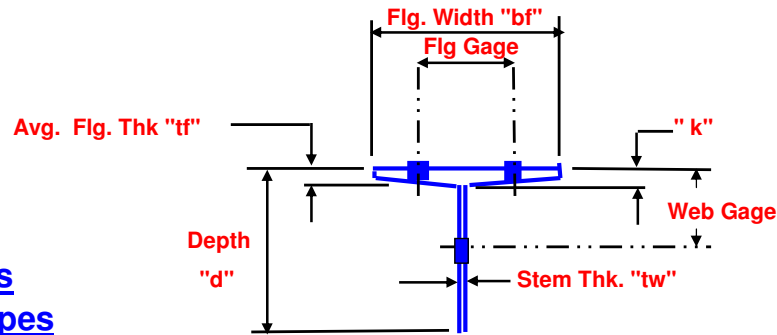
WTee Size	Depth d	Stem Thk. tw	Flg. Width bf	Flg. Thk. tf	k	k1	Flange Bolt Gage	Web Bolt Gage	Sq. Ft.- Surf Area - per Ft	Sq.M.- Surf Area - per M	Weight Kg per M
WT7x24	6 7/8	5/16	8	5/8	1 3/8	7/8	5 1/2	2 1/2	2.48	0.756	35.716
WT7x21.5	6 7/8	5/16	8	1/2	1 5/16	7/8	5 1/2	2 1/2	2.48	0.756	31.996
WT7x19	7	5/16	6 3/4	1/2	1 1/16	5/8	3 1/2	2 1/2	2.29	0.699	28.275
WT7x17	7	5/16	6 3/4	7/16	1	5/8	3 1/2	2 1/2	2.29	0.699	25.299
WT7x15	6 7/8	1/4	6 3/4	3/8	15/16	5/8	3 1/2	2 1/2	2.27	0.692	22.322
WT7x13	7	1/4	5	7/16	15/16	9/16	2 3/4	2 1/2	2.00	0.610	19.346
WT7x11	6 7/8	1/4	5	5/16	7/8	9/16	2 3/4	2 1/4	1.98	0.603	16.370
WT6x168	8 3/8	1 3/4	13 3/8	2 15/16	3 11/16	1 1/2	na	na	3.63	1.105	250.012
WT6x152.5	8 1/8	1 5/8	13 1/4	2 11/16	3 7/16	1 7/16	na	na	3.56	1.086	226.945
WT6x139.5	7 7/8	1 1/2	13 1/8	2 1/2	3 3/16	1 3/8	na	na	3.50	1.067	207.599
WT6x126	7 3/4	1 3/8	13	2 1/4	2 15/16	1 5/16	na	na	3.46	1.054	187.509
WT6x115	7 1/2	1 5/16	12 7/8	2 1/16	2 3/4	1 1/4	na	na	3.40	1.035	171.139
WT6x105	7 3/8	1 3/16	12 3/4	1 7/8	2 5/8	1 1/4	na	na	3.35	1.022	156.257
WT6x95	7 1/4	1 1/16	12 5/8	1 3/4	2 7/16	1 3/16	na	na	3.31	1.010	141.376
WT6x85	7	15/16	12 5/8	1 9/16	2 1/4	1 1/8	na	na	3.27	0.997	126.494
WT6x76	6 7/8	7/8	12 1/2	1 3/8	2 1/8	1 1/16	na	na	3.23	0.984	113.100
WT6x68	6 3/4	13/16	12 3/8	1 1/4	1 15/16	1	na	na	3.19	0.972	101.195
WT6x60	6 1/2	11/16	12 3/8	1 1/8	1 13/16	1	5 1/2	3	3.15	0.959	89.290
WT6x53	6 1/2	5/8	12 1/4	1	1 11/16	15/16	5 1/2	3	3.13	0.953	78.873
WT6x48	6 3/8	9/16	12 1/8	7/8	1 5/8	7/8	5 1/2	3	3.08	0.940	71.432
WT6x43.5	6 1/4	1/2	12 1/8	13/16	1 1/2	7/8	5 1/2	2 3/4	3.06	0.933	64.735
WT6x39.5	6 1/4	1/2	12 1/8	3/4	1 7/16	7/8	5 1/2	2 3/4	3.06	0.933	58.782
WT6x36	6 1/8	7/16	12	11/16	1 3/8	7/8	5 1/2	2 3/4	3.02	0.921	53.574
WT6x32.5	6	3/8	12	5/8	1 5/16	13/16	5 1/2	2 1/2	3.00	0.914	48.365
											0.000
WT6x29	6 1/8	3/8	10	5/8	1 3/8	13/16	5 1/2	2 3/4	2.69	0.819	43.157
WT6x26.5	6	3/8	10	9/16	1 1/4	13/16	5 1/2	2 1/2	2.67	0.813	39.436
WT6x25	6 1/8	3/8	8 1/8	5/8	1 3/8	13/16	5 1/2	2 3/4	2.38	0.724	37.204
WT6x22.5	6	5/16	8	9/16	1 1/4	13/16	5 1/2	2 1/2	2.33	0.711	33.484
WT6x20	6	5/16	8	1/2	1 1/4	3/4	5 1/2	2 1/2	2.33	0.711	29.763
WT6x17.5	6 1/4	5/16	6 1/2	1/2	1	9/16	3 1/2	2 1/2	2.13	0.648	26.043
WT6x15	6 1/8	1/4	6 1/2	7/16	15/16	1/2	3 1/2	2 1/2	2.10	0.641	22.322
WT6x13	6 1/8	1/4	6 1/2	3/8	7/8	1/2	3 1/2	2 1/2	2.10	0.641	19.346
WT6x11	6 1/8	1/4	4	7/16	7/8	1/2	2 1/4	2 1/2	1.69	0.514	16.370
WT6x9.5	6 1/8	1/4	4	3/8	13/16	1/2	2 1/4	2 1/4	1.69	0.514	14.138
WT6x8	6	1/4	4	1/4	3/4	1/2	2 1/4	2 1/4	1.67	0.508	11.905
WT6x7	6	3/16	4	1/4	11/16	1/2	2 1/4	2 1/4	1.67	0.508	10.417
WT5x56	5 5/8	3/4	10 3/8	1 1/4	1 7/8	15/16	na	na	2.67	0.813	83.337
WT5x50	5 1/2	11/16	10 3/8	1 1/8	1 3/4	7/8	na	na	2.65	0.806	74.408
WT5x44	5 3/8	5/8	10 1/4	1	1 5/8	13/16	5 1/2	2 3/4	2.60	0.794	65.479
WT5x38.5	5 1/4	1/2	10 1/4	7/8	1 1/2	13/16	5 1/2	2 3/4	2.58	0.787	57.294
WT5x34	5 1/4	1/2	10 1/8	3/4	1 3/8	3/4	5 1/2	2 3/4	2.56	0.781	50.598
WT5x30	5 1/8	7/16	10 1/8	11/16	1 5/16	3/4	5 1/2	2 1/2	2.54	0.775	44.645
WT5x27	5	3/8	10	5/8	1 1/4	11/16	5 1/2	2 1/2	2.50	0.762	40.180

Structural Tees Cut from W shapes



WTee Size	Depth d	Stem Thk. tw	Fig. Width bf	Fig. Thk. tf	k	k1	Flange Bolt Gage	Web Bolt Gage	Sq. Ft.- Surf Area - per Ft	Sq.M.- Surf Area - per M	Weight Kg per M
WT5x24.5	5	5/16	10	9/16	1 3/16	11/16	5 1/2	2 1/2	2.50	0.762	36.460
WT5x22.5	5	3/8	8	5/8	1 1/4	11/16	5 1/2	2 1/2	2.17	0.660	33.484
WT5x19.5	5	5/16	8	1/2	1 1/8	11/16	5 1/2	2 1/2	2.17	0.660	29.019
WT5x16.5	4 7/8	5/16	8	7/16	1 1/16	11/16	5 1/2	2 1/4	2.15	0.654	24.555
WT5x15	5 1/4	5/16	5 3/4	1/2	15/16	1/2	2 3/4	2 1/2	1.83	0.559	22.322
WT5x13	5 1/8	1/4	5 3/4	7/16	7/8	1/2	2 3/4	2 1/4	1.81	0.552	19.346
WT5x11	5 1/8	1/4	5 3/4	3/8	3/4	1/2	2 3/4	2 1/4	1.81	0.552	16.370
WT5x9.5	5 1/8	1/4	4	3/8	13/16	1/2	2 1/4	2 1/4	1.52	0.464	14.138
WT5x8.5	5	1/4	4	5/16	3/4	1/2	2 1/4	2 1/4	1.50	0.457	12.649
WT5x7.5	5	1/4	4	1/4	11/16	7/16	2 1/4	2 1/4	1.50	0.457	11.161
WT5x6	4 7/8	3/16	4	3/16	5/8	7/16	2 1/4	2	1.48	0.451	8.929
WT4x33.5	4 1/2	9/16	8 1/4	15/16	1 7/16	11/16	na	na	2.13	0.648	49.853
WT4x29	4 3/8	1/2	8 1/4	13/16	1 5/16	11/16	5 1/2	2 3/4	2.10	0.641	43.157
WT4x24	4 1/4	3/8	8 1/8	11/16	1 3/16	5/8	5 1/2	2 1/2	2.06	0.629	35.716
WT4x20	4 1/8	3/8	8 1/8	9/16	1 1/16	5/8	5 1/2	2 1/2	2.04	0.622	29.763
WT4x17.5	4	5/16	8	1/2	1	9/16	5 1/2	2 1/4	2.00	0.610	26.043
WT4x15.5	4	5/16	8	7/16	15/16	9/16	5 1/2	2 1/4	2.00	0.610	23.067
WT4x14	4	5/16	6 1/2	7/16	15/16	9/16	3 1/2	2 1/4	1.75	0.533	20.834
WT4x12	4	1/4	6 1/2	3/8	7/8	9/16	3 1/2	2 1/4	1.75	0.533	17.858
WT4x10.5	4 1/8	1/4	5 1/4	3/8	13/16	1/2	2 3/4	2 1/4	1.56	0.476	15.626
WT4x9	4 1/8	1/4	5 1/4	5/16	3/4	7/16	2 3/4	2 1/4	1.56	0.476	13.393
WT4x7.5	4	1/4	4	5/16	3/4	1/2	2 1/4	2 1/4	1.33	0.406	11.161
WT4x6.5	4	1/4	4	1/4	11/16	7/16	2 1/4	2 1/4	1.33	0.406	9.673
WT4x5	4	3/16	4	3/16	5/8	7/16	2 1/4	2	1.33	0.406	7.441
WT3x12.5	3 1/4	5/16	6 1/8	7/16	13/16	7/16	3 1/2	2 1/4	1.56	0.476	18.602
WT3x10	3 1/8	1/4	6	3/8	3/4	7/16	3 1/2	2 1/4	1.52	0.464	14.882
WT3x7.5	3	1/4	6	1/4	5/8	3/8	3 1/2	2 1/4	1.50	0.457	11.161
WT3x8	3 1/8	1/4	4	3/8	3/4	7/16	2 1/4	2 1/4	1.19	0.362	11.905
WT3x6	3	1/4	4	1/4	5/8	3/8	2 1/4	2 1/4	1.17	0.356	8.929
WT3x4.5	3	3/16	4	3/16	9/16	3/8	2 1/4	2	1.17	0.356	6.697
WT2.5x9.5	2 5/8	1/4	5	7/16	13/16	7/16	2 3/4	2 1/4	1.27	0.387	14.138
WT2.5x8	2 1/2	1/4	5	3/8	3/4	7/16	2 3/4	2 1/4	1.25	0.381	11.905
WT2x6.5	2 1/8	1/4	4	3/8	11/16	7/16	2 1/4	2	1.02	0.311	9.673

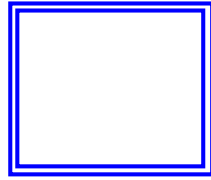
Structutal Tees Cut from S shapes



** Flg. Slope 1:6

STee Size	Depth d	Stem Thk. tw	Flg. Width bf	*Flg. Thk. tf	k	Grip	Max Bolt	Flange Bolt Gage	Web Bolt Gage	Sq. Ft.- Surf Area - per Ft	Sq.M.- Surf Area - per M	Weight Kg per M
ST12x60.5	12 1/4	13/16	8	1 1/16	2	1 1/8	1			3.38	1.029	90.034
ST12x53	12 1/4	5/8	7 7/8	1 1/16	2	1 1/8	1			3.35	1.022	78.873
ST12x50	12	3/4	7 1/4	7/8	1 3/4	7/8	1	4	3	3.21	0.978	74.408
ST12x45	12	5/8	7 1/8	7/8	1 3/4	7/8	1	4	3	3.19	0.972	66.967
ST12x40	12	1/2	7	7/8	1 3/4	7/8	1			3.17	0.965	59.527
ST10x48	10 1/8	13/16	7 1/4	15/16	1 3/4	15/16	1			2.90	0.883	71.432
ST10x43	10 1/8	11/16	7	15/16	1 3/4	15/16	1			2.85	0.870	63.991
ST10x37.5	10	5/8	6 3/8	13/16	1 5/8	13/16	7/8	3 1/2	3	2.73	0.832	55.806
ST10x33	10	1/2	6 1/4	13/16	1 5/8	13/16	7/8			2.71	0.826	49.109
ST9x35	9	11/16	6 1/4	11/16	1 1/2	11/16	7/8	3 1/2	2 3/4	2.54	0.775	52.086
ST9x27.35	9	7/16	6	11/16	1 1/2	11/16	7/8	3 1/2	2 3/4	2.50	0.762	40.701
ST7.5x25	7 1/2	9/16	5 5/8	5/8	1 3/8	9/16	3/4	3 1/2	2 3/4	2.19	0.667	37.204
ST7.5x21.45	7 1/2	7/16	5 1/2	5/8	1 3/8	9/16	3/4	3 1/2	2 3/4	2.17	0.660	31.921
ST6x25	6	11/16	5 1/2	11/16	1 7/16	11/16	3/4	3	2 3/4	1.92	0.584	37.204
ST6x20.4	6	7/16	5 1/4	11/16	1 7/16	5/8	3/4	3	2 3/4	1.88	0.572	30.359
ST6x17.5	6	7/16	5 1/8	9/16	1 3/16	1/2	3/4	3	2 1/2	1.85	0.565	26.043
ST6x15.9	6	3/8	5	9/16	1 3/16	1/2	3/4	3	2 1/2	1.83	0.559	23.662
ST5x17.5	5	5/8	5	1/2	1 1/8	1/2	3/4	2 3/4	2 1/2	1.67	0.508	26.043
ST5x12.7	5	5/16	4 5/8	1/2	1 1/8	1/2	3/4	2 3/4	2 1/2	1.60	0.489	18.900
ST4x11.5	4	7/16	4 1/8	7/16	1	7/16	3/4	2 1/4	2 1/2	1.35	0.413	17.114
ST4x9.2	4	1/4	4	7/16	1	7/16	3/4	2 1/4	2 1/2	1.33	0.406	13.691
ST3.5x10	3 1/2	7/16	3 7/8	3/8	15/16	3/8	5/8			1.23	0.375	14.882
ST3.5x7.65	3 1/2	1/4	3 5/8	3/8	15/16	3/8	5/8	2 1/4	2 1/2	1.19	0.362	96.731
ST3x8.625	3	7/16	3 5/8	3/8	7/8	3/8	5/8	2	2 1/4	1.10	0.337	12.835
ST3x6.25	3	1/4	3 3/8	3/8	7/8	3/8	-	-	2 1/4	1.06	0.324	9.301
ST2.5x7.375	2 1/2	1/2	3 1/4	5/16	13/16	5/16	-			0.96	0.292	10.975
ST2.5x5	2 1/2	3/16	3	5/16	13/16	5/16	-	-	2 1/4	0.92	0.279	7.441
ST2x4.75	2	5/16	2 3/4	5/16	3/4	5/16	-	-	2	0.79	0.241	7.069
ST2x3.85	2	3/16	2 5/8	5/16	3/4	5/16	-	-	2	0.77	0.235	5.729
ST1.5x3.75	1 1/2	3/8	2 1/2	1/4	11/16	1/4	-	-	-	0.67	0.203	5.581
ST1.5x2.85	1 1/2	3/16	2 3/8	1/4	11/16	1/4	-	-	-	0.65	0.197	

12/29/2008



Structural Tubing

Radius Equals two times wall thickness

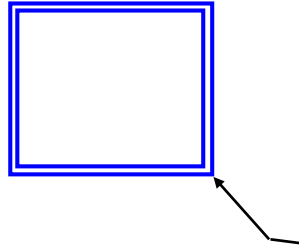
Square

Nominal Size	Wall Thk	Weight Lbs/Ft	Weight Kg/M	Surf. Area Sq.Ft-per Ft	Surf. Area Sq.M-per M
16x16	5/8	127.37	189.547	5.33	1.626
16x16	1/2	103.30	153.727	5.33	1.626
16x16	3/8	78.52	116.851	5.33	1.626
16x16	5/16	65.87	98.025	5.33	1.626
14x14	5/8	110.36	164.234	4.67	1.422
14x14	1/2	89.68	133.459	4.67	1.422
14x14	3/8	68.31	101.656	4.67	1.422
14x14	5/16	57.36	85.361	4.67	1.422
12x12	5/8	93.34	138.905	4.00	1.219
12x12	1/2	76.07	113.205	4.00	1.219
12x12	3/8	58.10	86.462	4.00	1.219
12x12	5/16	48.86	72.712	4.00	1.219
12x12	1/4	39.43	58.678	4.00	1.219
12x12	3/16	29.84	44.407	4.00	1.219
10x10	5/8	76.33	113.592	3.33	1.016
10x10	9/16	69.48	103.398	3.33	1.016
10x10	1/2	62.46	92.951	3.33	1.016
10x10	3/8	47.90	71.283	3.33	1.016
10x10	5/16	40.35	60.047	3.33	1.016
10x10	1/4	32.63	48.559	3.33	1.016
10x10	3/16	24.73	36.802	3.33	1.016
9x9	5/8	67.82	100.927	3.00	0.914
9x9	9/16	61.83	92.013	3.00	0.914
9x9	1/2	55.66	82.831	3.00	0.914
9x9	3/8	42.79	63.679	3.00	0.914
9x9	5/16	36.10	53.723	3.00	0.914
9x9	1/4	29.23	43.499	3.00	0.914
9x9	3/16	22.18	33.007	3.00	0.914
8x8	5/8	59.32	88.278	2.67	0.813
8x8	9/16	54.17	80.614	2.67	0.813
8x8	1/2	48.85	72.697	2.67	0.813
8x8	3/8	37.69	56.089	2.67	0.813
8x8	5/16	31.84	47.383	2.67	0.813
8x8	1/4	25.82	38.424	2.67	0.813
8x8	3/16	19.63	29.213	2.67	0.813
7x7	9/16	46.51	69.215	2.33	0.711
7x7	1/2	42.05	62.577	2.33	0.711
7x7	3/8	32.58	48.484	2.33	0.711
7x7	5/16	27.59	41.058	2.33	0.711
7x7	1/4	22.42	33.365	2.33	0.711
7x7	3/16	17.08	25.418	2.33	0.711
6x6	9/16	38.86	57.830	2.00	0.610
6x6	1/2	35.24	52.443	2.00	0.610
6x6	3/8	27.48	40.895	2.00	0.610
6x6	5/16	23.34	34.734	2.00	0.610
6x6	1/4	19.02	28.305	2.00	0.610
6x6	3/16	14.53	21.623	2.00	0.610
5x5	1/2	28.43	42.309	1.67	0.508
5x5	3/8	22.37	33.290	1.67	0.508
5x5	5/16	19.08	28.394	1.67	0.508
5x5	1/4	15.62	23.245	1.67	0.508

Rectangular

Nominal Size	Wall Thk	Weight Lbs/Ft	Weight Kg/M	Surf. Area Sq.Ft-per Ft	Surf. Area Sq.M-per M
20x12	1/2	103.30	153.727	5.33	1.626
20x12	3/8	78.52	116.851	5.33	1.626
20x12	5/16	65.87	98.025	5.33	1.626
20x8	1/2	89.68	133.459	4.67	1.422
20x8	3/8	68.31	101.656	4.67	1.422
20x8	5/16	57.36	85.361	4.67	1.422
20x4	1/2	76.07	113.205	4.00	1.219
20x4	3/8	58.10	86.462	4.00	1.219
20x4	5/16	48.86	72.712	4.00	1.219
18x6	1/2	76.07	113.205	4.00	1.219
18x6	3/8	58.10	86.462	4.00	1.219
18x6	5/16	48.86	72.712	4.00	1.219
16x12	5/8	110.36	164.234	4.67	1.422
16x12	1/2	89.68	133.459	4.67	1.422
16x12	3/8	68.31	101.656	4.67	1.422
16x12	5/16	57.36	85.361	4.67	1.422
16x8	1/2	76.07	113.205	4.00	1.219
16x8	3/8	58.10	86.462	4.00	1.219
16x8	5/16	48.86	72.712	4.00	1.219
16x4	1/2	62.46	92.951	3.33	1.016
16x4	3/8	47.90	71.283	3.33	1.016
16x4	5/16	40.35	60.047	3.33	1.016
14x10	5/8	93.34	138.905	4.00	1.219
14x10	1/2	76.07	113.205	4.00	1.219
14x10	3/8	58.10	86.462	4.00	1.219
14x10	5/16	48.86	72.712	4.00	1.219
14x6	1/2	62.46	92.951	3.33	1.016
14x6	3/8	47.90	71.283	3.33	1.016
14x6	5/16	40.35	60.047	3.33	1.016
14x6	1/4	32.63	48.559	3.33	1.016
14x4	1/2	55.66	82.831	3.00	0.914
14x4	3/8	42.79	63.679	3.00	0.914
14x4	5/16	36.10	53.723	3.00	0.914
14x4	1/4	29.23	43.499	3.00	0.914
12x8	5/8	76.33	113.592	3.33	1.016
12x8	9/16	69.48	103.398	3.33	1.016
12x8	1/2	62.46	92.951	3.33	1.016
12x8	3/8	47.90	71.283	3.33	1.016
12x8	5/16	40.35	60.047	3.33	1.016
12x8	1/4	32.63	48.559	3.33	1.016
12x8	3/16	24.73	36.802	3.33	1.016
12x6	5/8	67.82	100.927	3.00	0.914
12x6	9/16	61.83	92.013	3.00	0.914
12x6	1/2	55.66	82.831	3.00	0.914
12x6	3/8	42.79	63.679	3.00	0.914
12x6	5/16	36.10	53.723	3.00	0.914
12x6	1/4	29.23	43.499	3.00	0.914
12x6	3/16	22.18	33.007	3.00	0.914

12/29/2008



Structural Tubing

Radius Equals two times wall thickness

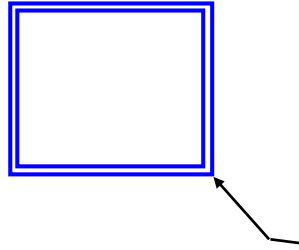
Square

Nominal Size	Wall Thk	Weight Lbs/Ft	Weight Kg./M	Surf. Area Sq.Ft-per Ft	Surf. Area Sq.M-per M
5x5	3/16	11.97	17.813	1.67	0.508
4.5x4.5	1/4	13.91	20.700	1.50	0.457
4.5x4.5	3/16	10.70	15.923	1.50	0.457
4x4	1/2	21.63	32.189	1.33	0.406
4x4	3/8	17.27	25.701	1.33	0.406
4x4	5/16	14.83	22.069	1.33	0.406
4x4	1/4	12.21	18.170	1.33	0.406
4x4	3/16	9.42	14.019	1.33	0.406
3.5x3.5	5/16	12.70	18.900	1.17	0.356
3.5x3.5	1/4	10.51	15.641	1.17	0.356
3.5x3.5	3/16	8.15	12.129	1.17	0.356
3x3	5/16	10.58	15.745	1.00	0.305
3x3	1/4	8.81	13.111	1.00	0.305
3x3	3/16	6.87	10.224	1.00	0.305
2.5x2.5	5/16	8.45	12.575	0.83	0.254
2.5x2.5	1/4	7.11	10.581	0.83	0.254
2.5x2.5	3/16	5.59	8.319	0.83	0.254
2x2	5/16	6.32	9.405	0.67	0.203
2x2	1/4	5.41	8.051	0.67	0.203
2x2	3/16	4.32	6.429	0.67	0.203

Rectangular

Nominal Size	Wall Thk	Weight Lbs/Ft	Weight Kg./M	Surf. Area Sq.Ft-per Ft	Surf. Area Sq.M-per M
12x4	5/8	59.32	88.278	2.67	0.813
12x4	9/16	54.17	80.614	2.67	0.813
12x4	1/2	48.85	72.697	2.67	0.813
12x4	3/8	37.69	56.089	2.67	0.813
12x4	5/16	31.84	47.383	2.67	0.813
12x4	1/4	25.82	38.424	2.67	0.813
12x4	3/16	19.63	29.213	2.67	0.813
12x2	1/4	22.42	33.365	2.33	0.711
12x2	3/16	17.08	25.418	2.33	0.711
10x8	5/8	67.82	100.927	3.00	0.914
10x8	9/16	61.83	92.013	3.00	0.914
10x8	1/2	55.66	82.831	3.00	0.914
10x8	3/8	42.79	63.679	3.00	0.914
10x8	5/16	36.10	53.723	3.00	0.914
10x8	1/4	29.23	43.499	3.00	0.914
10x8	3/16	22.18	33.007	3.00	0.914
10x6	5/8	59.32	88.278	2.67	0.813
10x6	9/16	54.17	80.614	2.67	0.813
10x6	1/2	48.85	72.697	2.67	0.813
10x6	3/8	37.69	56.089	2.67	0.813
10x6	5/16	31.84	47.383	2.67	0.813
10x6	1/4	25.82	38.424	2.67	0.813
10x6	3/16	19.63	29.213	2.67	0.813
10x5	5/8	55.06	81.938	2.50	0.762
10x5	9/16	50.34	74.914	2.50	0.762
10x5	1/2	45.45	67.637	2.50	0.762
10x5	3/8	35.13	52.279	2.50	0.762
10x5	5/16	29.72	44.228	2.50	0.762
10x5	1/4	24.12	35.895	2.50	0.762
10x5	3/16	18.35	27.308	2.50	0.762
10x4	9/16	46.51	69.215	2.33	0.711
10x4	1/2	42.05	62.577	2.33	0.711
10x4	3/8	32.58	48.484	2.33	0.711
10x4	5/16	27.59	41.058	2.33	0.711
10x4	1/4	22.42	33.365	2.33	0.711
10x4	3/16	17.08	25.418	2.33	0.711
10x2	3/8	27.48	40.895	2.00	0.610
10x2	5/16	23.34	34.734	2.00	0.610
10x2	1/4	19.02	28.305	2.00	0.610
10x2	3/16	14.53	21.623	2.00	0.610
9x7	5/8	59.32	88.278	2.67	0.813
9x7	9/16	54.17	80.614	2.67	0.813
9x7	1/2	48.85	72.697	2.67	0.813
9x7	3/8	37.69	56.089	2.67	0.813
9x7	5/16	31.84	47.383	2.67	0.813
9x7	1/4	25.82	38.424	2.67	0.813
9x7	3/16	19.63	29.213	2.67	0.813
9x6	5/8	55.06	81.938	2.50	0.762
9x6	9/16	50.34	74.914	2.50	0.762
9x6	1/2	45.45	67.637	2.50	0.762

12/29/2008



Structural Tubing

Radius Equals two times wall thickness

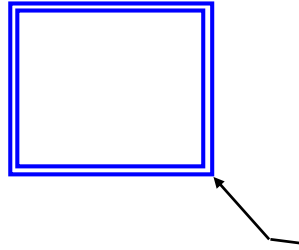
Square

Nominal Size	Wall Thk	Weight Lbs/Ft	Weight Kg./M	Surf. Area Sq.Ft-per Ft	Surf. Area Sq.M-per M
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Rectangular

Nominal Size	Wall Thk	Weight Lbs/Ft	Weight Kg./M	Surf. Area Sq.Ft-per Ft	Surf. Area Sq.M-per M
9x6	3/8	35.13	52.279	2.50	0.762
9x6	5/16	29.72	44.228	2.50	0.762
9x6	1/4	24.12	35.895	2.50	0.762
9x6	3/16	18.35	27.308	2.50	0.762
9x5	9/16	46.51	69.215	2.33	0.711
9x5	1/2	42.05	62.577	2.33	0.711
9x5	3/8	32.58	48.484	2.33	0.711
9x5	5/16	27.59	41.058	2.33	0.711
9x5	1/4	22.42	33.365	2.33	0.711
9x5	3/16	17.08	25.418	2.33	0.711
9x3	1/2	35.24	52.443	2.00	0.610
9x3	3/8	27.48	40.895	2.00	0.610
9x3	5/16	23.34	34.734	2.00	0.610
9x3	1/4	19.02	28.305	2.00	0.610
9x3	3/16	14.53	21.623	2.00	0.610
8x6	9/16	46.51	69.215	2.33	0.711
8x6	1/2	42.05	62.577	2.33	0.711
8x6	3/8	32.58	48.484	2.33	0.711
8x6	5/16	27.59	41.058	2.33	0.711
8x6	1/4	22.42	33.365	2.33	0.711
8x6	3/16	17.08	25.418	2.33	0.711
8x4	9/16	38.86	57.830	2.00	0.610
8x4	1/2	35.24	52.443	2.00	0.610
8x4	3/8	27.48	40.895	2.00	0.610
8x4	5/16	23.34	34.734	2.00	0.610
8x4	1/4	19.02	28.305	2.00	0.610
8x4	3/16	14.53	21.623	2.00	0.610
8x3	1/2	31.84	47.383	1.83	0.559
8x3	3/8	24.93	37.100	1.83	0.559
8x3	5/16	21.21	31.564	1.83	0.559
8x3	1/4	17.32	25.775	1.83	0.559
8x3	3/16	13.25	19.718	1.83	0.559
8x2	3/8	22.37	33.290	1.67	0.508
8x2	5/16	19.08	28.394	1.67	0.508
8x2	1/4	15.62	23.245	1.67	0.508
8x2	3/16	11.97	17.813	1.67	0.508
7x5	1/2	35.24	52.443	2.00	0.610
7x5	3/8	27.48	40.895	2.00	0.610
7x5	5/16	23.34	34.734	2.00	0.610
7x5	1/4	19.02	28.305	2.00	0.610
7x5	3/16	14.53	21.623	2.00	0.610
7x4	1/2	31.84	47.383	1.83	0.559
7x4	3/8	24.93	37.100	1.83	0.559
7x4	5/16	21.21	31.564	1.83	0.559
7x4	1/4	17.32	25.775	1.83	0.559
7x4	3/16	13.25	19.718	1.83	0.559
7x3	1/2	28.43	42.309	1.67	0.508
7x3	3/8	22.37	33.290	1.67	0.508
7x3	5/16	19.08	28.394	1.67	0.508
7x3	1/4	15.62	23.245	1.67	0.508

12/29/2008



Structural Tubing

Radius Equals two times wall thickness

Square

Nominal Size	Wall Thk	Weight Lbs/Ft	Weight Kg./M	Surf. Area Sq.Ft-per Ft	Surf. Area Sq.M-per M
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Rectangular

Nominal Size	Wall Thk	Weight Lbs/Ft	Weight Kg./M	Surf. Area Sq.Ft-per Ft	Surf. Area Sq.M-per M
7x3	3/16	11.97	17.813	1.67	0.508
7x2	1/4	13.91	20.700	1.50	0.457
7x2	3/16	10.70	15.923	1.50	0.457
6x5	1/2	31.84	47.383	1.83	0.559
6x5	3/8	24.93	37.100	1.83	0.559
6x5	5/16	21.21	31.564	1.83	0.559
6x5	1/4	17.32	25.775	1.83	0.559
6x5	3/16	13.25	19.718	1.83	0.559
6x4	1/2	28.43	42.309	1.67	0.508
6x4	3/8	22.37	33.290	1.67	0.508
6x4	5/16	19.08	28.394	1.67	0.508
6x4	1/4	15.62	23.245	1.67	0.508
6x4	3/16	11.97	17.813	1.67	0.508
6x3	3/8	19.82	29.495	1.50	0.457
6x3	5/16	16.96	25.239	1.50	0.457
6x3	1/4	13.91	20.700	1.50	0.457
6x3	3/16	10.70	15.923	1.50	0.457
6x2	3/8	17.27	25.701	1.33	0.406
6x2	5/16	14.83	22.069	1.33	0.406
6x2	1/4	12.21	18.170	1.33	0.406
6x2	3/16	9.42	14.019	1.33	0.406
5x4	3/8	19.82	29.495	1.50	0.457
5x4	5/16	16.96	25.239	1.50	0.457
5x4	1/4	13.91	20.700	1.50	0.457
5x4	3/16	10.70	15.923	1.50	0.457
5x3	1/2	21.63	32.189	1.33	0.406
5x3	3/8	17.27	25.701	1.33	0.406
5x3	5/16	14.83	22.069	1.33	0.406
5x3	1/4	12.21	18.170	1.33	0.406
5x3	3/16	9.42	14.019	1.33	0.406
5x2	5/16	12.70	18.900	1.17	0.356
5x2	1/4	10.51	15.641	1.17	0.356
5x2	3/16	8.15	12.129	1.17	0.356
4x3	5/16	12.70	18.900	1.17	0.356
4x3	1/4	10.51	15.641	1.17	0.356
4x3	3/16	8.15	12.129	1.17	0.356
4x2	5/16	10.58	15.745	1.00	0.305
4x2	1/4	8.81	13.111	1.00	0.305
4x2	3/16	6.87	10.224	1.00	0.305
3.5x2.5	1/4	8.81	13.111	1.00	0.305
3.5x2.5	3/16	6.87	10.224	1.00	0.305
3x2	1/4	7.11	10.581	0.83	0.254
3x2	3/16	5.59	8.319	0.83	0.254

Welded Steel Grating

Bearing Bars Spaced 1 3/16" on centers - Cross bars 4" on center

Note: Spans and loads in the shaded area exceeds a deflection of 1/4" for uniform loads of 100#/sq. ft. which provide safe pedestrian comfort, but can be exceeded for other types of loads at the discretion of the engineer.

U- Safe Uniform Load in lbs per sq. ft.

C- Safe Concentrated Loads in lbs per foot of Grating Width

D- Deflection in inches

Bearing Bar Size	Nominal Wt. lbs per sq.ft. Kg per sq. M	U D	←————— Span —————→												
			2'-0"	2'-6"	3'-0"	3'-6"	4'-0"	4'-6"	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	8'-0"	9'-0"
3/4" x 1/8"	4.0 19.530	U	386	247	172	126	96	76							
		D	0.095	0.151	0.216	0.295	0.374	0.486							
		C	386	308	258	220	194	171							
		D	0.076	0.119	0.173	0.234	0.308	0.389							
3/4"x3/16"	5.6 27.342	U	578	370	258	188	144	115							
		D	0.095	0.151	0.216	0.295	0.374	0.486							
		C	578	462	386	331	289	257							
		D	0.076	0.119	0.173	0.234	0.38	0.389							
1"x1/8"	5.1 24.900	U	686	439	304	224	171	135	109	91	76				
		D	0.072	0.111	0.159	0.219	0.288	0.366	0.451	0.547	0.673				
		C	686	549	457	392	343	305	275	250	228				
		D	0.057	0.09	0.129	0.176	0.231	0.293	0.36	0.434	0.518				
1"x3/16"	7.3 35.642	U	1029	659	459	338	257	203	164	135	114				
		D	0.072	0.111	0.159	0.219	0.288	0.366	0.451	0.547	0.673				
		C	1029	824	686	587	514	458	412	375	343				
		D	0.057	0.09	0.129	0.176	0.231	0.293	0.36	0.434	0.518				
1-1/4"x1/8"	6.1 29.783	U	1072	686	476	350	268	212	172	142	119	101	87		
		D	0.057	0.09	0.129	0.176	0.231	0.291	0.358	0.433	0.520	0.608	0.704		
		C	1072	858	716	613	536	477	430	390	358	330	306		
		D	0.046	0.072	0.104	0.141	0.183	0.233	0.288	0.349	0.416	0.487	0.565		
1-1/4"x3/16"	8.9 43.454	U	1608	1028	716	526	403	318	258	213	179	152	131		
		D	0.057	0.09	0.129	0.176	0.231	0.291	0.358	0.433	0.520	0.608	0.704		
		C	1608	1285	1073	918	803	716	644	585	536	495	459		
		D	0.046	0.072	0.104	0.141	0.183	0.233	0.288	0.349	0.416	0.487	0.565		
1-1/2"x1/8"	7.3 35.642	U	1544	987	686	505	387	306	248	205	172	149	128	96	75
		D	0.047	0.075	0.106	0.147	0.192	0.243	0.300	0.365	0.433	0.506	0.587	0.774	0.978
		C	1544	1235	1029	883	772	687	619	563	515	475	441	386	342
		D	0.038	0.059	0.087	0.117	0.154	0.195	0.241	0.289	0.347	0.406	0.470	0.614	0.777
1-1/2"x3/16"	10.6 51.754	U	2321	1485	1031	758	581	458	371	307	260	222	191	145	115
		D	0.047	0.075	0.106	0.147	0.192	0.243	0.300	0.365	0.433	0.506	0.587	0.774	0.978
		C	2321	1856	1547	1325	1159	1031	928	844	773	714	663	581	515
		D	0.038	0.059	0.087	0.117	0.154	0.195	0.241	0.289	0.347	0.406	0.470	0.614	0.777
1-3/4"x3/16"	12.2 59.566	U	3151	2016	1401	1029	788	622	505	416	351	299	259	197	155
		D	0.042	0.064	0.092	0.126	0.165	0.208	0.258	0.310	0.371	0.435	0.506	0.664	0.838
		C	3151	2521	2100	1800	1575	1400	1260	1145	1049	969	899	786	700
		D	0.033	0.052	0.074	0.101	0.132	0.167	0.206	0.249	0.297	0.347	0.403	0.527	0.667
2"x3/16"	13.9 67.866	U	4116	2633	1829	1344	1029	813	659	546	460	393	339	258	204
		D	0.036	0.056	0.081	0.111	0.144	0.183	0.226	0.273	0.325	0.384	0.447	0.580	0.732
		C	4116	3292	2745	2351	2058	1828	1646	1496	1370	1266	1175	1027	914
		D	0.029	0.045	0.064	0.088	0.115	0.145	0.180	0.217	0.259	0.303	0.353	0.460	0.583
2-1/4"x3/16"	15.4 75.189	U	5209	3332	2314	1670	1302	1028	835	689	583	496	428	327	259
		D	0.032	0.05	0.072	0.098	0.127	0.162	0.199	0.241	0.287	0.338	0.393	0.512	0.646
		C	5209	4167	3473	2916	2604	2314	2082	1892	1733	1601	1487	1301	1157
		D	0.026	0.039	0.057	0.079	0.102	0.129	0.160	0.194	0.230	0.270	0.314	0.410	0.518
2-1/2"x3/16'	17.1 83.490	U	6432	4115	2858	2099	1609	1271	1029	850	720	613	529	405	320
		D	0.028	0.044	0.064	0.088	0.116	0.145	0.180	0.217	0.260	0.305	0.354	0.465	0.586
		C	6432	5147	4286	3673	3214	2858	2571	2338	2141	1977	1836	1607	1429
		D	0.023	0.036	0.051	0.071	0.092	0.116	0.144	0.173	0.207	0.242	0.282	0.369	0.467

ASTM A325 & A490 BOLTS & NUTS

Bolt Length w/ Single Nut & Single Washer Calculator Enter Grip **Inches**

Bolt Dia.	Grip	1-Lg. Inches	2-Lg. Inches
1/2	1 1/4	2.09	2.25
5/8			
3/4			
7/8			
1			
1 1/8			
1 1/4			
1 3/8			
1 1/2			

1-With F.W.

2-With Bevel Washer

Bolt Length

In order to determine the required bolt length, the value in the table below should be added to the **Grip** (i.e., the total thickness of all connected material, exclusive of washers) or enter the Grip in the Calculator above for single nut with washer. The values are generalized with due allowance for manufacturing tolerances to provide for full thread engagement of heavy hex nut, when installed. For each hardened flat washer that is used add **5/32-inch**, and for each bevel washer add **5/16-inch**. The length is determined by use of the table/calculator should be **adjusted to the next 1/4-inch**.

★ Specify thread length when using Double Nuts

Nominal Bolt Size Inches (mm)	Single Nut To determine required Bolt Length Add to Grip		Double Nut To determine required Bolt Length Add to Grip		Double Nut Thread Length	
	Inches	millimeters	Inches	millimeters	Inches	millimeters
1/2" (13)	11/16"	17	1-3/16"	30	1-1/2"	38
5/8" (16)	7/8"	22	1-1/2"	38	1-7/8"	48
3/4" (19)	1"	25	1-3/4"	44	2-1/8"	54
7/8" (22)	1-1/8"	29	2"	51	2-3/8"	60
1" (25)	1-1/4"	32	2-1/4"	57	2-3/4"	70
1-1/8" (29)	1-1/2"	38	2-5/8"	67	3-1/8"	79
1-1/4" (32)	1-5/8"	41	2-7/8"	73	3-1/4"	83
1-3/8" (35)	1-3/4"	44	3-1/8"	79	3-5/8"	92
1-1/2" (38)	1-7/8"	48	3-3/8"	86	3-3/4"	95

Adjust Bolt Length to the Next 1/4" or 6mm

Bolt Length w/ Single Nut & Single Washer Calculator Enter Grip **millimeters**

Bolt Dia.	Grip	1-Lg. mm	2-Lg. mm
13	32	53.43	57.40
16			
19			
22			
25			
30			
32			
35			
38			

1-With F.W.

2-With Bevel Washer

Adjust Bolt Length to the Next 6mm

Bolted Parts (ASTM A325 & A490 BOLTS)

The slope of surfaces of bolted parts in contact with the bolt head and nut shall not exceed 1:20 with respect to plane normal to the bolt axis. Bolted steel parts shall not be separated by gaskets and shall fit solidly together after bolts are tightened. Holes may be punched, sub-punched and reamed, or drilled, as required by applicable code or specification. **Standard holes shall have a diameter 1/16" in excess of the nominal bolt diameter.**

★ **A490 bolts and galvanized A325 bolts shall not be reused. Retightening previous tightening of adjacent bolts shall not be considered as reuse.**

Oversized Holes may have nominal diameters up to: 3/16" larger than bolts 7/8" and less in diameter, 1/4" for 1" in diameter and 5/16" larger than bolts 1-1/8" and greater in diameter. Hardened washers shall be installed over oversized holes in an outer ply.

Oversized and Slotted Holes

Nominal Bolt Size	Max. Holes Size (Nominal) Inches		
	Oversize Holes	Short Slotted Holes	Long Slotted Holes
1/2"	5/8"	9/16"x11/16"	9/16"x1-1/4"
5/8"	13/16"	11/16"x7/8"	11/16"x1-9/16"
3/4"	15/16"	13/16"x1"	13/16"x1-7/8"
7/8"	1-1/16"	15/16"x1-1/8"	15/16"x2-3/16"
1"	1-1/4"	1-1/16"x1-5/16"	1-1/16"x2-1/2"
1-1/8"	1-7/16"	1-3/16"x1-1/2"	1-3/16"x2-13/16"
1-1/4"	1-9/16"	1-5/16"x1-5/8"	1-5/16"x3-1/8"
1-3/8"	1-11/16"	1-7/16"x1-3/4"	1-7/16"x3-7/16"
1-1/2"	1-13/16"	1-9/16"x1-7/8"	1-9/16"x3-3/4"
>1-1/2"	d+5/16"	(d+1/16")x(d+3/8")	(d+1/16")x(2.5xd)

Nominal Bolt Size	Max. Holes Size (Nominal) millimeters		
	Oversize Holes	Short Slotted Holes	Long Slotted Holes
13	16	14x17	14x32
16	21	17x22	17x40
19	24	21x25	21x48
22	27	24x29	24x56
25	32	27x33	27x64
29	37	30x38	30x71
32	40	33x41	33x79
35	43	37x44	37x87
38	46	40x58	40x95
>38	>46	(d+2)x(d+9)	(d+2)x(2.5xd)

Minimum Bolt Spacing

The distance between centers of standard, oversized, or slotted fasteners shall be not less than $2 \frac{2}{3}d$ where "d" is the nominal diameter of the fastener, inches. **A distance of 3d is preferred.**

Minimum Edge Distance Table Inches

(Center of Standard Hole to Edge of Connected Part)

Nominal Bolt Size	@ Sheared Edges	'@ Rolled Edges of Plates, Shapes or Bars or Gas Cut Edges
1/2"	7/8"	3/4"
5/8"	1-1/8"	7/8"
3/4"	1-1/4"	1"
7/8"	1-1/2" ★	1-1/8"
1"	1-3/4" ★	1-1/4"
1-1/8"	2"	1-1/2"
1-1/4"	2-1/4"	1-5/8"
>1-1/4"	1.75x Dia.	1.25x Dia.

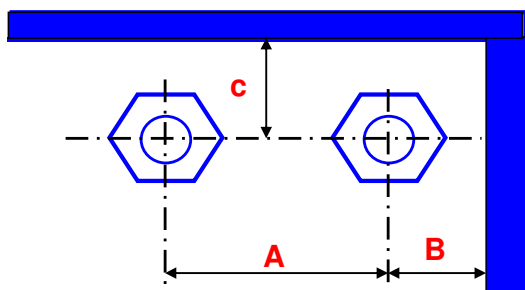
★ These may be 1-1/4" @ the ends of beam connection angles

Minimum Edge Distance Table millimeters

(Center of Standard Hole to Edge of Connected Part)

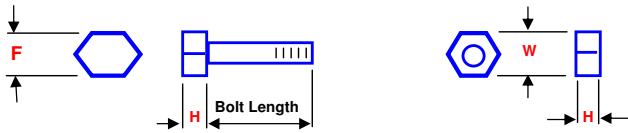
Nominal Bolt Size	@ Sheared Edges	'@ Rolled Edges of Plates, Shapes or Bars or Gas Cut Edges
13	22	19
16	29	22
19	32	25
22	38 ★	29
25	44 ★	32
29	51	38
32	57	41
>32	1.75x Dia.	1.25x Dia.

★ These may be 32 @ the ends of beam connection angles



Wrench Clearance Guide

Bolt Size Inches (mm)	Socket Wrench Inches (mm)			Open End Wrench Inches (mm)		
	A	B	C	A	B	C
1/4" (6)	3/4" (19)	1/2" (13)	1/2" (13)	1" (25)	7/8" (22)	5/8" (16)
3/8" (10)	1-1/8" (29)	3/4" (19)	3/4" (19)	1-3/8" (35)	1-1/8" (29)	3/4" (19)
1/2" (13)	1-3/8" (35)	7/8" (22)	7/8" (22)	1-3/4" (44)	1-3/8" (35)	7/8" (22)
5/8" (16)	1-5/8" (41)	1" (25)	1" (25)	2-1/8" (54)	1-1/2" (38)	1-1/8" (29)
3/4" (19)	1-7/8" (48)	1-1/8" (29)	1-1/8" (29)	2-5/8" (67)	1-5/8" (41)	1-1/4" (32)
7/8" (22)	2-1/8" (54)	1-1/4" (32)	1-1/4" (32)	3" (76)	1-7/8" (48)	1-5/8" (41)
1" (25)	2-3/8" (60)	1-3/8" (35)	1-3/8" (35)	3-1/4" (83)	2-1/8" (54)	1-7/8" (48)
1-1/8" (29)	2-5/8" (67)	1-1/2" (38)	1-1/2" (38)	3-3/8" (86)	2-1/4" (57)	1-7/8" (48)
1-1/4" (32)	2-7/8" (73)	1-5/8" (41)	1-5/8" (41)	3-7/8" (98)	2-1/2" (64)	2" (51)
1-3/8" (35)	3-1/8" (79)	1-3/4" (44)	1-3/4" (44)	4-3/8" (111)	2-3/4" (70)	2-1/4" (57)
1-1/2" (38)	3-3/8" (86)	1-7/8" (48)	1-7/8" (48)	4-7/8" (124)	2-7/8" (73)	2-1/2" (64)



ASTM A325 & A490 BOLTS & NUTS DIMENSIONS, Inches (mm)

Nominal Bolt Size D	Bolt Dimensions, Inches Heavy Hex Structural Bolts			Nut Dimensions, Inches Heavy Hex Nuts	
	Width across flats F	Height H	Thread Length	Width across flats W	Height H
1/2" (13)	7/8" (22)	5/16" (8)	1" (25)	7/8" (22)	31/64" (12)
5/8" (16)	1-1/16" (27)	25/64" (10)	1-1/4" (32)	1-1/16" (27)	39/64" (15)
3/4" (19)	1-1/4" (32)	15/32" (12)	1-3/8" (35)	1-1/4" (32)	47/64" (19)
7/8" (22)	1-7/16" (37)	35/64" (14)	1-1/2" (38)	1-7/16" (37)	55/64" (22)
1" (25)	1-5/8" (41)	39/64" (15)	1-3/4" (44)	1-5/8" (41)	63/64" (25)
1-1/8" (29)	1-13/16" (46)	11/16" (17)	2" (51)	1-13/16" (46)	1-7/64" (37)
1-1/4" (32)	2" (51)	25/32" (20)	2" (51)	2" (51)	1-7/32" (31)
1-3/8" (35)	2-3/16" (56)	27/32" (21)	2-1/4" (57)	2-3/16" (56)	1-11/32" (34)
1-1/2" (38)	2-3/8" (60)	15/16" (24)	2-1/4" (57)	2-3/8" (60)	1-15/32" (37)

WASHER DIMENSIONS, Inches (mm)

Nominal Bolt Size D	Circular Washers				Sq. or Rect. Bevel Washers		
	Nominal Outside Diameter	Nominal Diameter of Hole	Thickness		Minimum Side Dimension	Mean Thickness	Slope or Taper in Thickness
			Min.	Max.			
1/2" (13)	1-1/16" (27)	17/32" (13)	0.097 (2)	0.177 (4)	1-3/4" (44)	5/16" (8)	1:6'
5/8" (16)	1-5/16" (33)	11/16" (17)	0.122 (3)	0.177 (4)	1-3/4" (44)	5/16" (8)	1:6'
3/4" (19)	1-15/32" (37)	13/16" (21)	0.122 (3)	0.177 (4)	1-3/4" (44)	5/16" (8)	1:6'
7/8" (22)	1-3/4" (44)	15/16" (24)	0.136 (3)	0.177 (4)	1-3/4" (44)	5/16" (8)	1:6'
1" (25)	2" (51)	1-1/8" (29)	0.136 (3)	0.177 (4)	1-3/4" (44)	5/16" (8)	1:6'
1-1/8" (29)	2-1/4" (57)	1-1/4" (32)	0.136 (3)	0.177 (4)	2-1/4" (57)	5/16" (8)	1:6'
1-1/4" (32)	2-1/2" (64)	1-3/8" (35)	0.136 (3)	0.177 (4)	2-1/4" (57)	5/16" (8)	1:6'
1-3/8" (35)	2-3/4" (70)	1-1/2" (38)	0.136 (3)	0.177 (4)	2-1/4" (57)	5/16" (8)	1:6'
1-1/2" (38)	3" (76)	1-5/8" (41)	0.136 (3)	0.177 (4)	2-1/4" (57)	5/16" (8)	1:6'

Length Under Head	Weight of ASTM A325 or A490 high strength bolts									
	Heavy Hex Structural Bolts with Hex Nut in Pounds (Kg) per 100									
Inches (mm)	1/2" (13)	5/8" (16)	3/4" (19)	7/8" (22)	1" (25)	1-1/8" (29)	1-1/4" (32)	1-3/8" (35)	1-1/2" (38)	
1" (25)	16.5 (7.48)	29.4 (13.34)	47 (21.32)							
1-1/4" (32)	17.8 (8.07)	31.1 (14.11)	49.6 (22.5)	74.4 (33.75)	104 (47.17)					
1-1/2" (38)	19.2 (8.71)	33.1 (15.01)	52.2 (23.68)	78 (35.38)	109 (49.44)	148 (67.13)	197 (89.36)			
1-3/4" (44)	20.5 (9.29)	35.3 (16.01)	55.3 (25.08)	81.9 (37.15)	114 (51.71)	154 (69.85)	205 (92.99)	261 (118.39)	333 (151.05)	
2" (51)	21.9 (9.93)	37.4 (16.96)	58.4 (26.5)	86.1 (39.05)	119 (53.98)	160 (72.57)	212 (96.16)	270 (122.47)	344 (156.04)	
2-1/4" (57)	23.3 (10.57)	39.8 (18.05)	61.6 (27.94)	90.3 (40.96)	124 (56.25)	167 (75.75)	220 (99.79)	279 (126.55)	355 (161.03)	
2-1/2" (64)	24.7 (11.2)	41.7 (18.91)	64.7 (29.35)	94.6 (42.91)	130 (58.97)	174 (78.93)	229 (103.87)	290 (131.54)	366 (166.01)	
2-3/4" (70)	26.1 (11.84)	43.9 (19.91)	67.8 (30.75)	98.8 (44.81)	135 (61.23)	181 (82.10)	237 (107.50)	300 (136.08)	379 (171.91)	
3" (76)	27.4 (12.43)	46.1 (20.91)	70.9 (32.16)	103 (46.72)	141 (63.96)	188 (85.28)	246 (111.58)	310 (140.61)	391 (177.35)	
3-1/4" (83)	28.8 (13.06)	48.2 (21.86)	74 (33.57)	107 (48.53)	146 (66.22)	195 (88.45)	255 (115.67)	321 (145.60)	403 (182.80)	
3-1/2" (89)	30.2 (13.70)	50.4 (22.86)	77.1 (34.97)	111 (50.35)	151 (68.49)	202 (91.63)	263 (119.29)	332 (150.59)	416 (188.69)	
3-3/4" (95)	31.6 (14.33)	52.5 (23.81)	80.2 (36.38)	116 (52.62)	157 (71.21)	209 (94.80)	272 (123.38)	342 (155.13)	428 (194.14)	
4" (102)	33 (14.97)	54.7 (24.81)	83.3 (37.78)	120 (54.43)	162 (73.48)	216 (97.98)	280 (127.01)	353 (160.12)	441 (200.03)	
4-1/4" (108)	34.3 (15.56)	56.9 (25.81)	86.4 (39.19)	124 (56.25)	168 (76.2)	223 (101.15)	289 (131.09)	363 (164.65)	453 (205.48)	
4-1/2" (114)	35.7 (16.19)	59 (26.76)	89.5 (40.6)	128 (58.06)	173 (78.47)	230 (104.33)	298 (135.17)	374 (169.64)	465 (210.92)	
4-3/4" (121)	37.1 (16.83)	61.2 (27.76)	92.7 (42.05)	133 (60.33)	179 (81.19)	237 (107.50)	306 (138.80)	384 (174.18)	478 (216.82)	
5" (127)	38.5 (17.46)	63.3 (28.71)	95.8 (43.45)	137 (62.14)	184 (83.46)	244 (110.68)	315 (142.88)	395 (179.17)	490 (222.26)	
5-1/4" (133)	39.9 (18.10)	65.5 (29.71)	98.9 (44.86)	141 (63.96)	190 (86.18)	251 (113.85)	324 (146.96)	405 (183.70)	503 (228.16)	
5-1/2" (140)	41.2 (18.69)	67.7 (30.71)	102 (46.27)	146 (66.22)	196 (88.9)	258 (117.03)	332 (155.9)	416 (188.69)	515 (233.60)	
5-3/4" (146)	42.6 (19.32)	69.8 (31.66)	105 (47.63)	150 (68.04)	201 (91.17)	265 (120.20)	341 (154.68)	426 (193.23)	527 (239.04)	
6" (152)	44 (19.96)	71.9 (32.61)	108 (48.99)	154 (69.85)	207 (93.89)	272 (123.38)	349 (158.30)	437 (198.22)	540 (244.94)	
Per inch additional Add	5.5 (2.49)	8.6 (3.9)	12.4 (5.62)	16.9 (7.67)	22.1 (10.02)	28 (12.70)	34.4 (15.60)	42.5 (19.28)	49.7 (22.54)	
For Each 100 F.W. Add	2.1 (0.95)	3.6 (1.63)	4.8 (2.18)	7 (3.18)	9.4 (4.26)	11.3 (5.13)	13.8 (6.26)	16.8 (7.62)	20 (9.07)	
For Each 100 Bevel Washers Add	23.1 (10.48)	22.4 (10.16)	21 (9.53)	20.2 (9.16)	19.2 (8.71)	34 (15.42)	31.6 (14.33)			

Girder - Formulas

12/29/2008

Fig. Thk "tf"

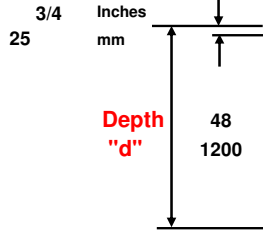


Plate - Formulas

Pl Thk. Inches	Pounds per Sq. Ft	Kilograms per Sq. Meter
3/4	30.625	149.524

KEY IN

Pl Thk. mm	Pounds per Sq. Ft	Kilograms per Sq. Meter
19.05	30.625	149.524

KEY IN

English

Depth Inches d	Web Thk. Inches tw	Fig. Width Inches bf	Fig. Thk. Inches tf	Girder Weight lbs/ft	Weight Weight Kg/M	Sq. Feet Surface Area Per Ft	Sq.Meters Surface Area Per Meter	Girder Length Feet	Fig. Plates Total Sq. Ft	Fig. Plates Total Sq. Meters	Web. Plate Total Sq. Ft	Web. Plate Total Sq. Meters
48	3/8	10	3/4	110.38	164.260	11.27	3.435	10.000	16.667	1.548	38.750	3.600
KEY IN	KEY IN	KEY IN	KEY IN	Lbs.	Kg	KEY IN			Lbs.	Kg	Lbs.	Kg
Total Weight x Length →				1103.78	500.66	Total Weight →			510.42	231.521	593.36	269.143

Metric

Depth mm d	Web Thk. mm tw	Fig. Width mm bf	Fig. Thk. mm tf	Weight lbs/ft	Weight Kg/M	Sq. Feet Surface Area Per Ft	Sq.Meters Surface Area Per Meter	Girder Length Meters	Fig. Plates Total Sq. Feet	Fig. Plates Total Sq. Meters	Web. Plate Total Sq. Feet	Web. Plate Total Sq. Meters	
1200.00	10.00	450.00	25.00	179.33	266.868	13.71	4.180	1.000	9.688	0.900	12.378	1.150	
KEY IN	KEY IN	KEY IN	KEY IN	Lbs.	Kg			KEY IN	Lbs.	Kg	Lbs.	Kg	
Total Weight x Length →				588.34	266.87	Total Weight →				389.34	176.604	199.00	90.264

Structural Plate Weight (Metric)

Qty.	Pl Thk. mm	Width mm	Length mm	Total Sq. Meters	Weight Kg	Weight Lbs
1	25.4	800	800	0.64	127.59	281.30

Key In

Key In

Key In

Key In

Structural Plate Weight (English)

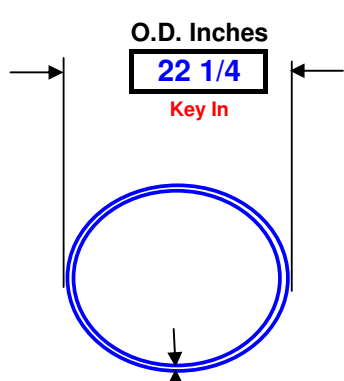
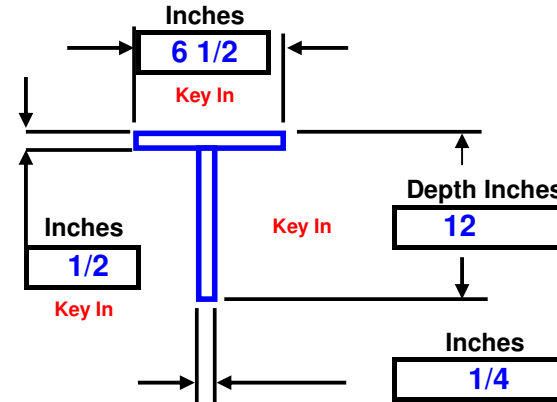
Qty.	Pl Thk. Inches	Width Inches	Length Inches	Total Sq. Ft	Weight Lbs	Weight Kg
1	3/4	18	15	1.88	57.42	26.05

Key In

Key In

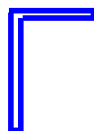
Key In

Key In

 O.D. Inches 22 1/4 Key In Pipe Weight Lbs/Ft. 227.17 Kg/Meter 338.06 Wall Thk. Inches 1 Key In	 Inches 6 1/2 Key In Inches 1/2 Key In Depth Inches 12 Key In Inches 1/4 Key In Tee Weight Lbs/Ft. 20.84 Kg/Meter 31.016
--	--

Steel Angle Properties - CG's & Lbs/Ft

Height = 4.00 inches
 Width = 6.00 inches
 Thickness = 0.7500 inches



Lbs/Ft = 23.61
 Kg/M = 35.136
 Xcg = 2.08 inches
 Ycg = 1.08 inches

Round Bar (dia in millimeters) - Lbs/Ft

Diameter = 12.7000 mm
 Lbs/Ft = 0.668
 Kg/M = 0.994



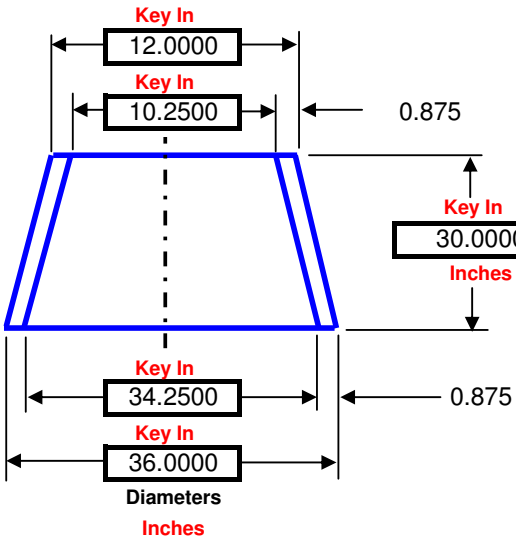
Round Bar (dia in inches) - Lbs/Ft

Diameter = 0.5000 inches
 Lbs/Ft = 0.668
 Kg/M = 0.994



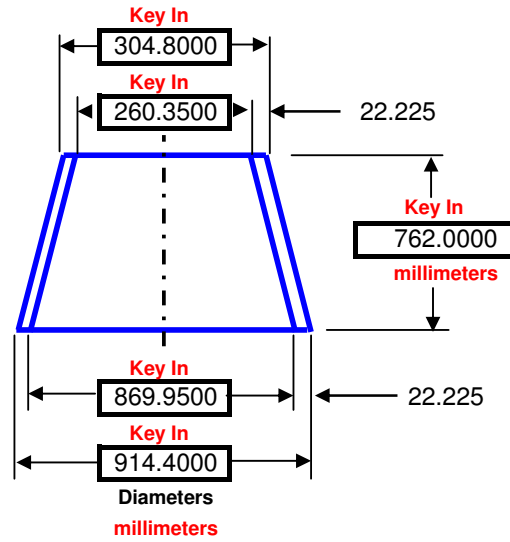
Steel Plate Properties - Lbs/Ft	Conversions - Metric / U.S.
Width = 12.00 inches Key In Thickness = 0.7500 inches Key In Lbs/Ft = 30.63	pounds X 0.45359 = kilograms kilograms X 2.2046 = pounds inches X 25.4 = millimeters millimeters X 0.03937 = inches feet X 0.3048 = meters meters X 3.28084 = feet cubic feet X 0.02832 = cubic meters cubic meters X 35.3107 = cubic feet kilograms/meter X 0.6720 = pounds/foot pounds/foot X 1.4881 = kilograms/meter
Steel Plate Properties - Lbs/Ft	
Width = 305.00 mm Key In Thickness = 3.0000 mm Key In Lbs/Ft = 4.83	
Steel Pipe Properties - Lbs/Ft	Miscellaneous - Metric / U.S.
Outside Dia = 12.75 inches Key In Wall Thck = 0.5000 inches Key In Lbs/Ft = 65.48 Kg/M = 97.445	Steel = 7850 kilograms per cubic meter Steel = 490 pounds per cubic foot
Steel Angle Properties - CG's & Kgs/M	Steel Tee Properties - CG's & Kgs/M
Height = 102 millimeters Key In Thickness = 10 millimeters Key In Width = 102 millimeters Key In Thickness = 10 millimeters Key In Kg/M = 15.23 Xcg = 29 millimeters Ycg = 29 millimeters 	Height = 305 millimeters Key In Thickness = 10 millimeters Key In Flange = 102 millimeters Key In Thickness = 15 millimeters Key In Kg/M = 34.78 Xcg = 51 millimeters Ycg = 107 millimeters 

Steel Cone Weight (English Units - Inches)



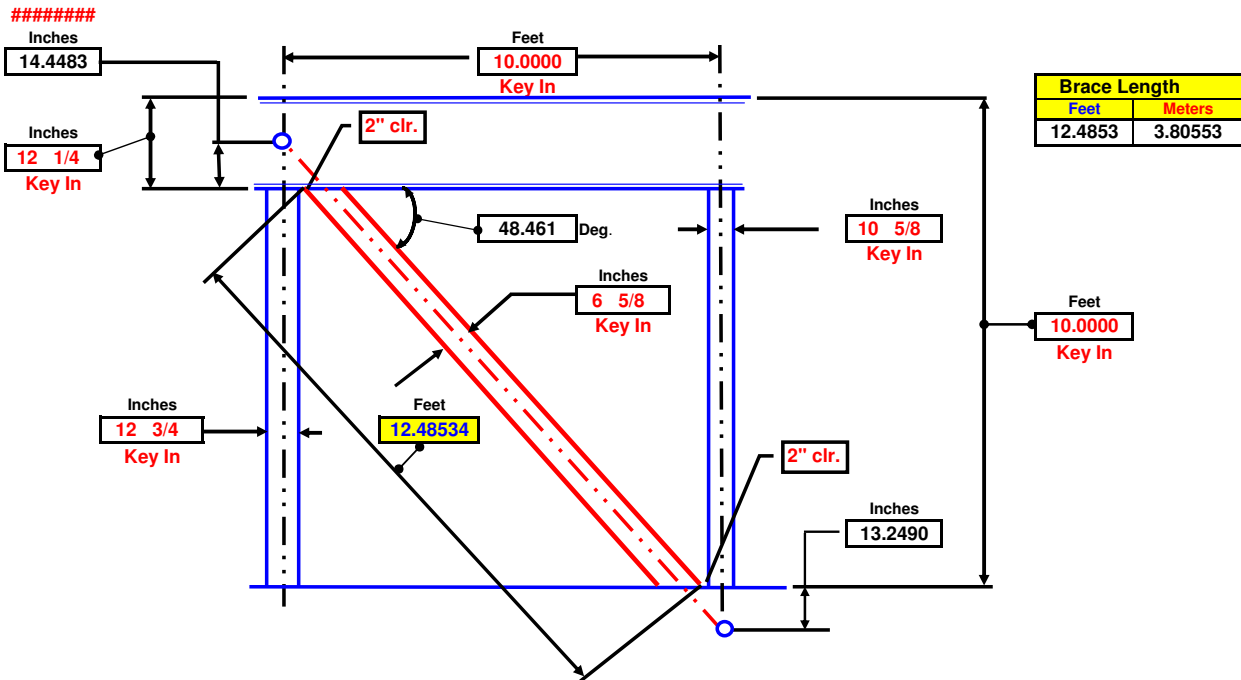
Lbs = 540.77209
Kg = 245.290

Steel Cone Weight (Metric Units - millimeters)

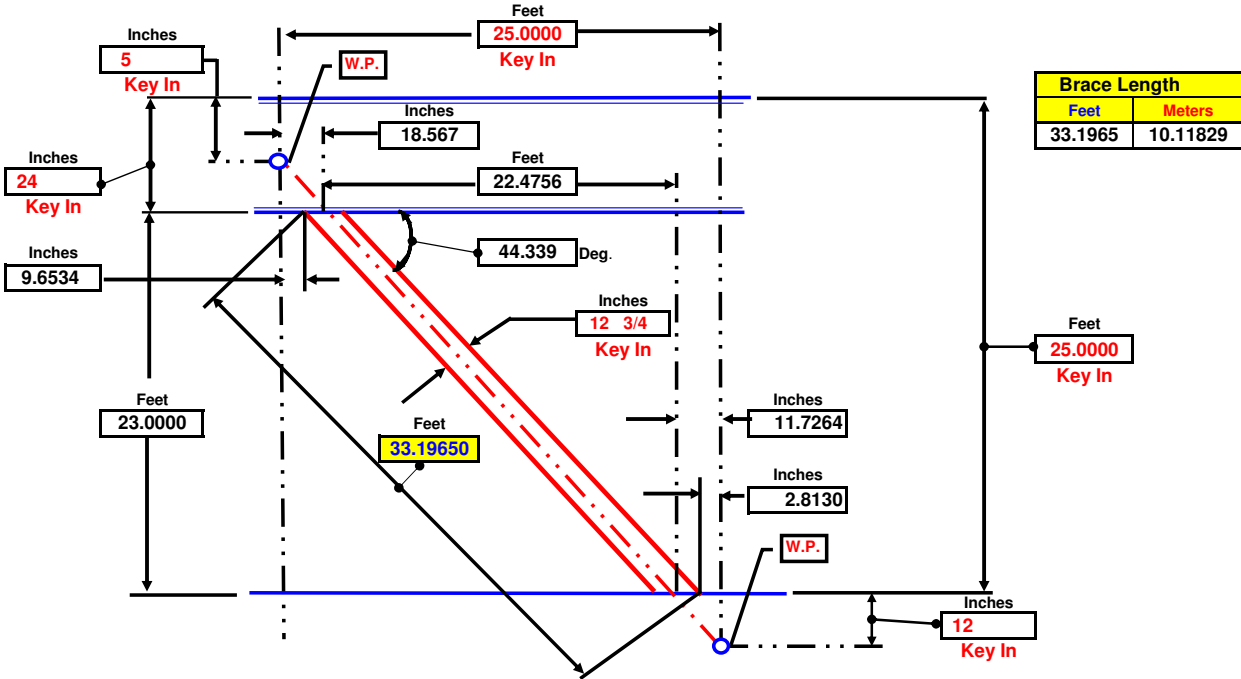


Lbs = 540.7721
Kg = 245.290

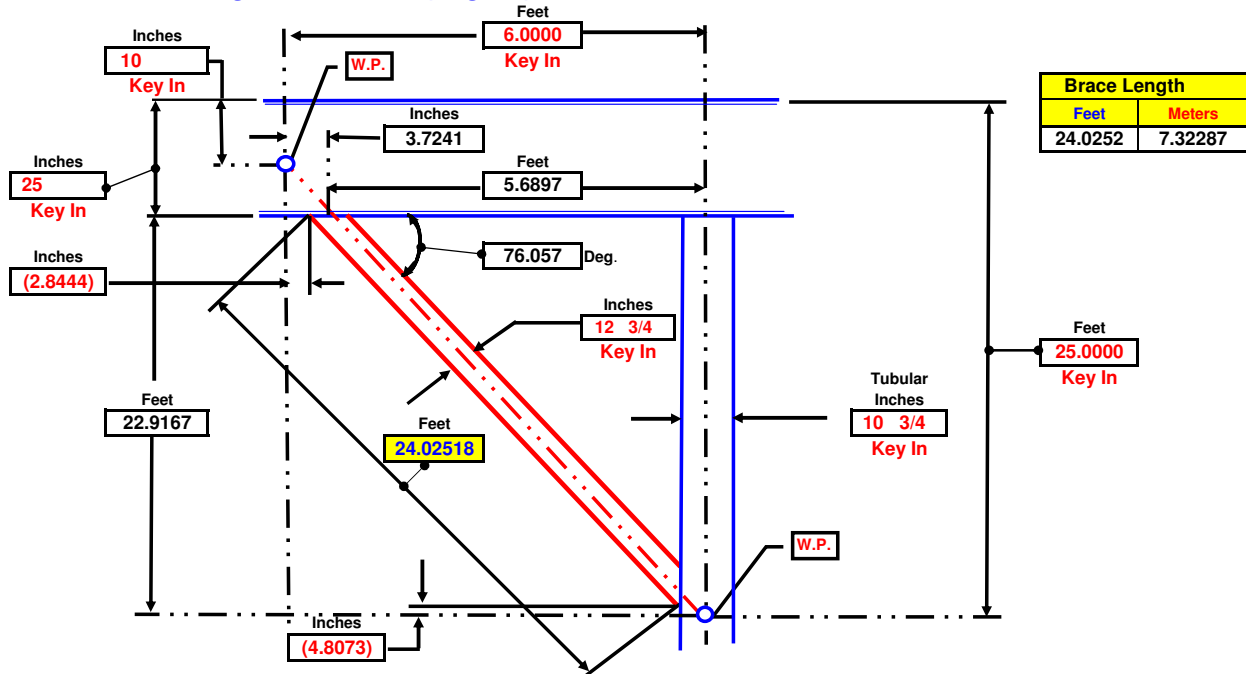
Brace Length w/ 2" Clr. (English)



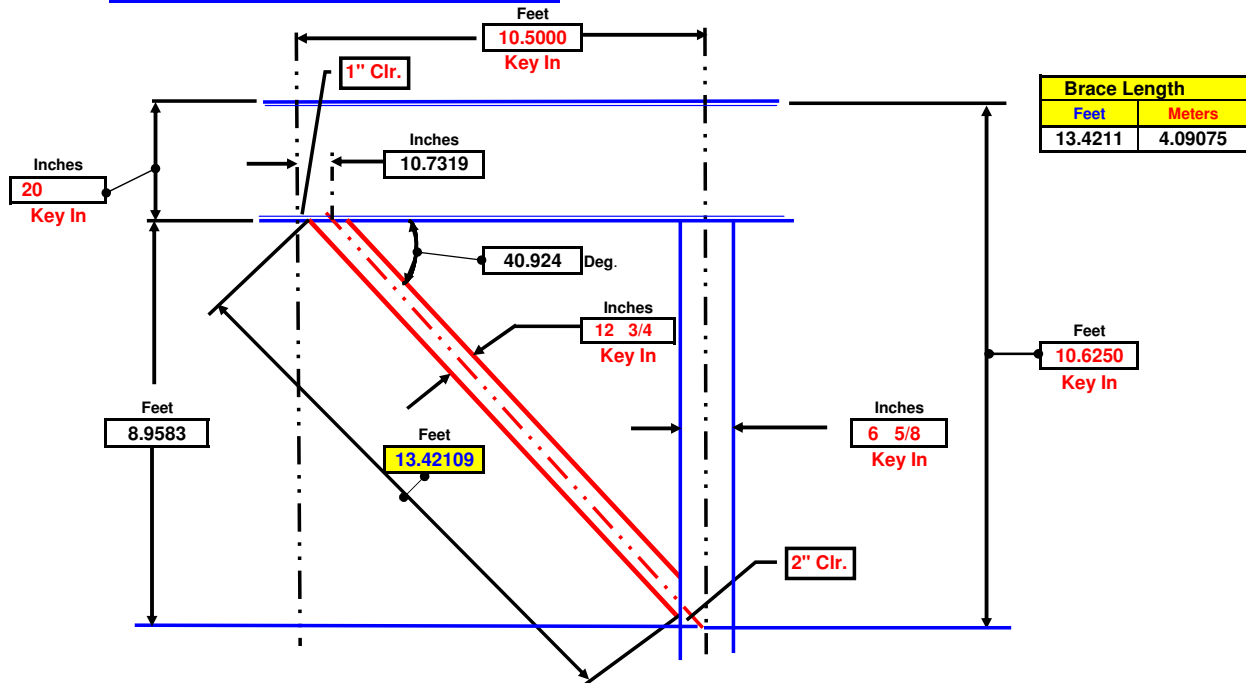
Brace Length W.P. to W.P. (English)



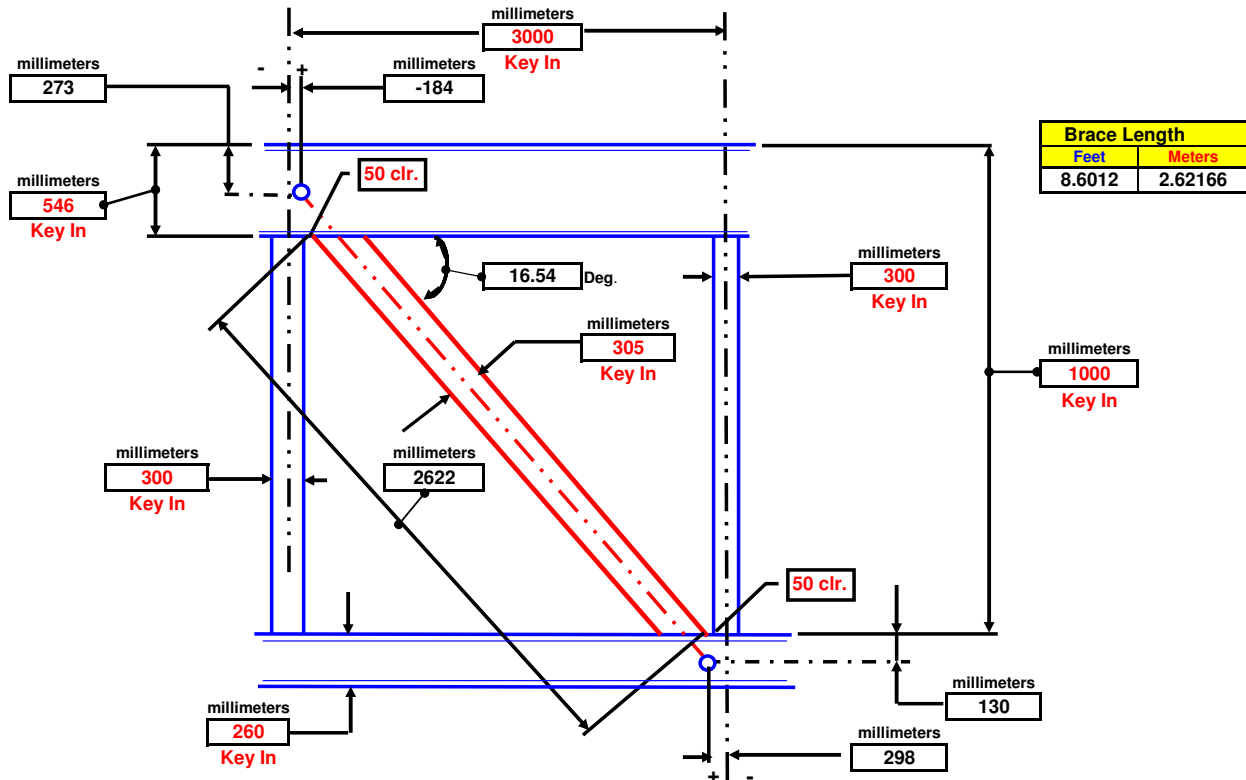
Brace Length W.P. to W.P. (English)



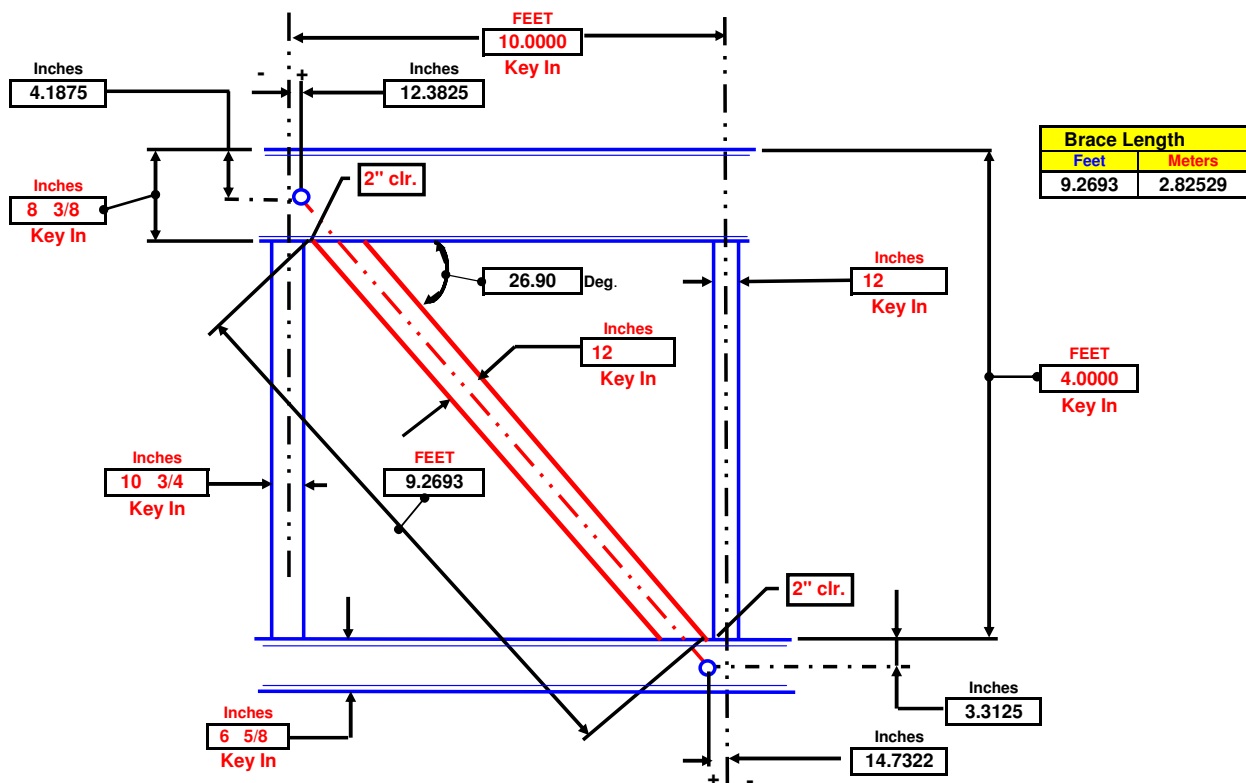
Brace Length 1"Clr. to 2"Clr. (English)



Brace Length w/ 50mm Clr. (Metric)

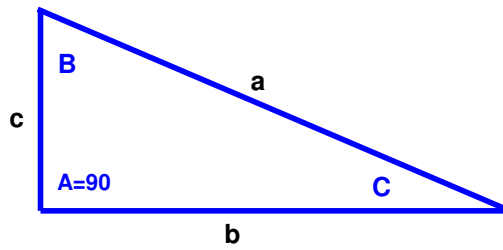


Brace Length w/ 2" Clr. (Metric)



Right Triangle

Angles are in Degrees



Converter

Feet + Inches = Total in Feet

	Feet	Inches	Total Feet	mm
Key In →	8	9 1/4	8.7708	2673.4

Known

Values Key in

Key In →	a	b
	50	10

Formulas →

c	A	B	C
48.9898	90	11.5370	78.4630

Key In →	a	c
	36	15

Formulas →

b	A	B	C
32.7261	90	65.3757	24.6243

Key In →	a	B
	0	0

Formulas →

b	A	c	C
0.0000	90	0.0000	90.0000

Key In →	a	C
	0	0

Formulas →

b	A	c	B
0.0000	90	0.0000	90.0000

Key In →	b	c
	350	700

Formulas →

a	A	B	C
782.6238	90	26.5651	63.4349

Key In →	b	B
	10	63.4349

Formulas →

a	A	c	C
11.1803	90	5.0000	26.5651

Key In →	b	C
	10	26.5651

Formulas →

a	A	c	B
11.1803	90	5.0000	63.4349

Key In →	c	C
	5	26.5651

Formulas →

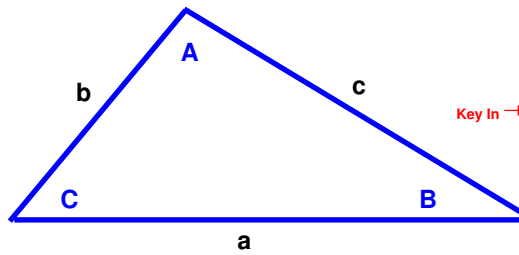
a	A	b	B
11.1803	90	10.0000	63.4349

Key In →	c	B
	5	63.4349

Formulas →

a	A	b	C
11.1803	90	10.0000	26.5651

Oblique Triangles
Angles are in Degrees



Converter

Feet + Inches = Total in Feet

Feet	Inches	Total Feet	mm
Key In → 12	Key In 10 3/4	12.8958	3930.7

Known

Values Key in

a	B	A
Key In → 119.3125	42.6088	97.7487

Formulas →

C	b	c
39.6425	81.517593	76.822288

a	A	C
Key In → 119.3125	97.7487	39.6425

Formulas →

B	b	c
42.6088	81.517593	76.822288

a	A	b
Key In → 119.3125	97.7487	81.5175

Formulas →

B	C	c
42.6087	39.6426	76.822385

a	C	b
Key In → 119.3125	39.6425	81.5175

Formulas →

A	B	c
FALSE	FALSE	#DIV/0!
97.7488	42.608731	76.822301

FALSE

Use this Line

a	b	c
Key In → 119.3125	83.3374	156.8052

Formulas →

A	B	C
48.5596	31.5748	99.8656

b	C	B
Key In → 81.5175	39.6425	42.6088

Formulas →

A	c	a
97.7487	76.822201	119.31236

Total	
Feet	Inches
1	0

1

0

304.80

0.3048

Total Millimeters	
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
11	11
12	12
13	13
14	14
15	15
16	16
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97	97
98	98
99	99
100	100

minimizers

Inches

: **ABB**

[illegible]

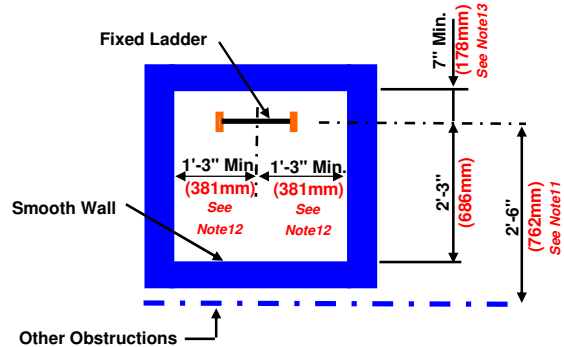
1

304.80

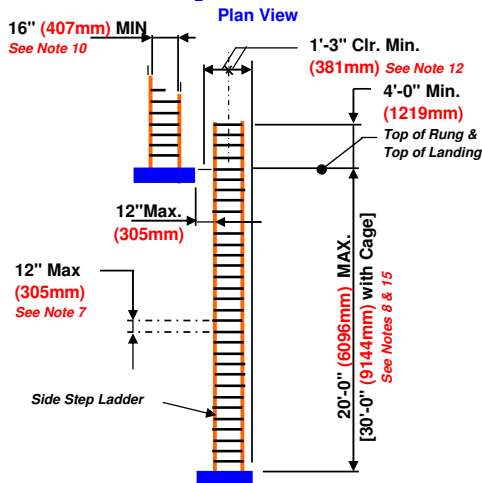
0.3048

[illegible]

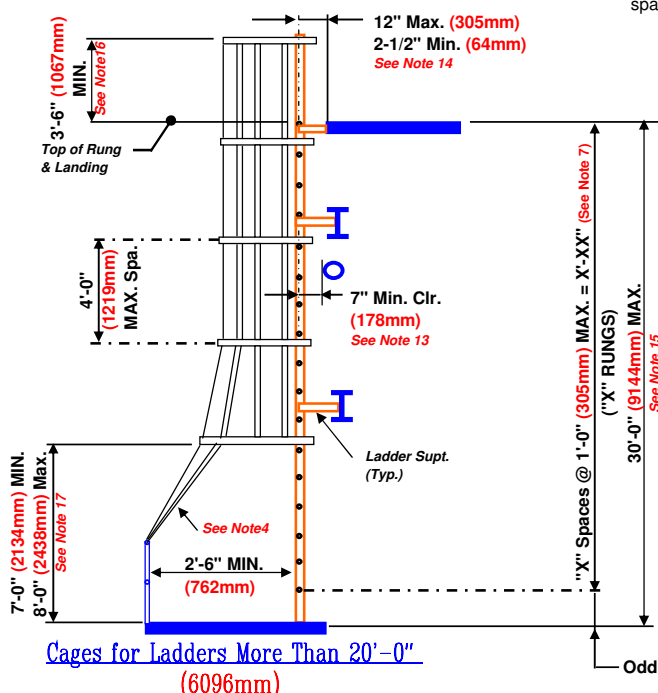
Note: Use Client Standards if project required
See Standards for member sizes & additional details



Clearance Diagram for Fixed Ladder in Well



Offset Fixed Ladder Sections Front Elevation



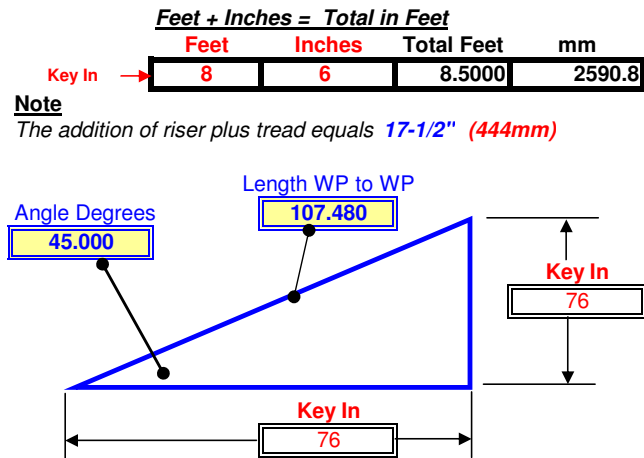
Stair Design Notes

1. Fixed stairs stringer size Should be determined as follows (unless specified by engineer) :

Horizontal Run Work Point to Work Point	Stringer Size
Less Than 10 Feet (3048mm)	C8 or C10
10 to 20 Feet (3048 to 6096mm)	C10X15.3 Min.
20 Feet (6096mm) or More	C12 Min.

2. Sway Bracing should be provided on all stairs with horizontal work point to work point of 15 Feet (4572mm) or more and all stairs with fiberglass treads.
3. Stair angles greater than 45 degrees should always be avoided if possible.
4. Stringer back to back dimension is typically 3'-0" (914mm) unless otherwise specified.
5. Any uniform combination of rise/tread dimensions may be used that will result in a stairway at the angle to horizontal within the permissible range. The table below gives rise/tread dimensions which will produce a stairway within the permissible range, stating the angle to the horizontal produced by each combination. However, the rise/tread combination are not limited to those given in the table below. Odd tread spacing should always occur at the bottom step

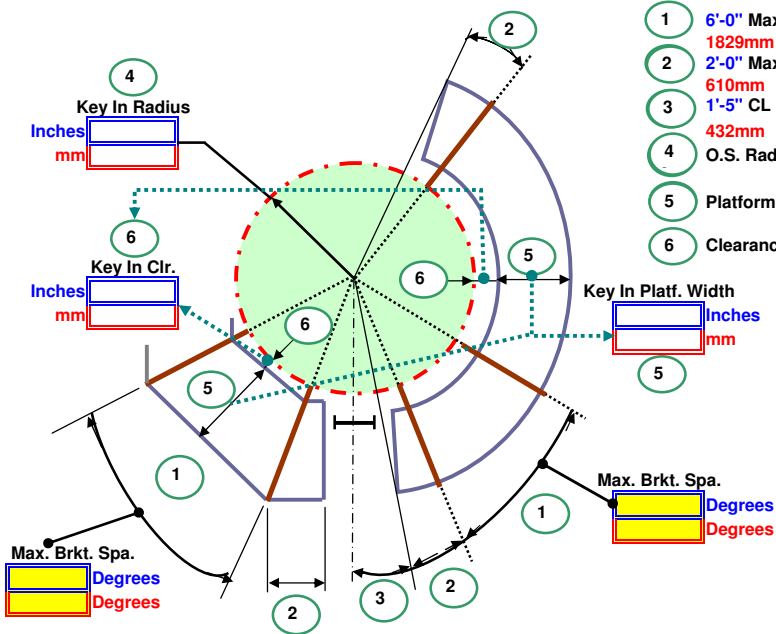
Angle To Horz.	Rise Inches (mm)	Tread Inches (mm)
30.58	6-1/2" (165)	11" (279)
32.00	6-3/4" (172)	10-3/4" (273)
33.68	7" (178)	10-1/2" (267)
35.27	7-1/4" (184)	10-1/4" (260)
36.87	7-1/2" (190)	10" (254)
38.48	7-3/4" (197)	9-3/4" (248)
40.00	8" (203)	9-1/2" (241)
41.73	8-1/4" (210)	9-1/4" (235)
43.37	8-1/2" (216)	9" (229)
45.00	8-3/4" (222)	8-3/4" (222)
46.63	9" (229)	8-1/2" (216)
48.00	9-1/4" (235)	8-1/4" (210)
49.90	9-1/2" (241)	8" (203)



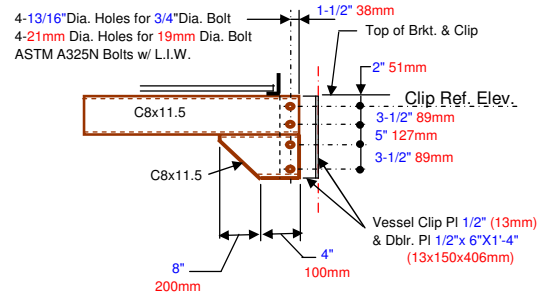
6. Vertical clearance above any stair to an overhead obstruction shall be at least 7 FEET (2134mm) measured from the leading edge of the tread.

29-Dec

Vertical Vessel Platform Max. Bracket Spacing



- 1 6'-0" Max. C/C Brkt. On O.S. Rad./Edge
1829mm
- 2 2'-0" Max. Cantilever on O.S. Rad./Edge
610mm
- 3 1'-5" CL Ladder to Platform on O.S. Rad.
432mm
- 4 O.S. Radius of Vessel plus Insulation (Key In)
- 5 Platform Width (Key In)
- 6 Clearance between vessel & platform - 3" min-4" Max. (Key in)
76mm to 102mm



Typical Hot Vessel Bracket Detail

****Pipe - Formulas**

Pipe Inches O.D.	Wall Inches Thk.	Weight Lbs/Ft	Weight Kg/M	Surface Area Sq. Ft-per Ft.	Wt. of Water per Ft - Lbs.	Wt of Pipe & Water-Lbs/Ft	Surface Area Sq. M-per M	Section Modulus =	7.9168	Z
10	0.104	11.00	16.37	2.62	32.63	43.63	0.80	Moment of Inertia =	39.5841	I
								Radius of Gyration=	3.4990	R

KEY IN KEY IN

Pipe Tables

** Use Pipe Weights if Given Below

Nominal Pipe Size	Outside Dia.	Designation	Wall Thickness	Weight Lbs/ Ft.	Weight Kg/M
1/8	0.405	10S	0.049	0.186	0.277
1/8	0.405	Std.	0.068	0.244	0.363
1/8	0.405	X-Stg.	0.095	0.314	0.467
1/4	0.540	10S	0.065	0.330	0.491
1/4	0.540	Std.	0.088	0.424	0.631
1/4	0.540	X-Stg.	0.119	0.535	0.796
3/8	0.675	10S	0.065	0.423	0.629
3/8	0.675	Std.	0.091	0.567	0.844
3/8	0.675	X-Stg.	0.126	0.738	1.098
1/2	0.840	10S	0.083	0.671	0.999
1/2	0.840	Std.	0.109	0.850	1.265
1/2	0.840	X-Stg.	0.147	1.087	1.618
1/2	0.840	160	0.138	1.310	1.949
1/2	0.840	XX-Stg.	0.294	1.714	2.551
3/4	1.050	10S	0.083	0.857	1.275
3/4	1.050	Std.	0.113	1.130	1.682
3/4	1.050	X-Stg.	0.154	1.473	2.192
3/4	1.050	160	0.219	1.940	2.887
3/4	1.050	XX-Stg.	0.308	2.440	3.631
1	1.315	10S	0.109	1.404	2.089
1	1.315	Std.	0.133	1.678	2.497
1	1.315	X-Stg.	0.179	2.171	3.231
1	1.315	160	0.250	2.850	4.241
1	1.315	XX-Stg.	0.358	3.659	5.445
1 1/4	1.660	10S	0.109	1.806	2.688
1 1/4	1.660	Std.	0.140	2.272	3.381
1 1/4	1.660	-	0.191	2.996	4.459
1 1/4	1.660	-	0.250	3.764	5.601
1 1/4	1.660	-	0.382	5.214	7.759
1 1/2	1.900	10S	0.109	2.085	3.103
1 1/2	1.900	Std.	0.145	2.717	4.043
1 1/2	1.900	X-Stg.	0.200	3.631	5.404
1 1/2	1.900	160	0.281	4.862	7.235
1 1/2	1.900	XX-Stg.	0.400	6.408	9.536
2	2.375	10S	0.109	2.638	3.926
2	2.375	Std.	0.154	3.652	5.435
2	2.375	X-Stg.	0.218	5.022	7.474
2	2.375	-	0.250	5.673	8.442
2	2.375	160	0.344	7.450	11.087
2	2.375	XX-Stg.	0.436	9.029	13.437
2 1/2	2.875	10S	0.120	3.530	5.253
2 1/2	2.875	Std.	0.203	5.790	8.616
2 1/2	2.875	X-Stg.	0.276	7.660	11.399
2 1/2	2.875	160	0.375	10.010	14.897
2 1/2	2.875	XX-Stg.	0.552	13.690	20.373
3	3.5	10S	0.120	4.33	6.444
3	3.5	API	0.125	4.52	6.727
3	3.5	API	0.156	5.58	8.304
3	3.5	API	0.188	6.65	9.896
3	3.5	Std.	0.216	7.57	11.265
3	3.5	API	0.250	8.68	12.917
3	3.5	API	0.281	9.65	14.361
3	3.5	X-Stg.	0.300	10.25	15.254

Nominal Pipe Size	Outside Dia.	Designation	Wall Thickness	Weight Lbs/ Ft.	Weight Kg/M
14	14.000	10 Ga.	0.134	20	29.763
14	14.000	8 Ga.	0.164	24	35.716
14	14.000	6 Ga.	0.194	29	43.157
14	14.000	API	0.210	31	46.133
14	14.000	API	0.219	32	47.621
14	14.000	3 Ga.	0.239	35	52.086
14	14.000	10	0.250	37	55.062
14	14.000	API	0.281	41	61.015
14	14.000	20	0.312	46	68.456
14	14.000	API	0.344	50	74.408
14	14.000	Std.	0.375	55	81.849
14	14.000	40	0.438	63	93.754
14	14.000	X-Stg.	0.500	72	107.148
14	14.000	60	0.594	85	126.494
14	14.000	--	0.625	89	132.447
14	14.000	80	0.750	107	159.234
14	14.000	--	0.875	123	183.044
14	14.000	100	0.938	131	194.949
14	14.000	--	1.000	139	206.855
14	14.000	120	1.094	151	224.713
14	14.000	--	1.125	155	230.665
14	14.000	140	1.250	171	254.476
14	14.000	--	1.375	186	276.799
14	14.000	160	1.406	190	282.751
14	14.000	--	1.500	200	297.633
16	16.000	10 Ga.	0.134	23	34.228
16	16.000	8 Ga.	0.164	28	41.669
16	16.000	--	0.188	32	47.621
16	16.000	6 Ga.	0.194	33	49.109
16	16.000	API	0.219	37	55.062
16	16.000	3 Ga.	0.239	40	59.527
16	16.000	10	0.250	42	62.503
16	16.000	API	0.281	47	69.944
16	16.000	20	0.312	52	77.385
16	16.000	API	0.344	57	84.825
16	16.000	Std.	0.375	63	93.754
16	16.000	API	0.438	73	108.636
16	16.000	X-Stg.	0.500	83	123.518
16	16.000	--	0.625	103	153.281
16	16.000	60	0.656	108	160.722
16	16.000	--	0.750	122	181.556
16	16.000	80	0.844	137	203.878
16	16.000	--	0.875	141	209.831
16	16.000	--	1.000	160	238.106
16	16.000	100	1.031	165	245.547
16	16.000	--	1.125	179	266.381
16	16.000	120	1.219	193	287.216
16	16.000	--	1.250	197	293.168
16	16.000	--	1.375	215	319.955
16	16.000	140	1.438	224	333.349
16	16.000	--	1.500	232	345.254
16	16.000	160	1.594	245	364.600
18	18.000	10 Ga.	0.134	26	38.692
18	18.000	8 Ga.	0.164	31	46.133
18	18.000	6 Ga.	0.194	37	55.062
18	18.000	3 Ga.	0.239	45	66.967
18	18.000	10	0.250	47	69.944
18	18.000	API	0.281	49	72.920
18	18.000	20	0.312	59	87.802
18	18.000	API	0.344	65	96.731
18	18.000	Std.	0.375	71	105.660

****Pipe - Formulas**

Pipe Inches O.D.	Wall Inches Thk.	Weight Lbs/Ft	Weight Kg/M	Surface Area Sq.Ft-per Ft.	Wt. of Water per Ft - Lbs.	Wt of Pipe & Water-Lbs/Ft	Surface Area Sq. M-per M	Section Modulus =	7.9168	Z
								Moment of Inertia =	39.5841	I
								Radius of Gyration=	3.4990	R
10	0.104	11.00	16.37	2.62	32.63	43.63	0.80			

KEY IN KEY IN

Pipe Tables

** Use Pipe Weights if Given Below

Nominal Pipe Size	Outside Dia.	Designation	Wall Thickness	Weight Lbs/ Ft.	Weight Kg/M
3	3.5	160	0.438	14.32	21.311
3	3.5	XX-Stg.	0.600	18.58	27.650
3 1/2	4	10S	0.120	4.97	7.396
3 1/2	4	API	0.125	5.18	7.709
3 1/2	4	API	0.156	6.41	9.539
3 1/2	4	API	0.188	7.71	11.474
3 1/2	4	Std.	0.226	9.11	13.557
3 1/2	4	API	0.250	10.02	14.911
3 1/2	4	API	0.281	11.17	16.623
3 1/2	4	X-Stg.	0.318	12.51	18.617
3 1/2	4	XX-Stg.	0.636	22.85	34.005
4	4.5	10S	0.120	5.61	8.349
4	4.5	API	0.125	5.84	8.691
4	4.5	API	0.156	7.24	10.774
4	4.5	API	0.188	8.56	12.739
4	4.5	API	0.219	10.02	14.911
4	4.5	Std.	0.237	10.79	16.057
4	4.5	API	0.250	11.35	16.891
4	4.5	API	0.281	12.67	18.855
4	4.5	API	0.312	14.00	20.834
4	4.5	X-Stg.	0.337	14.98	22.293
4	4.5	120	0.438	19.00	28.275
4	4.5	--	0.500	21.36	31.787
4	4.5	160	0.531	22.60	33.633
4	4.5	XX-Stg.	0.674	27.54	40.984
5	5.563	10S	0.134	7.77	11.563
5	5.563	API	0.156	9.02	13.423
5	5.563	API	0.188	10.80	16.072
5	5.563	API	0.219	12.51	18.617
5	5.563	Std.	0.258	14.62	21.757
5	5.563	API	0.281	15.86	23.602
5	5.563	API	0.312	17.51	26.058
5	5.563	API	0.344	19.19	28.558
5	5.563	X-Stg.	0.375	20.78	30.924
5	5.563	120	0.500	27.10	40.329
5	5.563	160	0.625	32.96	49.050
5	5.563	XX-Stg.	0.750	38.55	57.369
6	6.625	12 Ga.	0.104	7.25	10.789
6	6.625	10S	0.134	9.29	13.825
6	6.625	8 Ga.	0.164	11.33	16.861
6	6.625	API	0.188	12.93	19.242
6	6.625	6Ga.	0.194	13.34	19.852
6	6.625	API	0.219	15.02	22.352
6	6.625	API	0.250	17.02	25.329
6	6.625	API	0.277	18.86	28.067
6	6.625	Std.	0.280	18.97	28.230
6	6.625	API	0.312	21.05	31.326
6	6.625	API	0.344	23.09	34.362
6	6.625	API	0.375	25.10	37.353
6	6.625	X-Stg.	0.432	28.57	42.517
6	6.625	--	0.500	32.79	48.797
6	6.625	120	0.562	36.40	54.169
6	6.625	160	0.719	45.30	67.414
6	6.625	XX-Stg.	0.864	53.16	79.111
8	8.625	12 Ga.	0.104	9.47	14.093
8	8.625	10 Ga.	0.134	12.16	18.096
8	8.625	10S	0.148	13.40	19.941
8	8.625	8 Ga.	0.164	14.83	22.069

Nominal Pipe Size	Outside Dia.	Designation	Wall Thickness	Weight Lbs/ Ft.	Weight Kg/M
18	18.000	API	0.406	76	113.100
18	18.000	30	0.438	82	122.029
18	18.000	X-Stg.	0.500	93	138.399
18	18.000	40	0.562	105	156.257
18	18.000	--	0.625	116	172.627
18	18.000	60	0.750	138	205.367
18	18.000	--	0.875	160	238.106
18	18.000	80	0.938	171	254.476
18	18.000	--	1.000	182	270.846
18	18.000	--	1.125	203	302.097
18	18.000	100	1.156	208	309.538
18	18.000	--	1.250	224	333.349
18	18.000	120	1.375	244	363.112
18	18.000	--	1.500	265	394.363
18	18.000	140	1.562	275	409.245
18	18.000	160	1.781	309	459.843
20	20.000	10 Ga.	0.134	28	41.669
20	20.000	8 Ga.	0.164	35	52.086
20	20.000	6 Ga.	0.194	41	61.015
20	20.000	3 Ga.	0.239	50	74.408
20	20.000	10	0.250	53	78.873
20	20.000	API	0.281	59	87.802
20	20.000	API	0.312	66	98.219
20	20.000	API	0.344	72	107.148
20	20.000	Std.	0.375	79	117.565
20	20.000	API	0.406	85	126.494
20	20.000	API	0.438	92	136.911
20	20.000	X-Stg.	0.500	105	156.257
20	20.000	40	0.594	123	183.044
20	20.000	--	0.625	129	191.973
20	20.000	60	0.812	167	248.523
20	20.000	--	0.875	179	266.381
20	20.000	--	1.000	203	302.097
20	20.000	80	1.031	209	311.026
20	20.000	--	1.125	227	337.813
20	20.000	--	1.250	250	372.041
20	20.000	100	1.281	256	380.970
20	20.000	--	1.375	274	407.757
20	20.000	120	1.500	297	441.985
20	20.000	140	1.750	342	508.952
20	20.000	160	1.969	379	564.014
22	22.000	8 Ga.	0.164	38	56.550
22	22.000	6 Ga.	0.194	45	66.967
22	22.000	3 Ga.	0.239	56	83.337
22	22.000	API	0.250	58	86.314
22	22.000	API	0.281	65	96.731
22	22.000	API	0.312	72	107.148
22	22.000	API	0.344	80	119.053
22	22.000	API	0.375	87	129.470
22	22.000	API	0.406	94	139.887
22	22.000	API	0.438	101	150.305
22	22.000	API	0.500	115	171.139
22	22.000	--	0.625	143	212.807
22	22.000	--	0.750	170	252.988
22	22.000	--	0.875	198	294.656
22	22.000	--	1.000	224	333.349
22	22.000	--	1.125	251	373.529
22	22.000	--	1.250	277	412.221
22	22.000	--	1.375	303	450.914
22	22.000	--	1.500	329	489.606

****Pipe - Formulas**

Pipe Inches O.D.	Wall Inches Thk.	Weight Lbs/Ft	Weight Kg/M	Surface Area Sq.Ft-per Ft.	Wt. of Water per Ft - Lbs.	Wt of Pipe & Water-Lbs/Ft	Surface Area Sq. M-per M	Section Modulus =	7.9168	Z
10	0.104	11.00	16.37	2.62	32.63	43.63	0.80	Moment of Inertia =	39.5841	I
								Radius of Gyration=	3.4990	R

KEY IN KEY IN

Pipe Tables

** Use Pipe Weights if Given Below

Nominal Pipe Size	Outside Dia.	Designation	Wall Thickness	Weight Lbs/ Ft.	Weight Kg/M
8	8.625	API	0.188	16.90	25.150
8	8.625	6 Ga.	0.194	17.48	26.013
8	8.625	API	0.203	18.30	27.233
8	8.625	API	0.219	19.64	29.228
8	8.625	3 Ga.	0.239	21.42	31.876
8	8.625	20	0.250	22.40	33.335
8	8.625	30	0.277	24.70	36.758
8	8.625	API	0.312	27.72	41.252
8	8.625	Std.	0.322	28.55	42.487
8	8.625	API	0.344	30.40	45.240
8	8.625	API	0.375	33.10	49.258
8	8.625	60	0.406	35.70	53.127
8	8.625	API	0.438	38.33	57.041
8	8.625	X-Stg.	0.500	43.39	64.571
8	8.625	100	0.594	50.90	75.748
8	8.625	--	0.625	53.40	79.468
8	8.625	120	0.719	60.70	90.332
8	8.625	140	0.812	67.80	100.898
8	8.625	XX-Stg.	0.875	72.42	107.773
8	8.625	160	0.906	74.70	111.166
10	10.750	12 Ga.	0.104	11.83	17.605
10	10.750	10 Ga.	0.134	15.21	22.635
10	10.750	8 Ga.	0.164	18.56	27.620
10	10.750	10S	0.165	18.65	27.754
10	10.750	API	0.188	21.12	31.430
10	10.750	6 Ga.	0.194	21.89	32.576
10	10.750	API	0.203	22.86	34.019
10	10.750	API	0.219	24.60	36.609
10	10.750	3 Ga.	0.239	28.05	41.743
10	10.750	20	0.250	28.03	41.713
10	10.750	API	0.279	31.20	46.431
10	10.750	30	0.307	34.24	50.955
10	10.750	API	0.344	38.26	56.937
10	10.750	Std.	0.365	40.48	60.241
10	10.750	API	0.438	48.28	71.849
10	10.750	X-Stg.	0.500	54.74	81.462
10	10.750	80	0.594	64.40	95.838
10	10.750	100	0.719	77.00	114.589
10	10.750	--	0.750	80.10	119.202
10	10.750	120	0.844	89.20	132.744
10	10.750	140	1.000	104.20	155.067
10	10.750	160	1.125	116.00	172.627
12	12.750	12 Ga.	0.104	14.10	20.983
12	12.750	10 Ga.	0.134	18.10	26.936
12	12.750	8 Ga.	0.164	22.10	32.888
12	12.750	10S	0.180	24.20	36.014
12	12.750	6 Ga.	0.194	26.00	38.692
12	12.750	API	0.203	27.20	40.478
12	12.750	API	0.219	29.30	43.603
12	12.750	3 Ga.	0.239	32.00	47.621
12	12.750	20	0.250	33.40	49.705
12	12.750	API	0.281	37.40	55.657
12	12.750	API	0.312	41.50	61.759
12	12.750	30	0.330	43.80	65.182
12	12.750	API	0.344	45.50	67.711
12	12.750	Std.	0.375	49.60	73.813
12	12.750	40	0.406	53.60	79.766
12	12.750	API	0.438	57.50	85.569
12	12.750	X-Stg.	0.500	65.40	97.326
12	12.750	60	0.562	73.20	108.934
12	12.750	--	0.625	80.90	120.392

Nominal Pipe Size	Outside Dia.	Designation	Wall Thickness	Weight Lbs/ Ft.	Weight Kg/M
24	24.000	8 Ga.	0.164	42	62.503
24	24.000	6 Ga.	0.194	49	72.920
24	24.000	3 Ga.	0.239	61	90.778
24	24.000	10	0.250	63	93.754
24	24.000	API	0.281	71	105.660
24	24.000	API	0.312	79	117.565
24	24.000	API	0.344	87	129.470
24	24.000	Std.	0.375	95	141.376
24	24.000	API	0.406	102	151.793
24	24.000	API	0.438	110	163.698
24	24.000	X-Stg.	0.500	125	186.021
24	24.000	30	0.562	141	209.831
24	24.000	--	0.625	156	232.154
24	24.000	40	0.688	171	254.476
24	24.000	--	0.750	186	277.799
24	24.000	--	0.875	216	321.443
24	24.000	60	0.969	238	354.183
24	24.000	--	1.000	246	366.088
24	24.000	--	1.125	275	409.245
24	24.000	80	1.219	297	441.985
24	24.000	--	1.250	304	452.402
24	24.000	--	1.375	332	494.070
24	24.000	--	1.500	361	537.227
24	24.000	100	1.531	367	546.156
24	24.000	120	1.812	429	638.422
24	24.000	140	2.062	484	720.271
24	24.000	160	2.344	542	806.585
26	26.000	8 Ga.	0.164	45	66.967
26	26.000	6 Ga.	0.194	54	80.361
26	26.000	3 Ga.	0.239	66	98.219
26	26.000	API	0.250	67	99.707
26	26.000	API	0.281	77	114.589
26	26.000	API	0.312	84	125.006
26	26.000	API	0.344	94	139.887
26	26.000	API	0.375	103	153.281
26	26.000	API	0.406	111	165.186
26	26.000	API	0.438	120	178.580
26	26.000	API	0.500	136	202.390
26	26.000	--	0.625	169	251.500
26	26.000	--	0.750	202	300.609
26	26.000	--	0.875	235	349.719
26	26.000	--	1.000	267	397.340
26	26.000	--	1.125	299	444.961
26	26.000	--	1.375	362	538.715
26	26.000	--	1.500	393	584.848
30	30.000	8 Ga.	0.164	52	77.385
30	30.000	6 Ga.	0.194	62	92.266
30	30.000	3 Ga.	0.239	76	113.100
30	30.000	API	0.250	79	117.565
30	30.000	API	0.281	89	132.447
30	30.000	10	0.312	99	147.328
30	30.000	API	0.344	109	162.210
30	30.000	API	0.375	119	177.092
30	30.000	API	0.406	130	193.461
30	30.000	API	0.438	138	205.367
30	30.000	20	0.500	158	235.130
30	30.000	30	0.625	196	291.680
30	30.000	--	0.750	234	348.230
30	30.000	--	0.875	272	404.781
30	30.000	--	1.000	310	461.331
30	30.000	--	1.125	347	516.393

****Pipe - Formulas**

Pipe Inches O.D.	Wall Inches Thk.	Weight Lbs/Ft	Weight Kg/M	Surface Area Sq.Ft-per Ft.	Wt. of Water per Ft - Lbs.	Wt of Pipe & Water-Lbs/Ft	Surface Area Sq. M-per M	Section Modulus =	7.9168	Z
10	0.104	11.00	16.37	2.62	32.63	43.63	0.80	Moment of Inertia =	39.5841	I
								Radius of Gyration=	3.4990	R

KEY IN KEY IN

Pipe Tables

** Use Pipe Weights if Given Below

Nominal Pipe Size	Outside Dia.	Designation	Wall Thickness	Weight Lbs/ Ft.	Weight Kg/M
12	12.750	80	0.688	88.60	131.851
12	12.750	--	0.750	96.20	143.161
12	12.750	100	0.844	108.00	160.722
12	12.750	--	0.875	110.90	165.037
12	12.750	120	1.000	125.50	186.765
12	12.750	140	1.125	140.00	208.343
12	12.750	--	1.250	153.60	228.582
12	12.750	160	1.312	161.00	239.594
12	12.750	--	1.375	167.20	248.821
12	12.750	--	1.500	180.40	268.465

Nominal Pipe Size	Outside Dia.	Designation	Wall Thickness	Weight Lbs/ Ft.	Weight Kg/M
30	30.000	--	1.250	384	571.455
30	30.000	--	1.375	421	626.517
30	30.000	--	1.500	457	680.091
32	32.000	API	0.250	85	126.494
32	32.000	API	0.281	95	141.376
32	32.000	API	0.312	106	157.745
32	32.000	API	0.344	116	172.627
32	32.000	API	0.375	127	188.997
32	32.000	API	0.406	137	203.878
32	32.000	API	0.438	148	220.248
32	32.000	API	0.500	168	250.012
32	32.000	--	0.625	209	311.026
32	32.000	--	0.750	250	372.041
32	32.000	--	0.875	291	433.056
32	32.000	--	1.000	331	492.582
32	32.000	--	1.125	371	552.109
32	32.000	--	1.250	410	610.147
32	32.000	--	1.375	450	669.674
32	32.000	--	1.500	489	727.712
34	34.000	API	0.250	90	133.935
34	34.000	API	0.281	101	150.305
34	34.000	API	0.312	112	166.674
34	34.000	API	0.344	124	184.532
34	34.000	API	0.375	135	200.902
34	34.000	API	0.406	146	217.272
34	34.000	API	0.438	157	233.642
34	34.000	API	0.500	179	266.381
34	34.000	--	0.625	223	331.861
34	34.000	--	0.750	266	395.852
34	34.000	--	0.875	308	458.355
34	34.000	--	1.000	353	525.322
34	34.000	--	1.125	395	587.825
34	34.000	--	1.250	437	650.328
34	34.000	--	1.375	479	712.831
34	34.000	--	1.500	521	775.333
36	36.000	--	0.164	63	93.754
36	36.000	--	0.194	74	110.124
36	36.000	--	0.239	91	135.423
36	36.000	API	0.250	96	142.864
36	36.000	API	0.281	107	159.234
36	36.000	API	0.312	119	177.092
36	36.000	API	0.344	131	194.949
36	36.000	API	0.375	143	212.807
36	36.000	API	0.406	154	229.177
36	36.000	API	0.438	166	247.035
36	36.000	API	0.500	190	282.751
36	36.000	--	0.625	236	351.207
36	36.000	--	0.750	282	419.662
36	36.000	--	0.875	329	489.606
36	36.000	--	1.000	374	556.573
36	36.000	--	1.125	419	623.541
36	36.000	--	1.250	464	690.508
36	36.000	--	1.375	509	757.475
36	36.000	--	1.500	553	822.955
42	42.000	--	0.250	112	166.674
42	42.000	--	0.375	167	248.523
42	42.000	--	0.500	222	330.372
42	42.000	--	0.625	276	410.733
42	42.000	--	0.750	331	492.582

****Pipe - Formulas**

Pipe Inches O.D.	Wall Inches Thk.	Weight Lbs/Ft	Weight Kg/M	Surface Area Sq.Ft-per Ft.	Wt. of Water per Ft - Lbs.	Wt of Pipe & Water-Lbs/Ft	Surface Area Sq. M-per M	Section Modulus =	7.9168	Z
10	0.104	11.00	16.37	2.62	32.63	43.63	0.80	Moment of Inertia =	39.5841	I
								Raduis of Gyration=	3.4990	R

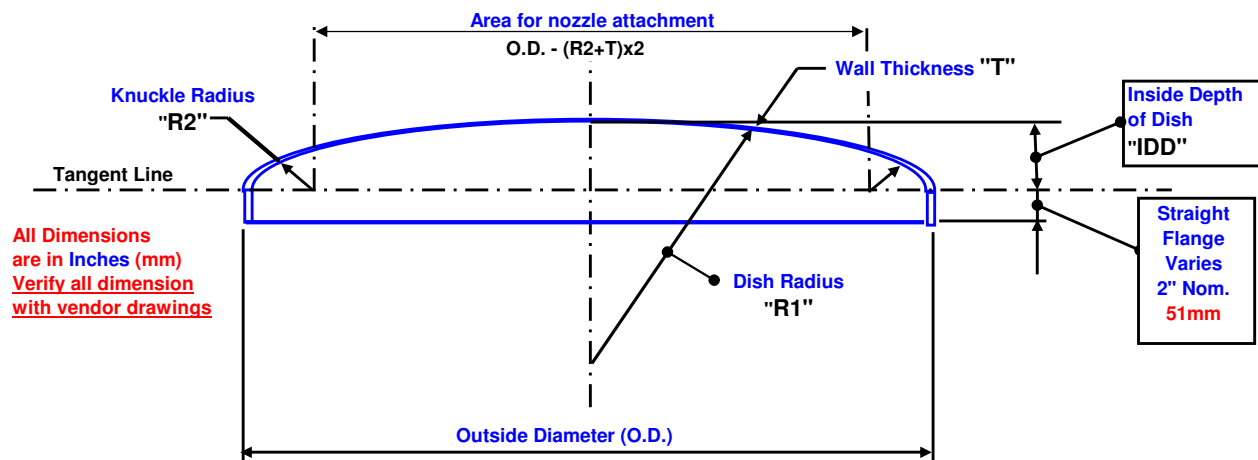
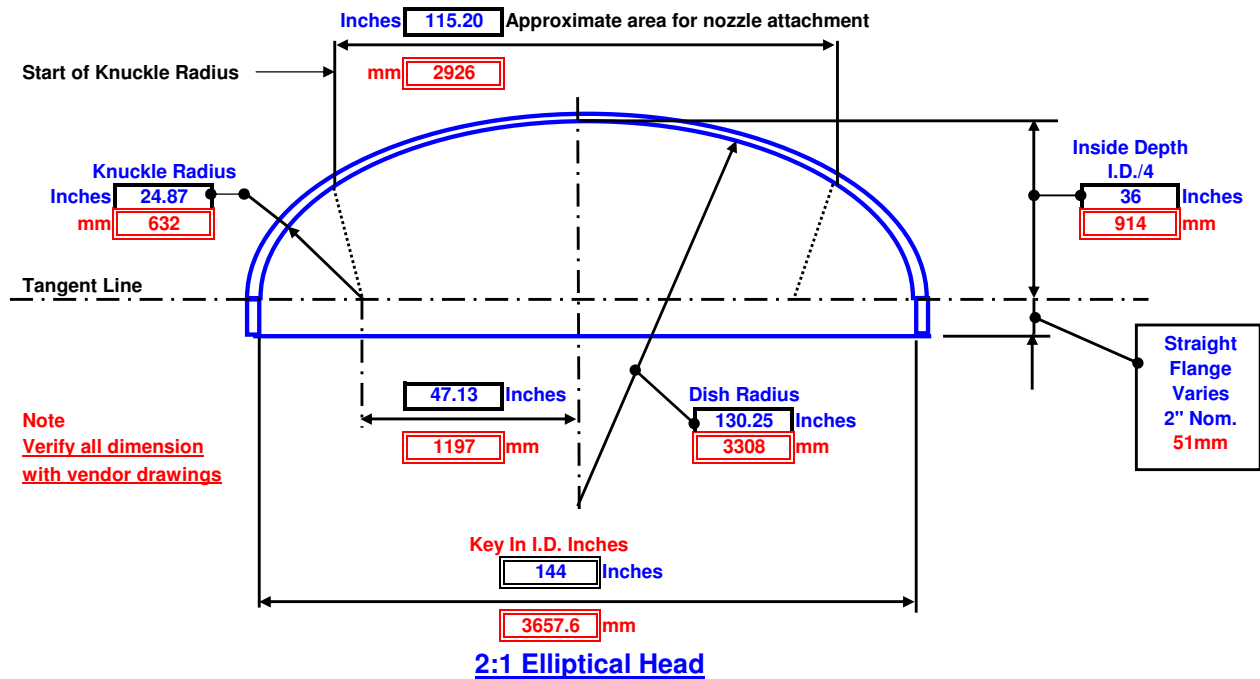
KEY IN KEY IN

Pipe Tables

** Use Pipe Weights if Given Below

Nominal Pipe Size	Outside Dia.	Desig- nation	Wall Thickness	Weight Lbs/ Ft.	Weight Kg/M
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Nominal Pipe Size	Outside Dia.	Desig- nation	Wall Thickness	Weight Lbs/ Ft.	Weight Kg/M
42	42.000	--	0.875	385	572.943
42	42.000	--	1.000	438	651.816
42	42.000	--	1.125	492	732.177
42	42.000	--	1.250	544	809.561
42	42.000	--	1.375	597	888.434
42	42.000	--	1.500	649	965.818



Not all wall thicknesses are shown. Interpolate for approximate inside depth O.D. dish IDD
Inches (Flanged & Dished Head Table)

O.D.	"T"	"R1"	"R2"	IDD
26"	0.38	24	1.13	4.25
	0.50	24	1.50	4.38
	0.63	24	1.88	4.50
	0.75	24	2.25	4.69
28"	0.38	26	4.50	4.50
	0.50	26	4.63	4.63
	0.63	26	4.75	4.75
	0.75	26	4.94	4.94
30"	0.38	30	4.50	4.50
	0.50	30	4.63	4.63
	0.63	30	4.81	4.81
	0.75	30	5.00	5.00
32"	0.38	30	5.00	5.00
	0.50	30	5.19	5.19
	0.63	30	5.31	5.31
	0.75	30	5.50	5.50
34"	0.38	34	5.00	5.00
	0.50	34	5.19	5.19
	0.63	33	5.44	5.44
	0.75	30	6.06	6.06

Millimeters (Flanged & Dished Head Table)

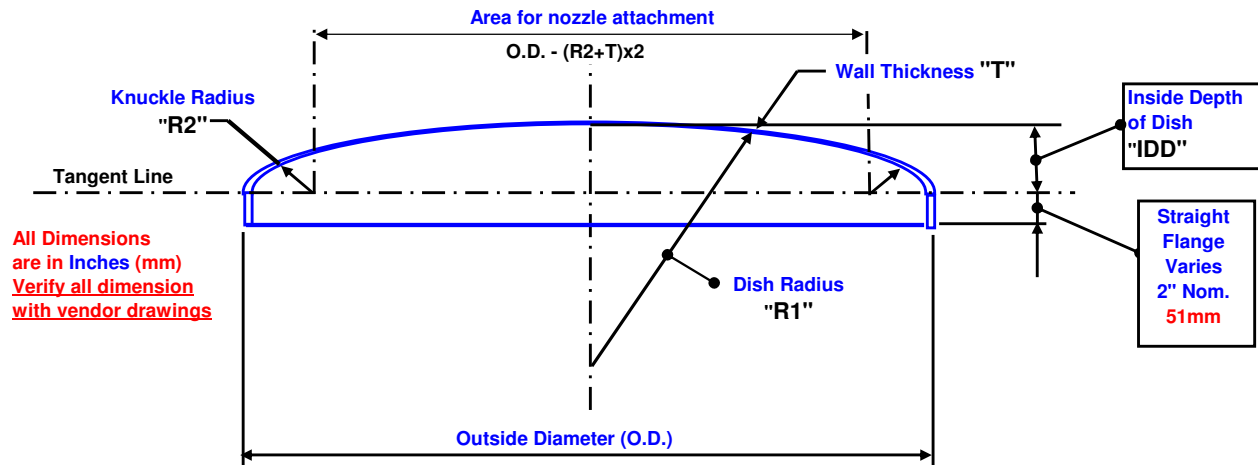
O.D.	"T" (mm)	"R1" (mm)	"R2" (mm)	IDD (mm)
26" 660	10	610	29	108
	13	610	38	111
	16	610	48	114
	19	610	57	119
28" 711	10	660	114	114
	13	660	118	118
	16	660	121	121
	19	660	125	125
30" 762	10	762	114	114
	13	762	118	118
	16	762	122	122
	19	762	127	127
32" 813	10	762	127	127
	13	762	132	132
	16	762	135	135
	19	762	140	140
34 864	10	864	127	127
	13	864	132	132
	16	838	138	138
	19	762	154	154

36"	0.38	36	5.25	5.25
	0.50	36	5.44	5.44
	0.63	36	5.63	5.63
	0.75	36	5.75	5.75
O.D	"T"	"R1"	"R2"	IDD
38"	0.38	36	1.13	5.81
	0.50	36	1.50	6.00
	0.63	36	1.88	6.13
	0.75	36	2.25	6.31
40"	0.38	40	1.13	5.81
	0.50	40	1.50	5.94
	0.63	36	1.88	6.69
	0.75	36	2.25	6.88
42"	0.38	42	1.13	6.06
	0.50	42	1.50	6.25
	0.63	42	1.88	6.38
	0.75	40	2.25	6.81
48"	0.38	48	1.13	6.88
	0.50	48	1.50	7.00
	0.63	48	1.88	7.19
	0.75	48	2.25	7.38
54"	0.38	54	1.13	7.69
	0.50	54	1.50	7.81
	0.63	54	1.88	8.00
	0.75	54	2.25	8.19
60"	0.38	60	1.13	8.50
	0.50	60	1.50	8.63
	0.63	60	1.88	8.81
	0.75	60	2.25	8.94
O.D	"T"	"R1"	"R2"	IDD
66"	0.38	66	1.13	9.31
	0.50	66	1.50	9.44
	0.63	66	1.88	9.63
	0.75	66	2.25	9.75
72"	0.38	72	1.13	10.06
	0.63	72	1.88	10.38
	0.88	72	2.63	10.69
	1.13	72	3.38	11.00
78"	0.38	78	1.13	10.88
	0.63	78	1.88	11.19
	0.88	78	2.63	11.50
	1.13	78	3.38	11.81
84"	0.38	84	1.13	11.69
	0.63	84	1.88	11.88
	0.88	84	2.63	12.31
	1.13	84	3.38	12.63
90"	0.38	90	1.13	12.50
	0.63	84	1.88	13.75
	0.88	84	2.63	14.00
	1.13	84	3.38	14.31
96"	0.50	96	1.50	13.44
	0.75	96	2.25	13.75
	1.00	96	3.00	14.06
	1.25	96	3.75	14.38
O.D	"T"	"R1"	"R2"	IDD
102"	0.50	102	1.50	14.25
	0.75	96	2.25	15.50
	1.00	96	3.00	15.75
	1.25	96	3.75	16.06
108"	0.50	108	1.50	15.06
	0.75	108	2.25	15.38
	1.00	102	3.00	16.56
	1.25	102	3.75	16.81
114"	0.50	114	1.50	15.88
	0.75	114	2.25	16.19
	1.00	108	3.00	17.38
	1.25	108	3.75	17.63
120"	0.50	120	1.50	16.69
	0.88	120	2.63	17.13
	1.25	120	3.75	17.59
	1.63	120	4.88	18.06

36" 914	10	914	133	133
	13	914	138	138
	16	914	143	143
	19	914	146	146
O.D	"T" (mm)	"R1" (mm)	"R2" (mm)	IDD (mm)
38" 965	10	914	29	148
	13	914	38	152
	16	914	48	156
	19	914	57	160
40" 1016	10	1016	29	148
	13	1016	38	151
	16	914	48	170
	19	914	57	175
42" 1067	10	1067	29	154
	13	1067	38	159
	16	1067	48	162
	19	1016	57	173
42" 1219	10	1219	29	175
	13	1219	38	178
	16	1219	48	183
	19	1219	57	187
54" 1372	10	1372	29	195
	13	1372	38	198
	16	1372	48	203
	19	1372	57	208
60" 1524	10	1524	29	216
	13	1524	38	219
	16	1524	48	224
	19	1524	57	227
O.D	"T" (mm)	"R1" (mm)	"R2" (mm)	IDD (mm)
66" 1676	10	1676	29	236
	13	1676	38	240
	16	1676	48	245
	19	1676	57	248
72" 1829	10	1829	29	256
	16	1829	48	264
	22	1829	67	272
	29	1829	86	279
78" 1981	10	1981	29	276
	16	1981	48	284
	22	1981	67	292
	29	1981	86	300
84" 2134	10	2134	29	297
	16	2134	48	302
	22	2134	67	313
	29	2134	86	321
90" 2286	10	2286	29	318
	16	2134	48	349
	22	2134	67	356
	29	2134	86	363
96" 2438	13	2438	38	341
	19	2438	57	349
	25	2438	76	357
	32	2438	95	365
O.D	"T" (mm)	"R1" (mm)	"R2" (mm)	IDD (mm)
102" 2591	13	2591	38	362
	19	2438	57	394
	25	2438	76	400
	32	2438	95	408
108" 2743	13	2743	38	383
	19	2743	57	391
	25	2591	76	421
	32	2591	95	427
114" 2896	13	2896	38	403
	19	2896	57	411
	25	2743	76	441
	32	2743	95	448
120" 3048	13	3048	38	424
	22	3048	67	435
	32	3048	95	447
	41	3048	124	459

126"	0.50	126	1.50	17.50
	0.88	120	2.63	18.81
	1.25	120	3.75	19.25
	1.63	120	4.88	19.69
132"	0.63	132	1.88	18.44
	0.88	132	2.63	18.75
	1.13	132	3.38	19.00
	1.50	132	4.50	19.50
O.D	"T"	"R1"	"R2"	IDD
138"	0.63	132	1.88	20.13
	0.88	132	2.63	20.44
	1.13	132	3.38	20.69
	1.50	132	4.50	21.13
144"	0.63	144	1.88	20.00
	0.88	144	2.63	20.31
	1.13	144	3.38	20.63
	1.50	144	4.50	21.13
156"	0.75	144	2.25	23.69
	1.13	144	3.38	24.13
	1.50	144	4.50	24.50
	1.88	144	5.63	24.94
168"	0.75	170	2.25	23.13
	1.13	170	3.38	23.56
	1.50	170	4.50	24.06
	1.88	170	5.63	24.19
180"	0.88	170	2.63	26.69
	1.25	170	3.75	27.13
	1.63	170	4.88	27.56
	2.00	170	6.00	28.00
192"	0.88	170	2.63	30.50
	1.25	170	3.75	30.88
	1.63	170	4.88	31.25
	2.00	170	6.00	31.63
O.D	"T"	"R1"	"R2"	IDD
204"	0.88	170	2.63	34.63
	1.25	170	3.75	35.00
	1.63	170	4.88	35.31
	2.00	170	6.00	35.63
216"	0.88	170	2.63	39.06
	1.25	170	3.75	39.50
	1.63	170	4.88	39.75
	2.00	170	6.00	40.00
228"	0.88	180	2.63	41.25
	1.25	180	3.75	41.50
	1.63	180	4.88	41.75
	2.00	180	6.00	42.00
240"	0.88	180	2.63	46.25
	1.25	180	3.75	46.44
	1.63	180	4.88	46.63
	2.00	180	6.00	46.81

126" 3200	13	3200	38	445
	22	3048	67	478
	32	3048	95	489
	41	3048	124	500
132" 3353	16	3353	48	468
	22	3353	67	476
	29	3353	86	483
	38	3353	114	495
O.D	"T" (mm)	"R1" (mm)	"R2" (mm)	IDD (mm)
138" 3505	16	3353	48	511
	22	3353	67	519
	29	3353	86	526
	38	3353	114	537
144" 3658	16	3658	48	508
	22	3658	67	516
	29	3658	86	524
	38	3658	114	537
156" 3962	19	3658	57	602
	29	3658	86	613
	38	3658	114	622
	48	3658	143	633
168" 4267	19	4318	57	588
	29	4318	86	598
	38	4318	114	611
	48	4318	143	614
180 4572	22	4318	67	678
	32	4318	95	689
	41	4318	124	700
	51	4318	152	711
192" 4877	22	4318	67	775
	32	4318	95	784
	41	4318	124	794
	51	4318	152	803
O.D	"T" (mm)	"R1" (mm)	"R2" (mm)	IDD (mm)
204" 5182	22	4318	67	880
	32	4318	95	889
	41	4318	124	897
	51	4318	152	905
216" 5486	22	4318	67	992
	32	4318	95	1003
	41	4318	124	1010
	51	4318	152	1016
228" 5791	22	4572	67	1048
	32	4572	95	1054
	41	4572	124	1060
	51	4572	152	1067
240" 6096	22	4572	67	1175
	32	4572	95	1180
	41	4572	124	1184
	51	4572	152	1189



Not all wall thicknesses are shown. Interpolate for approximate inside depth O.D. dish IDD

Flanged and Dished Head (ASME)

Inches (Flanged & Dished Head ASME Table)

O.D	"T"	"R1"	"R2"	IDD
26"	0.38	24	1.63	4.50
	0.50	24	1.63	4.44
	0.63	24	1.88	4.50
	0.75	24	2.25	4.69
28"	0.38	26	1.75	4.81
	0.50	26	1.75	4.75
	0.63	26	1.88	4.75
	0.75	26	2.25	4.94
30"	0.38	30	1.88	4.88
	0.50	30	1.88	4.81
	0.63	30	1.88	4.81
	0.75	30	2.25	5.00
32"	0.38	30	2.00	5.56
	0.50	30	2.00	5.50
	0.63	30	2.00	5.38
	0.75	30	2.25	5.50
34"	0.38	34	2.13	5.56
	0.50	34	2.13	5.50
	0.63	30	2.13	6.00
	0.75	30	2.25	6.06
36"	0.38	36	2.25	5.94
	0.50	36	2.25	5.88
	0.63	36	2.25	5.81
	0.75	36	2.25	5.75
O.D	"T"	"R1"	"R2"	IDD
38"	0.38	36	2.38	6.50
	0.50	36	2.38	6.44
	0.63	36	2.38	6.38
	0.75	36	2.38	6.38
40"	0.38	40	2.50	6.63
	0.50	40	2.50	6.56
	0.63	36	2.50	6.94
	0.75	36	2.50	7.00
42"	0.38	40	2.63	7.19
	0.50	40	2.63	7.13
	0.63	40	2.63	7.06
	0.75	40	2.63	7.00
48"	0.38	42	3.00	8.00
	0.50	42	3.00	8.75
	0.63	42	3.00	8.69
	0.75	42	3.00	8.63
54"	0.38	54	3.25	8.94
	0.50	48	3.25	9.75
	0.63	48	3.25	9.75
	0.75	48	3.25	9.63
60"	0.38	60	3.63	10.00
	0.50	60	3.63	9.88
	0.63	54	3.63	10.69
	0.75	54	3.63	10.63

ASME

Millimeters (Flanged & Dished Head ASME Table)

O.D	"T" (mm)	"R1" (mm)	"R2" (mm)	IDD (mm)
26" 660	10	610	41	114
	13	610	41	113
	16	610	48	114
	19	610	57	119
28" 711	10	660	44	122
	13	660	44	121
	16	660	48	121
	19	660	57	125
30" 762	10	762	48	124
	13	762	48	122
	16	762	48	122
	19	762	57	127
32" 813	10	762	51	141
	13	762	51	140
	16	762	51	137
	19	762	57	140
34 864	10	864	54	141
	13	864	54	140
	16	762	54	152
	19	762	57	154
36" 914	10	914	57	151
	13	914	57	149
	16	914	57	148
	19	914	57	146
O.D	"T" (mm)	"R1" (mm)	"R2" (mm)	IDD (mm)
38" 965	10	914	60	165
	13	914	60	164
	16	914	60	162
	19	914	60	162
40" 1016	10	1016	64	168
	13	1016	64	167
	16	914	64	176
	19	914	64	178
42" 1067	10	1016	67	183
	13	1016	67	181
	16	1016	67	179
	19	1016	67	178
42" 1219	10	1067	76	203
	13	1067	76	222
	16	1067	76	221
	19	1067	76	219
54" 1372	10	1372	83	227
	13	1219	83	248
	16	1219	83	248
	19	1219	83	245
60" 1524	10	1524	92	254
	13	1524	92	251
	16	1372	92	272
	19	1372	92	270

ASME

O.D	"T"	"R1"	"R2"	IDD
66"	0.38	66	4.00	11.00
	0.50	60	4.00	10.94
	0.63	60	4.00	11.75
	0.75	60	4.00	11.63
72"	0.38	72	4.38	12.00
	0.63	72	4.38	11.88
	0.75	72	4.38	11.88
	0.88	66	4.38	12.63
78"	0.38	78	4.75	13.00
	0.50	72	4.75	13.81
	0.75	72	4.75	13.69
	1.00	72	4.75	13.50
84"	0.38	84	5.13	14.00
	0.63	84	5.13	13.88
	0.88	84	5.13	13.75
	1.00	84	5.13	13.69
90"	0.38	90	5.50	15.13
	0.50	84	5.50	15.81
	0.75	84	5.50	15.69
	1.00	84	5.50	15.56
96"	0.38	96	5.88	16.13
	0.50	90	5.88	16.88
	0.88	90	5.88	16.63
	1.25	90	5.88	16.44
O.D	"T"	"R1"	"R2"	IDD
102"	0.50	96	6.13	17.88
	0.75	96	6.13	17.69
	1.00	96	6.13	17.56
	1.13	90	6.13	18.50
108"	0.50	102	6.50	18.88
	0.75	102	6.50	18.75
	1.00	102	6.50	18.56
	1.13	96	6.50	19.44
114"	0.50	108	6.88	19.88
	0.75	108	6.88	19.75
	1.00	108	6.88	19.63
	1.25	108	6.88	19.50
120"	0.50	114	7.25	20.88
	0.88	114	7.25	20.69
	1.25	108	7.25	21.44
	1.63	108	7.25	21.25
126"	0.50	120	7.63	21.88
	0.88	120	7.63	21.69
	1.25	120	7.63	21.50
	1.38	114	7.63	22.31
132"	0.75	126	8.00	22.81
	0.88	120	8.00	23.69
	1.25	120	8.00	23.44
	1.63	120	8.00	23.25
O.D	"T"	"R1"	"R2"	IDD
138"	0.63	132	8.38	23.94
	1.00	132	8.38	23.75
	1.38	132	8.38	23.56
	1.75	132	8.38	23.38
144"	0.63	132	8.75	25.88
	1.00	132	8.75	25.63
	1.38	132	8.75	25.44
	1.75	132	8.75	25.19
156"	0.75	144	9.38	27.75
	1.13	144	9.38	27.50
	1.50	144	9.38	27.31
	1.88	144	9.38	27.06
168"	0.75	144	10.13	31.81
	1.13	144	10.13	31.50
	1.50	144	10.13	31.31
	1.88	144	10.13	31.13
180"	0.88	170	10.88	31.44
	1.25	170	10.88	31.25
	1.63	170	10.88	31.00
	2.00	170	10.88	30.81

ASME

O.D	"T" (mm)	"R1" (mm)	"R2" (mm)	IDD (mm)
66" 1676	10	1676	102	279
	13	1524	102	278
	16	1524	102	298
	19	1524	102	295
72" 1829	10	1829	111	305
	16	1829	111	302
	19	1829	111	302
	22	1676	111	321
78" 1981	10	1981	121	330
	13	1829	121	351
	19	1829	121	348
	25	1829	121	343
84" 2134	10	2134	130	356
	16	2134	130	353
	22	2134	130	349
	25	2134	130	348
90" 2286	10	2286	140	384
	13	2134	140	402
	19	2134	140	399
	25	2134	140	395
96" 2438	10	2438	149	410
	13	2286	149	429
	22	2286	149	422
	32	2286	149	418
O.D	"T" (mm)	"R1" (mm)	"R2" (mm)	IDD (mm)
102" 2591	13	2438	156	454
	19	2438	156	449
	25	2438	156	446
	29	2286	156	470
108" 2743	13	2591	165	480
	19	2591	165	476
	25	2591	165	471
	29	2438	165	494
114" 2896	13	2743	175	505
	19	2743	175	502
	25	2743	175	499
	32	2743	175	495
120" 3048	13	2896	184	530
	22	2896	184	526
	32	2743	184	545
	41	2743	184	540
126" 3200	13	3048	194	556
	22	3048	194	551
	32	3048	194	546
	35	2896	194	567
132" 3353	19	3200	203	579
	22	3048	203	602
	32	3048	203	595
	41	3048	203	591
O.D	"T" (mm)	"R1" (mm)	"R2" (mm)	IDD (mm)
138" 3505	16	3353	213	608
	25	3353	213	603
	35	3353	213	598
	44	3353	213	594
144" 3658	16	3353	222	657
	25	3353	222	651
	35	3353	222	646
	44	3353	222	640
156" 3962	19	3658	238	705
	29	3658	238	699
	38	3658	238	694
	48	3658	238	687
168" 4267	19	3658	257	808
	29	3658	257	800
	38	3658	257	795
	48	3658	257	791
180" 4572	22	4318	276	799
	32	4318	276	794
	41	4318	276	787
	51	4318	276	783

ASME

ASME

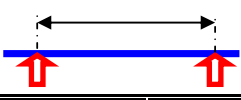
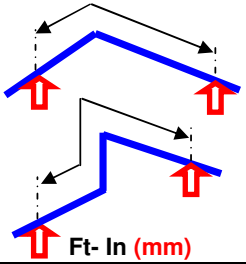
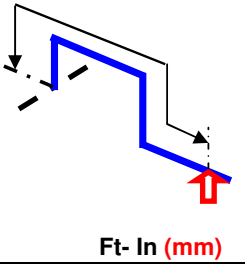
192"	0.88	170	11.63	35.44
	1.25	170	11.63	35.19
	1.63	170	11.63	34.94
	2.00	170	11.63	34.75
O.D	"T"	"R1"	"R2"	IDD
204"	0.88	170	12.25	39.56
	1.25	170	12.25	39.38
	1.63	170	12.25	39.19
	2.00	170	12.25	38.94
210"	1.00	170	12.63	41.81
	1.38	170	12.63	41.63
	1.75	170	12.63	41.31
	2.25	170	12.63	41.00
216"	1.00	170	13.00	44.25
	1.38	170	13.00	44.00
	1.75	170	13.00	43.69
	2.00	170	13.00	43.50
228"	1.00	180	13.75	46.56
	1.38	180	13.75	46.31
	1.75	180	13.75	46.06
	2.00	180	13.75	45.69

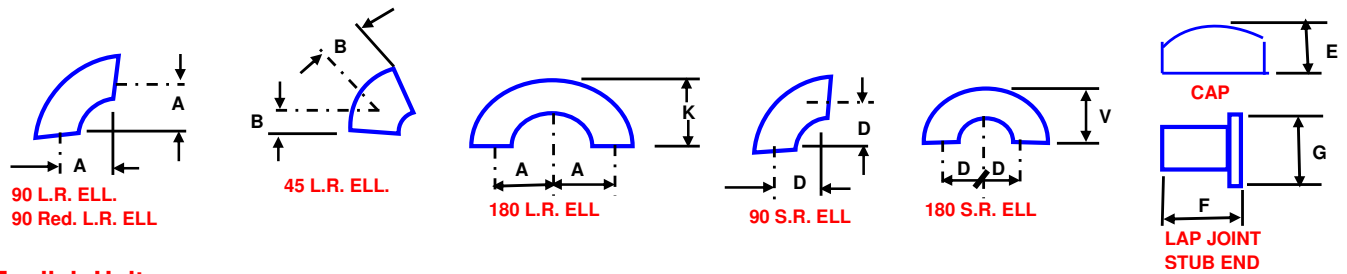
ASME

192" 4877	22	4318	295	900
	32	4318	295	894
	41	4318	295	887
	51	4318	295	883
O.D	"T" (mm)	"R1" (mm)	"R2" (mm)	IDD (mm)
204" 5182	22	4318	311	1005
	32	4318	311	1000
	41	4318	311	995
	51	4318	311	989
210" 5334	25	4318	321	1062
	35	4318	321	1057
	44	4318	321	1049
	57	4318	321	1041
216" 5486	25	4318	330	1124
	35	4318	330	1118
	44	4318	330	1110
	51	4318	330	1105
228" 5791	25	4572	349	1183
	35	4572	349	1176
	44	4572	349	1170
	51	4572	349	1161

ALLOWABLE PIPING SPANS

Based on using carbon steel pipe (A53-GR B) filled with water
Spans are based on insulated and uninsulated lines below 650 Deg. F

Nom. Pipe Size	Pipe Sch.	Wall Thickness Inches				
			Allowable Span	Deflection	Ft- In (mm)	Ft- In (mm)
			Ft- In (mm)	Inches (mm)		
3/4"	80	0.154	10'-0" (3048)	3/8" (10)	7'-0" (2134)	8'-0" (2438)
1"	80	0.179	13'-0" (3962)	1/2" (13)	9'-0" (2743)	10'-0" (3048)
1-1/2"	80	0.200	17'-0" (5182)	3/4" (19)	11'-0" (3353)	13'-0" (3962)
2"	40	0.154	19'-0" (5791)	1" (25)	12'-0" (3658)	14'-0" (4267)
2"	80	0.218	20'-0" (6096)	1" (25)	13'-0" (3962)	15'-0" (4572)
3"	40	0.216	25'-0" (7620)	1" (25)	16'-0" (4877)	18'-0" (5486)
4"	40	0.237	27'-0" (8230)	1" (25)	18'-0" (5486)	20'-0" (6096)
6"	40	0.280	33'-0" (10058)	1" (25)	22'-0" (6706)	25'-0" (7620)
8"	30	0.277	36'-0" (10973)	1" (25)	24'-0" (7315)	27'-0" (8230)
10"	20	0.250	38'-0" (11582)	1" (25)	25'-0" (7620)	29'-0" (8839)
10"	40	0.365	41'-0" (12497)	1" (25)	27'-0" (8230)	31'-0" (9449)
12"	20	0.250	40'-0" (12192)	1" (25)	26'-0" (7925)	30'-0" (9144)
12"	Std	0.375	44'-0" (13411)	1" (25)	29'-0" (8839)	33'-0" (10058)
14"	10	0.250	42'-0" (12802)	1" (25)	27'-0" (8230)	31'-0" (9449)
14"	Std	0.375	46'-0" (14021)	1" (25)	30'-0" (9144)	34'-0" (10363)
16"	10	0.250	44'-0" (13411)	1" (25)	28'-0" (8534)	33'-0" (10058)
16"	Std	0.375	48'-0" (14630)	1" (25)	31'-0" (9449)	36'-0" (10973)
18"	10	0.250	45'-0" (13716)	1" (25)	29'-0" (8839)	34'-0" (10363)
18"	Std	0.375	50'-0" (15240)	1" (25)	33'-0" (10058)	38'-0" (11582)
20"	10	0.250	47'-0" (14326)	1" (25)	31'-0" (9449)	36'-0" (10973)
20"	20	0.375	52'-0" (15850)	1" (25)	34'-0" (10363)	39'-0" (11887)
24"	10	0.250	49'-0" (14935)	1" (25)	32'-0" (9754)	37'-0" (11278)
24"	20	0.375	55'-0" (16764)	1" (25)	36'-0" (10973)	41'-0" (12497)

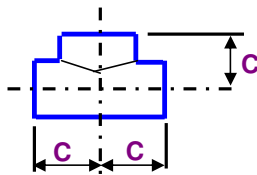


English Units

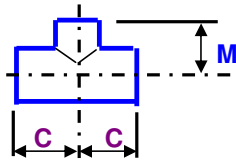
Nom. Pipe Size	Wall Thickness INCHES														A IN	B IN	K IN	D IN	V IN	E IN	F INCHES		G IN
	Pipe O.D.	Light Wall	Sch. 20	Sch. 30	Sch. Std	Sch. 40	Sch. 60	Sch. X-Stg.	Sch. 80	Sch. 100	Sch. 120	Sch. 140	Sch. 160	Sch. XX-Stg.									
																					ASA	MSS	
1/2	0.840	0.083			0.109	0.109		0.147	0.147				0.188	0.294	1 1/2	5/8	1 7/8			1	3	2	1 3/8
3/4	1.050	0.083			0.113	0.113		0.154	0.154				0.219	0.308	1 1/8	7/16	1 11/16			1	3	2	1 11/16
1	1.315	0.109			0.133	0.133		0.179	0.179				0.250	0.358	1 1/2	7/8	2 3/16	1	1 5/8	1 1/2	4	2	2
1 1/4	1.660	0.109			0.140	0.140		0.191	0.191				0.250	0.382	1 7/8	1	2 3/4	1 1/4	2 1/16	1 1/2	4	2	2 1/2
1 1/2	1.900	0.109			0.145	0.145		0.200	0.200				0.281	0.400	2 1/4	1 1/8	3 1/4	1 1/2	2 7/16	1 1/2	4	2	2 7/8
2	2.375	0.109			0.154	0.154		0.218	0.218				0.344	0.436	3	1 3/8	4 3/16	2	3 3/16	1 1/2	6	2 1/2	3 5/8
2 1/2	2.875	0.120			0.203	0.203		0.276	0.276				0.375	0.552	3 3/4	1 3/4	5 3/16	2 1/2	3 15/16	1 1/2	6	2 1/2	4 1/8
3	3.500				0.216	0.216		0.300	0.300				0.438	0.600	4 1/2	2	6 1/4	3	4 3/4	2	6	2 1/2	5
3 1/2	4.000				0.226	0.226		0.318	0.318				0.636	0.875	5 1/4	2 1/4	7 1/4	3 1/2	5 1/2	2 1/2	6	3	5 1/2
4	4.500				0.237	0.237		0.337	0.337		0.438		0.531	0.674	6	2 1/2	8 1/4	4	6 1/4	2 1/2	6	3	6 3/16
5	5.563				0.258	0.258		0.375	0.375		0.500		0.625	0.750	7 1/2	3 1/8	10 5/16	5	7 3/4	3	8	3	7 5/16
6	6.625				0.280	0.280		0.432	0.432		0.562		0.719	0.864	9	3 3/4	12 5/16	6	9 1/2	3 1/2	8	3	8 1/2
8	8.625		0.250	0.277	0.322	0.322	0.406	0.500	0.500	0.594	0.719	0.812	0.906	0.875	12	5	16 5/16	8	12 5/16	4	8	4	10 5/8
10	10.750		0.250	0.307	0.365	0.365	0.500	0.500	0.594	0.719	0.844	1.000	1.125	1.000	15	6 1/4	20 3/8	10	15 3/8	5	10	5	12 3/4
12	12.750		0.250	0.330	0.375	0.406	0.562	0.500	0.688	0.844	1.000	1.125	1.312	1.000	18	7 1/2	24 3/8	12	18 3/8	6	10	6	15
14	14.000		0.312	0.375	0.375	0.438	0.594	0.500	0.750	0.938	1.094	1.250	1.406		21	8 3/4	28	14	21	6 1/2	12		16 1/4
16	16.000		0.312	0.375	0.375	0.500	0.656	0.500	0.844	1.031	1.219	1.438	1.594		24	10	32	16	24	7	12		18 1/2
18	18.000		0.312	0.438	0.375	0.562	0.750	0.500	0.938	1.156	1.375	1.562	1.781		27	11 1/4	36	18	27	8	12		21
20	20.000		0.375	0.500	0.375	0.594	0.812	0.500	1.031	1.281	1.500	1.750	1.969		30	12 1/2	40	20	30	9	12		23
24	24.000		0.375	0.562	0.375	0.688	0.969	0.500	1.219	1.531	1.812	2.062	2.344		36	15	48	24	36	10 1/2	12		27 1/4
30	30.000		0.500	0.625	0.375	0.750		0.500							45	18 1/2	60	30	45	10 1/2			
36	36.000		0.500	0.625	0.375	0.750		0.500							54	22 1/4		36	54	10 1/2			
42	42.000				0.375			0.500							63	26		42		12			
48	48.000				0.375			0.500							72	29 3/4		48		13 1/2			

Metric Units

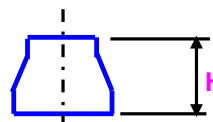
Nom. Pipe Size	Wall Thickness millimeters														A mm	B mm	K mm	D mm	V mm	E mm	F millimeters		G mm
	Pipe O.D.	Light Wall	Sch. 20	Sch. 30	Sch. Std	Sch. 40	Sch. 60	Sch. X-Stg.	Sch. 80	Sch. 100	Sch. 120	Sch. 140	Sch. 160	Sch. XX-Stg.									
																					ASA	MSS	
13	21.34	2.11			2.77	2.77		3.73	3.73				4.78	7.47	38.1	15.9	47.6			25.4	76.2	50.8	34.9
19	26.67	2.11			2.87	2.87		3.91	3.91				5.56	7.82	28.6	11.1	42.9			25.4	76.2	50.8	42.9
25	33.40	2.77			3.38	3.38		4.55	4.55				6.35	9.09	38.1	22.2	55.6	25.4	41.3	38.1	101.6	50.8	50.8
32	42.16	2.77			3.56	3.56		4.85	4.85				6.35	9.70	47.6	25.4	69.9	31.8	52.4	38.1	101.6	50.8	63.5
38	48.26	2.77			3.68	3.68		5.08	5.08				7.14	10.16	57.2	28.6	82.6	38.1	61.9	38.1	101.6	50.8	73.0
51	60.33	2.77			3.91	3.91		5.54	5.54				8.74	11.07	76.2	34.9	106.4	50.8	81.0	38.1	152.4	63.5	92.1
64	73.03	3.05			5.16	5.16		7.01	7.01				9.53	14.02	95.3	44.5	131.8	63.5	100.0	38.1	152.4	63.5	104.8
76	88.90				5.49	5.49		7.62	7.62				11.13	15.24	114.3	50.8	158.8	76.2	120.7	50.8	152.4	63.5	127.0
89	101.60				5.74	5.74		8.08	8.08				16.15		133.4	57.2	184.2	88.9	139.7	63.5	152.4	76.2	139.7
102	114.30				6.02	6.02		8.56	8.56		11.13		13.49	17.12	152.4	63.5	209.6	101.6	158.8	63.5	152.4	76.2	157.2
127	141.30				6.55	6.55		9.53	9.53		12.70		15.88	19.05	190.5	79.4	261.9	127.0	196.9	76.2	203.2	76.2	185.7
152	168.28				7.11	7.11		10.97	10.97		14.27		18.26	21.95	228.6	95.3	312.7	152.4	236.5	88.9	203.2	88.9	215.9
203	219.08		6.35	7.04	8.18	8.18	10.31	12.70	12.70	15.09	18.26	20.62	23.01	22.23	304.8	127.0	414.3	203.2	312.7	101.6	203.2	101.6	269.9
254	273.05		6.35	7.80	9.27	9.27	12.70	12.70	15.09	18.26	21.44	25.40	28.58	25.40	381.0	158.8	517.5	254.0	390.5	127.0	254.0	127.0	323.9
305	323.85		6.35	8.38	9.53	10.31	14.27	12.70	17.48	21.44	25.40	28.58	33.32	25.40	457.2	190.5	619.1	304.8	466.7	152.4	254.0	152.4	381.0
356	356.60		7.92	9.53	9.53	11.13	15.09	12.70	19.05	23.83	27.79	31.75	35.71		533.4	222.3	711.2	355.6	533.4	165.1	304.8		412.8
406	406.40		7.92	9.53	9.53	12.70	16.66	12.70	21.44	26.19	30.96	36.53	40.49		609.6	254.0	812.8	406.4	609.6	177.8	304.8		469.9
457	457.20		7.92	11.13	9.53	14.27	19.05	12.70	23.83	29.36	34.93	39.67	45.24		685.8	285.8	914.4	457.2	685.8	203.2	304.8		533.4
508	508.00		9.53	12.70	9.53	15.09	20.62	12.70	26.19	32.54	38.10	44.45	50.01		762.0	317.5	1016.0	508.0	762.0	228.6	304.8		584.2
610	609.60		9.53	14.27	9.53	17.48	24.61	12.70	30.96	38.89	46.02	52.37	59.54		914.4	381.0	1219.2	609.6	914.4	266.7	304.8		692.2
762	762.00		12.70	15.88	9.53			12.70							1143.0	469.9	1524.0	762.0	1143.0	266.7			
914	914.40		12.70	15.88	9.53	19.05		12.70							1371.6	565.2		914.4	1371.6	266.7			
1067	1066.80				9.53			12.70							1600.2	660.4		1066.8		304.8			
1219	1219.20				9.53			12.70							1828.8	755.7		1219.2		342.9			



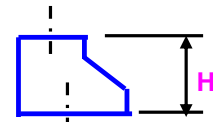
Straight Tee



Reducing Tee



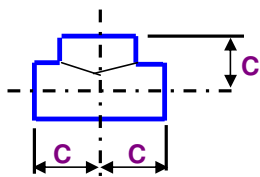
Conc. Reducer



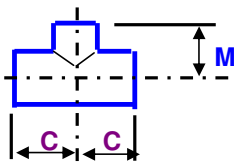
Eccentric Reducer

English Units

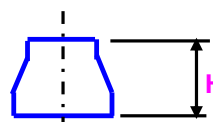
Nom. Pipe Size IN	Outlet IN	C IN	M IN	H IN	Nom. Pipe Size IN	Outlet IN	C IN	M IN	H IN	Nom. Pipe Size IN	Outlet IN	C IN	M IN	H IN	Nom. Pipe Size IN	Outlet IN	C IN	M IN	H IN	
3/4	3/4	1 1/8			3 1/2	3 1/2	3 3/4			10	10	8 1/2			20	20	15			
3/4	1/2	1 1/8	1 1/8	1 1/2		3	3 3/4	3 5/8	4		8	8 1/2	8	7		18	15	14 1/2	20	
1	1	1 1/2				2 1/2	3 3/4	3 1/2	4		6	8 1/2	7 5/8	7		16	15	14	20	
1	3/4	1 1/2	1 1/2	2		2	3 3/4	3 1/4	4		5	8 1/2	7 1/2	7		14	15	14	20	
1	1/2	1 1/2	1 1/2	2	3 1/2	1 1/2	3 3/4	3 1/8	4	10	4	8 1/2	7 1/4	7		12	15	13 5/8	20	
1 1/4	1 1/4	1 7/8			4	4	4 1/8			12	12	10				10	15	13 1/8	20	
	1	1 7/8	1 7/8	2		3 1/2	4 1/8	4	4		10	10	9 1/2	8	20	8	15	12 3/4	20	
	3/4	1 7/8	1 7/8	2		3	4 1/8	3 7/8	4		8	10	9	8	24	24	17			
1 1/4	1/2	1 7/8	1 7/8	2		2 1/2	4 1/8	3 3/4	4		6	10	8 5/8	8		20	17	17	20	
1 1/2	1 1/2	2 1/4				2	4 1/8	3 1/2	4	12	5	10	8 1/2	8		18	17	16 1/2	20	
	1 1/4	2 1/4	2 1/4	2 1/2	4	1 1/2	4 1/8	3 3/8	4	14	14	11				16	17	16	20	
	1	2 1/4	2 1/4	2 1/2	5	5	4 7/8				12	11	10 5/8	13		14	17	16	20	
	3/4	2 1/4	2 1/4	2 1/2		4	4 7/8	4 5/8	5		10	11	10 1/8	13		12	17	15 5/8	20	
1 1/2	1/2	2 1/4	2 1/4	2 1/2		3 1/2	4 7/8	4 1/2	5		8	11	9 3/4	13	24	10	17	15 1/8	20	
2	2	2 1/2				3	4 7/8	4 3/8	5	14	6	11	9 3/8	13	30	30	22			
	1 1/2	2 1/2	2 3/8	3		2 1/2	4 7/8	4 1/4	5	16	16	12				24	22	21	24	
	1 1/4	2 1/2	2 1/4	3	5	2	4 7/8	4 1/8	5		14	12	12	14		20	22	20	24	
	1	2 1/2	2	3	6	6	5 5/8				12	12	11 5/8	14		18	22	19 1/2	24	
2	3/4	2 1/2	1 3/4	3		5	5 5/8	5 3/8	5 1/2		10	12	11 1/8	14		16	22	19	24	
2 1/2	2 1/2	3				4	5 5/8	5 1/8	5 1/2		8	12	10 3/4	14	30	14	22	19	24	
	2	3	2 3/4	3 1/2		3 1/2	5 5/8	5	5 1/2	16	6	12	10 3/8	14	36	36	26 1/2			
	1 1/2	3	2 5/8	3 1/2		3	5 5/8	4 7/8	5 1/2	18	18	13 1/2				30	26 1/2	25	24	
	1 1/4	3	2 1/2	3 1/2	6	2 1/2	5 5/8	4 3/4	5 1/2		16	13 1/2	13	15		24	26 1/2	24	24	
2 1/2	1	3	2 1/4	3 1/2	8	8	7				14	13 1/2	13	15		20	26 1/2	23	24	
3	3	3 3/8				6	7	6 5/8	6		12	13 1/2	12 5/8	15		18	26 1/2	22 1/2	24	
	2 1/2	3 3/8	3 1/4	3 1/2		5	7	6 3/8	6		10	13 1/2	12 1/8	15	36	16	26 1/2	22	24	
	2	3 3/8	3	3 1/2		4	7	6 1/8	6	18	8	13 1/2	11 3/4	15	42	42	30	28		
	1 1/2	3 3/8	2 7/8	3 1/2	8	3 1/2	7	6	6							36	30	28	24	
3	1 1/4	3 3/8	2 3/4	3 1/2												30	30	28	24	
																24	30	26	24	
																42	20	30	26	24
																48	48	35	33	
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																36	36	35	31	28
																48	30	35	30	28



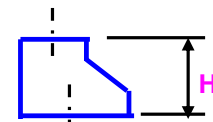
Straight Tee



Reducing Tee



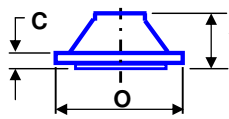
Conc. Reducer



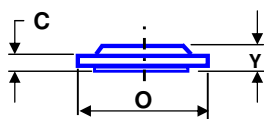
Eccentric Reducer

Metric Units

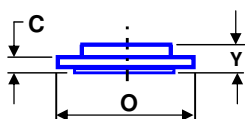
Nom. Pipe Size mm	Outlet mm	C mm	M mm	H mm	Nom. Pipe Size mm	Outlet mm	C mm	M mm	H mm	Nom. Pipe Size mm	Outlet mm	C mm	M mm	H mm	Nom. Pipe Size mm	Outlet mm	C mm	M mm	H mm	
19	19	29			89	89	95			254	254	216			508	508	381			
19	13	29	29	38		76	95	92	102		203	216	203	178		457	381	368	508	
25	25	38				64	95	89	102		152	216	194	178		406	381	356	508	
25	19	38	38	51		51	95	83	102		127	216	191	178		356	381	356	508	
25	13	38	38	51	89	38	95	79	102	254	102	216	184	178		305	381	346	508	
32	32	48			102	102	105			305	305	254				254	381	333	508	
	25	48	48	51		89	105	102	102		254	254	241	203	508	203	381	324	508	
	19	48	48	51		76	105	98	102		203	254	229	203	610	610	432			
32	13	48	48	51		64	105	95	102		152	254	219	203		508	432	432	508	
38	38	57				51	105	89	102	305	127	254	216	203		457	432	419	508	
	32	57	57	64	102	38	105	86	102	356	356	279				406	432	406	508	
	25	57	57	64	127	127	124				305	279	270	330		356	432	406	508	
	19	57	57	64		102	124	117	127		254	279	257	330		305	432	397	508	
38	13	57	57	64		89	124	114	127		203	279	248	330	610	254	432	384	508	
51	51	64				76	124	111	127	356	152	279	238	330	762	762	559			
	38	64	60	76		64	124	108	127	406	406	305				610	559	533	610	
	32	64	57	76	127	51	124	105	127		356	305	305	356		508	559	508	610	
	25	64	51	76	152	152	143				305	305	295	356		457	559	495	610	
51	19	64	44	76		127	143	137	140		254	305	283	356		406	559	483	610	
64	64	76				102	143	130	140		203	305	273	356	762	356	559	483	610	
	51	76	70	89		89	143	127	140	406	152	305	264	356	914	914	673			
	38	76	67	89		76	143	124	140	457	457	343				762	673	635	610	
	32	76	64	89	152	64	143	121	140		406	343	330	381		610	673	610	610	
64	25	76	57	89	203	203	178				356	343	330	381		508	673	584	610	
76	76	86				152	178	168	152		305	343	321	381		457	673	572	610	
	64	86	83	89		127	178	162	152		254	343	308	381	914	406	673	559	610	
	51	86	76	89		102	178	156	152	457	203	343	298	381	1067	1067	762	711		
	38	86	73	89	203	89	178	152	152							914	762	711	610	
76	32	86	70	89												762	762	711	610	
																610	762	660	610	
																1067	508	762	660	610
																1219	1219	889	838	
																	1067	889	813	711
																	914	889	787	711
																1219	762	889	762	711



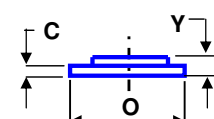
Weld Neck Flange



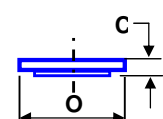
Slip-On Flange



Threaded Flange



Lap Joint Flange



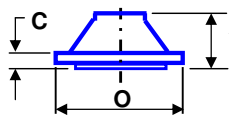
Blind Flange

Nom. Pipe Size	150 LB. FLANGES - INCHES						
	O INCHES	C INCHES	Y				No. & Size of Holes
			Weld Neck	Slip on Thrd.	Lap Joint	Bolt Circle	
1/2	3 1/2	7/16	1 7/8	5/8	5/8	2 3/8	4 - 5/8
3/4	3 7/8	1/2	2 1/16	5/8	5/8	2 3/4	4 - 5/8
1	4 1/4	9/16	2 3/16	11/16	11/16	3 1/8	4 - 5/8
1 1/4	4 5/8	5/8	2 1/4	13/16	13/16	3 1/2	4 - 5/8
1 1/2	5	11/16	2 7/16	7/8	7/8	3 7/8	4 - 5/8
2	6	3/4	2 1/2	1	1	4 3/4	4-3/4
2 1/2	7	7/8	2 3/4	1 1/8	1 1/8	5 1/2	4-3/4
3	7 1/2	15/16	2 3/4	1 3/16	1 3/16	6	4-3/4
3 1/2	8 1/2	15/16	2 13/16	1 1/4	1 1/4	7	8-3/4
4	9	15/16	3	1 5/16	1 5/16	7 1/2	8-3/4
5	10	15/16	3 1/2	1 7/16	1 7/16	8 1/2	8-7/8
6	11	1	3 1/2	1 9/16	1 9/16	9 1/2	8-7/8
8	13 1/2	1 1/8	4	1 3/4	1 3/4	11 3/4	8-7/8
10	16	1 3/16	4	1 15/16	1 15/16	14 1/4	12-1
12	19	1 1/4	4 1/2	2 3/16	2 3/16	17	12-1
14	21	1 3/8	5	2 1/4	3 1/8	18 3/4	12-1 1/8
16	23 1/2	1 7/16	5	2 1/2	3 7/16	21 1/4	16-1 1/8
18	25	1 9/16	5 1/2	2 11/16	3 13/16	22 3/4	16-1 1/4
20	27 1/2	1 11/16	5 11/16	2 7/8	4 1/16	25	20-1 1/4
24	32	1 7/8	6	3 1/4	4 3/8	29 1/2	20-1 3/8

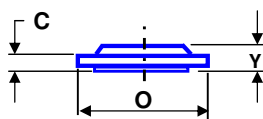
Nom. Pipe Size	150 LB. FLANGES - MILLIMETERS						No. & Size of Holes
	O mm	C mm	Y				
			Weld Neck	Slip on Thrd.	Lap Joint	Bolt Circle	
12.70	89	11	48	16	16	60	4 - 16
19.05	98	13	52	16	16	70	4 - 16
25.40	108	14	56	17	17	79	4 - 16
31.75	117	16	57	21	21	89	4 - 16
38.10	127	17	62	22	22	98	4 - 16
50.80	152	19	64	25	25	121	4-19
63.50	178	22	70	29	29	140	4-19
76.20	191	24	70	30	30	152	4-19
88.90	216	24	71	32	32	178	8-19
101.60	229	24	76	33	33	191	8-19
127.00	254	24	89	37	37	216	8-22
152.40	279	25	89	40	40	241	8-22
203.20	343	29	102	44	44	298	8-22
254.00	406	30	102	49	49	362	12-25
304.80	483	32	114	56	56	432	12-25
355.60	533	35	127	57	79	476	12-29
406.40	597	37	127	64	87	540	16-29
457.20	635	40	140	68	97	578	16-32
508.00	699	43	144	73	103	635	20-32
609.60	813	48	152	83	111	749	20-35

Nom. Pipe Size	300 LB. FLANGES - INCHES						No. & Size of Holes
	O INCHES	C INCHES	Y				
			Weld Neck	Slip on Thrd.	Lap Joint	Bolt Circle	
1/2	3 3/4	9/16	2 1/16	7/8	7/8	2 5/8	4 - 5/8
3/4	4 5/8	5/8	2 1/4	1	1	3 1/4	4 - 3/4
1	4 7/8	11/16	2 7/16	1 1/16	1 1/16	3 1/2	4 - 3/4
1 1/4	5 1/4	3/4	2 9/16	1 1/16	1 1/16	3 7/8	4 - 3/4
1 1/2	6 1/8	13/16	2 11/16	1 3/16	1 3/16	4 1/2	4 - 7/8
2	6 1/2	7/8	2 3/4	1 5/16	1 5/16	5	8-3/4
2 1/2	7 1/2	1	3	1 1/2	1 1/2	5 7/8	8-7/8
3	8 1/4	1 1/8	3 1/8	1 11/16	1 11/16	6 5/8	8-7/8
3 1/2	9	1 3/16	3 3/16	1 3/4	1 3/4	7 1/4	8-7/8
4	10	1 1/4	3 3/8	1 7/8	1 7/8	7 7/8	8-7/8
5	11	1 3/8	3 7/8	2	2	9 1/4	8-7/8
6	12 1/2	1 7/16	3 7/8	2 1/16	2 1/16	10 5/8	12-7/8
8	15	1 5/8	4 3/8	2 7/16	2 7/16	13	12-1
10	17 1/2	1 7/8	4 5/8	2 5/8	3 3/4	15 1/4	16-1 1/8
12	20 1/2	2	5 1/8	2 7/8	4	17 3/4	16-1 1/4
14	23	2 1/8	5 5/8	3	4 3/8	20 1/4	20-1 1/4
16	25 1/2	2 1/4	5 3/4	3 1/4	4 3/4	22 1/2	20-1 3/8
18	28	2 3/8	6 1/4	3 1/2	5 1/8	24 3/4	24-1 3/8
20	30 1/2	2 1/2	6 3/8	3 3/4	5 1/2	27	24-1 3/8
24	36	2 3/4	6 5/8	4 3/16	6	32	24-1 5/8

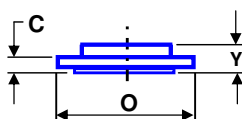
Nom. Pipe Size	300 LB. FLANGES - MILLIMETERS						No. & Size of Holes
	O mm	C mm	Y				
			Weld Neck	Slip on Thrd.	Lap Joint	Bolt Circle	
12.70	95	14	52	22	22	67	4 - 16
19.05	117	16	57	25	25	83	4 - 19
25.40	124	17	62	27	27	89	4 - 19
31.75	133	19	65	27	27	98	4 - 19
38.10	156	21	68	30	30	114	4 - 22
50.80	165	22	70	33	33	127	8-19
63.50	191	25	76	38	38	149	8-22
76.20	210	29	79	43	43	168	8-22
88.90	229	30	81	44	44	184	8-22
101.60	254	32	86	48	48	200	8-22
127.00	279	35	98	51	51	235	8-22
152.40	318	37	98	52	52	270	12-22
203.20	381	41	111	62	62	330	12-25
254.00	445	48	117	67	95	387	16-29
304.80	521	51	130	73	102	451	16-32
355.60	584	54	143	76	111	514	20-32
406.40	648	57	146	83	121	572	20-35
457.20	711	60	159	89	130	629	24-35
508.00	775	64	162	95	140	686	24-35
609.60	914	70	168	106	152	813	24-41



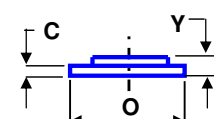
Weld Neck Flange



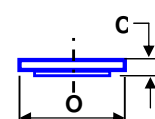
Slip-On Flange



Threaded Flange



Lap Joint Flange



Blind Flange

** The 1/4" raised face is not included in the thickness "Y" or "C"

400 LB. FLANGES - INCHES							
Nom. Pipe Size	O INCHES	C INCHES	**Y (See Note Above)				No. & Size of Holes
			Weld Neck	Slip on Thrd.	Lap Joint	Bolt Circle	
1/2	3 3/4	9/16	2 1/16	7/8	7/8	2 5/8	4 - 5/8
3/4	4 5/8	5/8	2 1/4	1	1	3 1/4	4 - 3/4
1	4 7/8	11/16	2 7/16	1 1/16	1 1/16	3 1/2	4 - 3/4
1 1/4	5 1/4	13/16	2 5/8	1 1/8	1 1/8	3 7/8	4 - 3/4
1 1/2	6 1/8	7/8	2 3/4	1 1/4	1 1/4	4 1/2	4 - 7/8
2	6 1/2	1	2 7/8	1 7/16	1 7/16	5	8-3/4
2 1/2	7 1/2	1 1/8	3 1/8	1 5/8	1 5/8	5 7/8	8-7/8
3	8 1/4	1 1/4	3 1/4	1 13/16	1 13/16	6 5/8	8-7/8
3 1/2	9	1 3/8	3 5/8	1 15/16	1 15/16	7 1/4	8-1
4	10	1 3/8	3 1/2	2	2	7 7/8	8-1
5	11	1 1/2	4	2 1/8	2 1/8	9 1/4	8-1
6	12 1/2	1 5/8	4 1/16	2 1/4	2 1/4	10 5/8	12-1
8	15	1 7/8	4 5/8	2 11/16	2 11/16	13	12-1 1/8
10	17 1/2	2 1/8	4 7/8	2 7/8	4	15 1/4	16-1 1/4
12	20 1/2	2 1/4	5 3/8	3 1/8	4 1/4	17 3/4	16-1 3/8
14	23	2 3/8	5 7/8	3 5/16	4 5/8	20 1/4	20-1 3/8
16	25 1/2	2 1/2	6	3 11/16	5	22 1/2	20-1 1/2
18	28	2 5/8	6 1/2	3 7/8	5 3/8	24 3/4	24-1 1/2
20	30 1/2	2 3/4	6 5/8	4	5 3/4	27	24-1 5/8
24	36	3	6 7/8	4 1/2	6 1/4	32	24-1 7/8

** The 6mm raised face is not included in the thickness "Y" or "C"

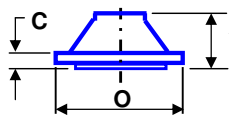
400 LB. FLANGES - MILLIMETERS							
Nom. Pipe Size	O mm	C mm	**Y (See Note Above)				No. & Size of Holes
			Weld Neck	Slip on Thrd.	Lap Joint	Bolt Circle	
12.70	95	14	52	22	22	67	4 - 16
19.05	117	16	57	25	25	83	4 - 19
25.40	124	17	62	27	27	89	4 - 19
31.75	133	21	67	29	29	98	4 - 19
38.10	156	22	70	32	32	114	4 - 22
50.80	165	25	73	37	37	127	8-19
63.50	191	29	79	41	41	149	8-22
76.20	210	32	83	46	46	168	8-22
88.90	229	35	92	49	49	184	8-25
101.60	254	35	89	51	51	200	8-25
127.00	279	38	102	54	54	235	8-25
152.40	318	41	103	57	57	270	12-25
203.20	381	48	117	68	68	330	12-29
254.00	445	54	124	73	102	387	16-32
304.80	521	57	137	79	108	451	16-35
355.60	584	60	149	84	117	514	20-35
406.40	648	64	152	94	127	572	20-38
457.20	711	67	165	98	137	629	24-38
508.00	775	70	168	102	146	686	24-41
609.60	914	76	175	114	159	813	24-48

** The 1/4" raised face is not included in the thickness "Y" or "C"

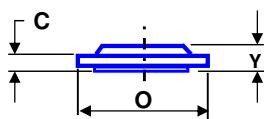
600 LB. FLANGES - INCHES							
Nom. Pipe Size	O INCHES	C INCHES	**Y (See Note Above)				No. & Size of Holes
			Weld Neck	Slip on Thrd.	Lap Joint	Bolt Circle	
1/2	3 3/4	9/16	2 1/16	7/8	7/8	2 5/8	4 - 5/8
3/4	4 5/8	5/8	2 1/4	1	1	6 1/4	4 - 3/4
1	4 7/8	11/16	2 7/16	1 1/16	1 1/16	3 1/2	4 - 3/4
1 1/4	5 1/4	13/16	2 5/8	1 1/8	1 1/8	3 7/8	4 - 3/4
1 1/2	6 1/8	7/8	2 3/4	1 1/4	1 1/4	4 1/2	4 - 7/8
2	6 1/2	1	2 7/8	1 7/16	1 7/16	5	8-3/4
2 1/2	7 1/2	1 1/8	3 1/8	1 5/8	1 5/8	5 7/8	8-7/8
3	8 1/4	1 1/4	3 1/4	1 13/16	1 13/16	6 5/8	8-7/8
3 1/2	9	1 3/8	3 3/8	1 15/16	1 15/16	7 1/4	8-1
4	10 3/4	1 1/2	4	2 1/8	2 1/8	8 1/2	8-1
5	13	1 3/4	4 1/2	2 3/8	2 3/8	10 1/2	8-1 1/8
6	14	1 7/8	4 5/8	2 5/8	2 5/8	11 1/2	12-1 1/8
8	16 1/2	2 3/16	5 1/4	3	3	13 3/4	12-1 1/4
10	20	2 1/2	6	3 3/8	4 3/8	17	16-1 3/8
12	22	2 5/8	6 1/8	3 5/8	4 5/8	19 1/4	20-1 3/8
14	23 3/4	2 3/4	6 1/2	3 11/16	5	20 3/4	20-1 1/2
16	27	3	7	4 3/16	5 1/2	23 3/4	20-1 5/8
18	29 1/4	3 1/4	7 1/4	4 5/8	6	25 3/4	20-1 3/4
20	32	3 1/2	7 1/2	5	6 1/2	28 1/2	24-1 3/4
24	37	4	8	5 1/2	7 1/4	33	24-2

** The 6mm raised face is not included in the thickness "Y" or "C"

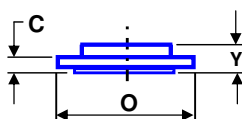
600 LB. FLANGES - MILLIMETERS							
Nom. Pipe Size	O mm	C mm	**Y (See Note Above)				No. & Size of Holes
			Weld Neck	Slip on Thrd.	Lap Joint	Bolt Circle	
12.70	95	14	52	22	22	67	4 - 16
19.05	117	16	57	25	25	159	4 - 19
25.40	124	17	62	27	27	89	4 - 19
31.75	133	21	67	29	29	98	4 - 19
38.10	156	22	70	32	32	114	4 - 22
50.80	165	25	73	37	37	127	8-19
63.50	191	29	79	41	41	149	8-22
76.20	210	32	83	46	46	168	8-22
88.90	229	35	86	49	49	184	8-25
101.60	273	38	102	54	54	216	8-25
127.00	330	44	114	60	60	267	8-29
152.40	356	48	117	67	67	292	12-29
203.20	419	56	133	76	76	349	12-32
254.00	508	64	152	86	111	432	16-35
304.80	559	67	156	92	117	489	20-35
355.60	603	70	165	94	127	527	20-38
406.40	686	76	178	106	140	603	20-41
457.20	743	83	184	117	152	654	20-44
508.00	813	89	191	127	165	724	24-44
609.60	940	102	203	140	184	838	24-51



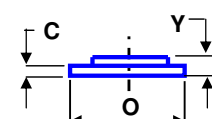
Weld Neck Flange



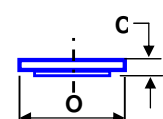
Slip-On Flange



Threaded Flange



Lap Joint Flange



Blind Flange

** The 1/4" raised face is not included in the thickness "Y" or "C"

Nom. Pipe Size	900 LB. FLANGES - INCHES						
	O INCHES	C INCHES	**Y (See Note Above)				No. & Size of Holes
			Weld Neck	Slip on Thrd.	Lap Joint	Bolt Circle	
1/2	4 3/4	7/8	2 3/8	1 1/4	1 1/4	3 1/4	4 - 7/8
3/4	5 1/8	1	2 3/4	1 3/8	1 3/8	3 1/2	4 - 7/8
1	5 7/8	1 1/8	2 7/8	1 5/8	1 5/8	4	4 - 1
1 1/4	6 1/4	1 1/8	2 7/8	1 5/8	1 5/8	4 3/8	4 - 1
1 1/2	7	1 1/4	3 1/4	1 3/4	1 3/4	4 7/8	4 - 1 1/8
2	8 1/2	1 1/2	4	2 1/4	2 1/4	6 1/2	8-1
2 1/2	9 5/8	1 5/8	4 1/8	2 1/2	2 1/2	7 1/2	8-1 1/8
3	9 1/2	1 1/2	4	2 1/8	2 1/8	7 1/2	8-1
3 1/2							
4	11 1/2	1 3/4	4 1/2	2 3/4	2 3/4	9 1/4	8-1 1/4
5	13 3/4	2	5	3 1/8	3 1/8	11	8-1 3/8
6	15	2 3/16	5 1/2	3 3/8	3 3/8	12 1/2	12-1 1/4
8	18 1/2	2 1/2	6 3/8	4	4 1/2	15 1/2	12-1 1/2
10	21 1/2	2 3/4	7 1/4	4 1/4	5	18 1/2	16-1 1/2
12	24	3 1/8	7 7/8	4 5/8	5 5/8	21	20-1 1/2
14	25 1/4	3 3/8	8 3/8	5 1/8	6 1/8	22	20-1 5/8
16	27 3/4	3 1/2	8 1/2	5 1/4	6 1/2	24 1/4	20-1 3/4
18	31	4	9	6	7 1/2	27	20-2
20	33 3/4	4 1/4	9 3/4	6 1/4	8 1/4	29 1/2	24-2 1/8
24	41	5 1/2	11 1/2	8	10 1/2	35 1/2	24-2 5/8

** The 1/4" raised face is not included in the thickness "Y" or "C"

Nom. Pipe Size	1500 LB. FLANGES - INCHES						
	O INCHES	C INCHES	**Y (See Note Above)				No. & Size of Holes
			Weld Neck	Slip on Thrd.	Lap Joint	Bolt Circle	
1/2	4 3/4	7/8	2 3/8	1 1/4	1 1/4	3 1/4	4 - 7/8
3/4	5 1/8	1	2 3/4	1 3/8	1 3/8	3 1/2	4 - 7/8
1	5 7/8	1 1/8	2 7/8	1 5/8	1 5/8	4	4 - 1
1 1/4	6 1/4	1 1/8	2 7/8	1 5/8	1 5/8	4 3/8	4 - 1
1 1/2	7	1 1/4	3 1/4	1 3/4	1 3/4	4 7/8	4 - 1 1/8
2	8 1/2	1 1/2	4	2 1/4	2 1/4	6 1/2	8-1
2 1/2	9 5/8	1 5/8	4 1/8	2 1/2	2 1/2	7 1/2	8-1 1/8
3	10 1/2	1 7/8	4 5/8	2 7/8	2 7/8	8	8-1 1/4
3 1/2							
4	12 1/4	2 1/8	4 7/8	3 9/16	3 9/16	9 1/2	8-1 3/8
5	14 3/4	2 7/8	6 1/8	4 1/8	4 1/8	11 1/2	8-1 5/8
6	15 1/2	3 1/4	6 3/4	4 11/16	4 11/16	12 1/2	12-1 1/2
8	19	3 5/8	8 3/8	5 5/8	5 5/8	15 1/2	12-1 3/4
10	23	4 1/4	10	6 1/4	7	19	12-2
12	26 1/2	4 7/8	11 1/8	7 1/8	8 5/8	22 1/2	16-2 1/8
14	29 1/2	5 1/4	11 3/4		9 1/2	25	16-2 3/8
16	32 1/2	5 3/4	12 1/4		10 1/4	27 3/4	16 - 2 5/8
18	36	6 3/8	12 7/8		10 7/8	30 1/2	16 - 2 7/8
20	38 3/4	7	14		11 1/2	32 3/4	16-3 1/8
24	46	8	16		13	39	16 - 3 5/8

** The 6mm raised face is not included in the thickness "Y" or "C"

Nom. Pipe Size	900 LB. FLANGES - MILLIMETERS						
	O mm	C mm	**Y (See Note Above)				No. & Size of Holes
			Weld Neck	Slip on Thrd.	Lap Joint	Bolt Circle	
12.70	121	22	60	32	32	83	4 - 22
19.05	130	25	70	35	35	89	4 - 22
25.40	149	29	73	41	41	102	4 - 25
31.75	159	29	73	41	41	111	4 - 25
38.10	178	32	83	44	44	124	4 - 29
50.80	216	38	102	57	57	165	8-25
63.50	244	41	105	64	64	191	8-29
76.20	241	38	102	54	54	191	8-25
88.90							
101.60	292	44	114	70	70	235	8-32
127.00	349	51	127	79	79	279	8-35
152.40	381	56	140	86	86	318	12-32
203.20	470	64	162	102	114	394	12-38
254.00	546	70	184	108	127	470	16-38
304.80	610	79	200	117	143	533	20-38
355.60	641	86	213	130	156	559	20-41
406.40	705	89	216	133	165	616	20-44
457.20	787	102	229	152	191	686	20-51
508.00	857	108	248	159	210	749	24-54
609.60	1041	140	292	203	267	902	24-67

** The 6mm raised face is not included in the thickness "Y" or "C"

Nom. Pipe Size	1500 LB. FLANGES - MILLIMETERS						
	O mm	C mm	**Y (See Note Above)				No. & Size of Holes
			Weld Neck	Slip on Thrd.	Lap Joint	Bolt Circle	
12.70	121	22	60	32	32	83	4 - 22
19.05	130	25	70	35	35	89	4 - 22
25.40	149	29	73	41	41	102	4 - 25
31.75	159	29	73	41	41	111	4 - 25
38.10	178	32	83	44	44	124	4 - 29
50.80	216	38	102	57	57	165	8-25
63.50	244	41	105	64	64	191	8-29
76.20	267	48	117	73	73	203	8-32
88.90							
101.60	311	54	124	90	90	241	8-35
127.00	375	73	156	105	105	292	8-41
152.40	394	83	171	119	119	318	12-38
203.20	483	92	213	143	143	394	12-44
254.00	584	108	254	159	178	483	12-51
304.80	673	124	283	181	219	572	16-54
355.60	749	133	298	0	241	635	16-60
406.40	826	146	311	0	260	705	16 - 67
457.20	914	162	327	0	276	775	16- 73
508.00	984	178	356	0	292	832	16-79
609.60	1168	203	406	0	330	991	16 - 92

150 # FLANGE WEIGHTS

English

Nom. Pipe Size Inches	150# WNFLG. Wt. Lbs	150# SO-FLG. Wt. Lbs	150# Thrd-FLG. Wt. Lbs	150# Lap-FLG. Wt. Lbs	150# Blind-FLG. Wt. Lbs
1/2	2	1	1	1	2
3/4	2	1.5	1.5	1.5	2
1	2.5	2	2	2	2
1 1/4	2.5	2.5	2.5	2.5	3
1 1/2	4	3	3	3	3
2	6	5	5	5	4
2 1/2	10	8	8	8	7
3	11.5	9	10	9	9
3 1/2	12	11	12	11	13
4	16.5	13	13	12	17
5	21	15	15	13	20
6	26	17	19.5	18	27
8	42	28	30	28	47
10	54	40	41	36	67
12	88	61	65	60	102
14	114	83	85	77	139
16	142	106	93	104	187
18	165	109	120	146	217
20	197	148	155	159	283
24	268	204	210	195	415

Metric

Nom. Pipe Size mm	150# WNFLG. Wt. Kg	150# SO-FLG. Wt. Kg	150# Thrd-FLG. Wt. Kg	150# Lap-FLG. Wt. Kg	150# Blind-FLG. Wt. Kg
12.70	0.9	0.5	0.5	0.5	0.9
19.05	0.9	0.7	0.7	0.7	0.9
25.40	1.1	0.9	0.9	0.9	0.9
31.75	1.1	1.1	1.1	1.1	1.4
38.10	1.8	1.4	1.4	1.4	1.4
63.50	2.7	2.3	2.3	2.3	1.8
63.50	4.5	3.6	3.6	3.6	3.2
76.20	5.2	4.1	4.5	4.1	4.1
88.90	5.4	5.0	5.4	5.0	5.9
101.60	7.5	5.9	5.9	5.4	7.7
152.40	9.5	6.8	6.8	5.9	9.1
152.40	11.8	7.7	8.8	8.2	12.2
203.20	19.1	12.7	13.6	12.7	21.3
254.00	24.5	18.1	18.6	16.3	30.4
304.80	39.9	27.7	29.5	27.2	46.3
406.40	51.7	37.6	38.6	34.9	63.0
406.40	64.4	48.1	42.2	47.2	84.8
457.20	74.8	49.4	54.4	66.2	98.4
508.00	89.4	67.1	70.3	72.1	128.4
609.60	121.6	92.5	95.3	88.5	188.2

300 # FLANGE WEIGHTS

English

Nom. Pipe Size Inches	300# WNFLG. Wt. Lbs	300# SO-FLG. Wt. Lbs	300# Thrd-FLG. Wt. Lbs	300# Lap-FLG. Wt. Lbs	300# Blind-FLG. Wt. Lbs
1/2	2	1.5	1.5	1.5	2
3/4	3	2.5	2.5	2.5	3
1	4	3	3	3	4
1 1/4	5	4.5	4.5	4.5	6
1 1/2	7	6.5	6.5	6.5	7
2	8	7	7	7	8
2 1/2	12	10	10	10	12
3	18	13	14	14.5	16
3 1/2	20	16	16	16	21
4	26.5	23.5	24	24	28
5	36	29	31	26	37
6	45	36	36	38	48
8	69	56	56	55	79
10	100	77	80	88	122
12	142	113	110	139	183
14	206	159	164	184	241
16	249	210	220	234	315
18	306	253	280	305	414
20	369	307	325	375	515
24	519	490	490	530	800

Metric

Nom. Pipe Size mm	300# WNFLG. Wt. Kg	300# SO-FLG. Wt. Kg	300# Thrd-FLG. Wt. Kg	300# Lap-FLG. Wt. Kg	300# Blind-FLG. Wt. Kg
12.70	0.9	0.7	0.7	0.7	0.9
19.05	1.4	1.1	1.1	1.1	1.4
25.40	1.8	1.4	1.4	1.4	1.8
31.75	2.3	2.0	2.0	2.0	2.7
38.10	3.2	2.9	2.9	2.9	3.2
63.50	3.6	3.2	3.2	3.2	3.6
63.50	5.4	4.5	4.5	4.5	5.4
76.20	8.2	5.9	6.4	6.6	7.3
88.90	9.1	7.3	7.3	7.3	9.5
101.60	12.0	10.7	10.9	10.9	12.7
152.40	16.3	13.2	14.1	11.8	16.8
152.40	20.4	16.3	16.3	17.2	21.8
203.20	31.3	25.4	25.4	24.9	35.8
254.00	45.4	34.9	36.3	39.9	55.3
304.80	64.4	51.3	49.9	63.0	83.0
406.40	93.4	72.1	74.4	83.5	109.3
406.40	112.9	95.3	99.8	106.1	142.9
457.20	138.8	114.8	127.0	138.3	187.8
508.00	167.4	139.3	147.4	170.1	233.6
609.60	235.4	222.3	222.3	240.4	362.9

400 # FLANGE WEIGHTS

English

Nom. Pipe Size Inches	400 # WNFLG. Wt. Lbs	400 # SO-FLG. Wt. Lbs	400 # Thrd-FLG. Wt. Lbs	400 # Lap-FLG. Wt. Lbs	400 # Blind-FLG. Wt. Lbs
1/2	3	2	2	2	2
3/4	3.5	3	3	3	3
1	4	3.5	3.5	3.5	4
1 1/4	5.5	4.5	4.5	4.5	6
1 1/2	8	6.5	6.5	6.5	8
2	10	8	8	8	10
2 1/2	14	12	12	11	15
3	18	15	15	14	20
3 1/2	26	21	21	20	29
4	30	24	24	22	33
5	39	31	31	29	44
6	49	39	39	37	61
8	78	63	63	59	100
10	110	91	91	95	155
12	160	129	129	152	226
14	233	191	191	210	310
16	294	253	253	280	398
18	260	310	310	345	502
20	445	378	378	420	621
24	640	539	539	615	936

Metric

Nom. Pipe Size mm	400 # WNFLG. Wt. Kg	400 # SO-FLG. Wt. Kg	400 # Thrd-FLG. Wt. Kg	400 # Lap-FLG. Wt. Kg	400 # Blind-FLG. Wt. Kg
12.70	1.4	0.9	0.9	0.9	0.9
19.05	1.6	1.4	1.4	1.4	1.4
25.40	1.8	1.6	1.6	1.6	1.8
31.75	2.5	2.0	2.0	2.0	2.7
38.10	3.6	2.9	2.9	2.9	3.6
63.50	4.5	3.6	3.6	3.6	4.5
63.50	6.4	5.4	5.4	5.0	6.8
76.20	8.2	6.8	6.8	6.4	9.1
88.90	11.8	9.5	9.5	9.1	13.2
101.60	13.6	10.9	10.9	10.0	15.0
152.40	17.7	14.1	14.1	13.2	20.0
152.40	22.2	17.7	17.7	16.8	27.7
203.20	35.4	28.6	28.6	26.8	45.4
254.00	49.9	41.3	41.3	43.1	70.3
304.80	72.6	58.5	58.5	68.9	102.5
406.40	105.7	86.6	86.6	95.3	140.6
406.40	133.4	114.8	114.8	127.0	180.5
457.20	117.9	140.6	140.6	156.5	227.7
508.00	201.8	171.5	171.5	190.5	281.7
609.60	290.3	244.5	244.5	279.0	424.6

600 # FLANGE WEIGHTS

English

Nom. Pipe Size Inches	600 # WNFLG. Wt. Lbs	600 # SO-FLG. Wt. Lbs	600 # Thrd-FLG. Wt. Lbs	600 # Lap-FLG. Wt. Lbs	600 # Blind-FLG. Wt. Lbs
1/2	3	2	2	2	2
3/4	3.5	3	3	3	3
1	4	3.5	3.5	3.5	4
1 1/4	5.5	4.5	4.5	4.5	6
1 1/2	8	6.5	6.5	6.5	8
2	10	8	8	8	10
2 1/2	14	12	12	11	15
3	18	15	15	14	20
3 1/2	26	21	21	20	29
4	37	33	33	31	41
5	68	56	63	63	68
6	73	66	80	78	86
8	112	97	97	112	139
10	189	177	177	195	231
12	226	215	215	240	295
14	268	259	259	290	378
16	481	366	366	400	527
18	555	476	476	469	665
20	690	612	612	604	855
24	977	876	876	866	1175

Metric

Nom. Pipe Size mm	600 # WNFLG. Wt. Kg	600 # SO-FLG. Wt. Kg	600 # Thrd-FLG. Wt. Kg	600 # Lap-FLG. Wt. Kg	600 # Blind-FLG. Wt. Kg
12.70	1.4	0.9	0.9	0.9	0.9
19.05	1.6	1.4	1.4	1.4	1.4
25.40	1.8	1.6	1.6	1.6	1.8
31.75	2.5	2.0	2.0	2.0	2.7
38.10	3.6	2.9	2.9	2.9	3.6
63.50	4.5	3.6	3.6	3.6	4.5
63.50	6.4	5.4	5.4	5.0	6.8
76.20	8.2	6.8	6.8	6.4	9.1
88.90	11.8	9.5	9.5	9.1	13.2
101.60	16.8	15.0	15.0	14.1	18.6
152.40	30.8	25.4	28.6	28.6	30.8
152.40	33.1	29.9	36.3	35.4	39.0
203.20	50.8	44.0	44.0	50.8	63.0
254.00	85.7	80.3	80.3	88.5	104.8
304.80	102.5	97.5	97.5	108.9	133.8
406.40	121.6	117.5	117.5	131.5	171.5
406.40	218.2	166.0	166.0	181.4	239.0
457.20	251.7	215.9	215.9	212.7	301.6
508.00	313.0	277.6	277.6	274.0	387.8
609.60	443.2	397.3	397.3	392.8	533.0

900 # FLANGE WEIGHTS

English

Nom. Pipe Size Inches	900 # WNFLG. Wt. Lbs	900 # SO-FLG. Wt. Lbs	900 # Thrd-FLG. Wt. Lbs	900 # Lap-FLG. Wt. Lbs	900 # Blind-FLG. Wt. Lbs
1/2	7	6	6	6	4
3/4	7	6	6	6	6
1	8.5	7.5	7.5	7.5	9
1 1/4	10	10	10	10	10
1 1/2	14	14	14	14	14
2	24	22	22	21	25
2 1/2	36	36	36	29	35
3	29	31	31	25	32
4	51	53	53	51	54
5	86	83	83	81	87
6	110	108	108	105	113
8	187	172	172	188	197
10	268	245	245	277	290
12	372	326	326	371	413
14	562	380	380	397	494
16	685	459	459	488	619
18	924	647	647	670	880
20	1164	792	792	868	1107
24	2107	1480	1480	1659	2099

Metric

Nom. Pipe Size mm	900 # WNFLG. Wt. Kg	900 # SO-FLG. Wt. Kg	900 # Thrd-FLG. Wt. Kg	900 # Lap-FLG. Wt. Kg	900 # Blind-FLG. Wt. Kg
12.70	3.2	2.7	2.7	2.7	1.8
19.05	3.2	2.7	2.7	2.7	2.7
25.40	3.9	3.4	3.4	3.4	4.1
31.75	4.5	4.5	4.5	4.5	4.5
38.10	6.4	6.4	6.4	6.4	6.4
63.50	10.9	10.0	10.0	9.5	11.3
63.50	16.3	16.3	16.3	13.2	15.9
76.20	13.2	14.1	14.1	11.3	14.5
101.60	23.1	24.0	24.0	23.1	24.5
152.40	39.0	37.6	37.6	36.7	39.5
152.40	49.9	49.0	49.0	47.6	51.3
203.20	84.8	78.0	78.0	85.3	89.4
254.00	121.6	111.1	111.1	125.6	131.5
304.80	168.7	147.9	147.9	168.3	187.3
406.40	254.9	172.4	172.4	180.1	224.1
406.40	310.7	208.2	208.2	221.4	280.8
457.20	419.1	293.5	293.5	303.9	399.2
508.00	528.0	359.2	359.2	393.7	502.1
609.60	955.7	671.3	671.3	752.5	952.1

1500 # FLANGE WEIGHTS

English

Nom. Pipe Size Inches	1500 # WNFLG. Wt. Lbs	1500 # SO-FLG. Wt. Lbs	1500 # Thrd-FLG. Wt. Lbs	1500 # Lap-FLG. Wt. Lbs	1500 # Blind-FLG. Wt. Lbs
1/2	7	6	6	6	7
3/4	7	6	6	6	10
1	8.5	7.5	7.5	7.5	12
1 1/4	10	10	10	10	18
1 1/2	14	14	14	14	25
2	24	22	23	21	39
2 1/2	36	36	36	29	56
3	48	48	48	38	86
4	69	73	73	75	133
5	132	132	132	138	223
6	164	164	164	170	345
8	273	258	258	286	533
10	454	436	436	485	1025
12	690	667	667	749	1464
14					
16					
18					
20					
24					

Metric

Nom. Pipe Size mm	1500 # WNFLG. Wt. Kg	1500 # SO-FLG. Wt. Kg	1500 # Thrd-FLG. Wt. Kg	1500 # Lap-FLG. Wt. Kg	1500 # Blind-FLG. Wt. Kg
12.70	3.2	2.7	2.7	2.7	3.2
19.05	3.2	2.7	2.7	2.7	4.5
25.40	3.9	3.4	3.4	3.4	5.4
31.75	4.5	4.5	4.5	4.5	8.2
38.10	6.4	6.4	6.4	6.4	11.3
63.50	10.9	10.0	10.4	9.5	17.7
63.50	16.3	16.3	16.3	13.2	25.4
76.20	21.8	21.8	21.8	17.2	39.0
101.60	31.3	33.1	33.1	34.0	60.3
152.40	59.9	59.9	59.9	62.6	101.2
152.40	74.4	74.4	74.4	77.1	156.5
203.20	123.8	117.0	117.0	129.7	241.8
254.00	205.9	197.8	197.8	220.0	464.9
304.80	313.0	302.5	302.5	339.7	664.1
406.40					
406.40					
457.20					
508.00					
609.60					

2500 # FLANGE WEIGHTS

English

Nom. Pipe Size Inches	2500 # WNFLG. Wt. Lbs	2500 # SO-FLG. Wt. Lbs	2500 # Thrd-FLG. Wt. Lbs	2500 # Lap-FLG. Wt. Lbs	2500 # Blind-FLG. Wt. Lbs
1/2	8	7	7	7	7
3/4	9	9	9	8	10
1	13	12	12	12	12
1 1/4	20	18	18	17	18
1 1/2	28	25	25	24	25
2	42	38	38	37	39
2 1/2	52	55	55	53	56
3	94	83	83	80	86
4	146	127	127	122	133
5	244	210	210	204	223
6	378	323	323	314	345
8	576	485	485	471	533
10	1068	925	925	897	1025
12	1608	1300	1300	1262	1464
14					
16					
18					
20					
24					

Metric

Nom. Pipe Size mm	2500 # WNFLG. Wt. Kg	2500 # SO-FLG. Wt. Kg	2500 # Thrd-FLG. Wt. Kg	2500 # Lap-FLG. Wt. Kg	2500 # Blind-FLG. Wt. Kg
12.70	3.6	3.2	3.2	3.2	3.2
19.05	4.1	4.1	4.1	3.6	4.5
25.40	5.9	5.4	5.4	5.4	5.4
31.75	9.1	8.2	8.2	7.7	8.2
38.10	12.7	11.3	11.3	10.9	11.3
63.50	19.1	17.2	17.2	16.8	17.7
63.50	23.6	24.9	24.9	24.0	25.4
76.20	42.6	37.6	37.6	36.3	39.0
101.60	66.2	57.6	57.6	55.3	60.3
152.40	110.7	95.3	95.3	92.5	101.2
152.40	171.5	146.5	146.5	142.4	156.5
203.20	261.3	220.0	220.0	213.6	241.8
254.00	484.4	419.6	419.6	406.9	464.9
304.80	729.4	589.7	589.7	572.4	664.1
406.40					
406.40					
457.20					
508.00					
609.60					